From Potentials and Chemicals to the Mind

A Study of the Human Mind through Neurology, Psychiatry, Psychedelics, and the Philosophy of Mind

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Introduction

The title of these notes is meant literally. They begin with ions crossing a membrane and they intend to arrive, by stages and without any step that is not physical, at seeing, remembering, wanting, fearing, and knowing oneself to be a self. Whether that journey can be completed is a real question. What is not in doubt is that a great deal of it has already been made, and the point of writing these notes is to see how far the completed part actually goes.

That journey is the first part of a larger project. These notes are a study of the human mind, drawing on neurology, psychiatry, psychedelics, and the philosophy of mind, and they are the place in which I study the mind, awareness, and consciousness. Each of the four contributes something the others cannot. Neurology supplies the machinery: the tissue and chemistry out of which everything else is built. Psychiatry shows what happens when that machinery goes wrong in ways that are lived rather than merely measured, as altered mood, belief, and perception. Psychedelics come close to an experiment on experience itself, since a single molecule acting at a known receptor can reshape perception, the sense of self, and the felt significance of things, in ways that people often count among the most important experiences of their lives. The philosophy of mind asks what all of this amounts to, and in particular whether a complete description of the machinery would be a complete description of a mind.

The study is rooted in science and in physics, which is why it begins with neurology and, within neurology, with the physical chemistry of a membrane. Whatever else the mind turns out to be, it is bound to this tissue closely enough that an account which ignores the physical ground is not worth trusting. But the study does not stop where the physical account runs out. It will take up the hard problem of consciousness, and it will take seriously the deeper questions usually called spiritual: what the self is, what awareness is in itself, and what the states reported in meditation and in mystical and psychedelic experience are experiences of. These belong here because they are part of the lived experience of having a mind, and a study of the mind that left them out would be leaving out part of its subject.

It is worth being precise, once, about which question is being asked, because two quite different ones travel under the same name. Chalmers drew the distinction that has stuck. The *easy* problems of consciousness are the ones that ask how a physical system manages to do the things minds do: how it discriminates one stimulus from another, integrates separate streams into a single scene, stores what happened and retrieves it, attends to one thing rather than another, assigns value, controls behaviour in the service of a goal, reports on its own states, and builds a model of itself. They are called easy not because they are simple, since several of them have taken a century and are not finished, but because it is clear what an answer would look like: a mechanism, specified well enough that one could see how it does the job. The *hard* problem asks something else. It asks why any of that machinery, however completely described, should be accompanied by experience at all, why there is something it is like to be the system rather than nothing.

The neurological sections that open these notes pursue the easy problems. They do so because that is where the neuroscience is, and because the progress there is genuinely remarkable and deserves to be laid out in order. The hard problem is named here and then deliberately held back: nothing in the neurological sections should be read as an attempt on it, and a reader who thinks it dissolves once the easy problems are solved, and a reader who thinks it survives untouched, can both use them, because they are an account of mechanism either way. Holding it back is a matter of order, not of dismissal. The hard problem cannot be posed sharply until it is clear exactly what a mechanistic account does deliver, and the neurology is what establishes that. Once it has, the later parts take the question up directly, along with the spiritual questions that sit close to it, and approach them with the same care as everything else: keeping clear what is established, what is argued, and what remains open, and not allowing the depth of a question to stand in for evidence about its answer.

The method is to work up through levels. An ion has a concentration gradient; a channel has a conductance; a synapse has a strength; a neuron has a firing rate; a circuit has a computation; a nucleus has a function; a system has a capacity. Each level has its own vocabulary, and the vocabulary of one level does not appear at the level below: there is no fear in a sodium channel and no attention in a vesicle. What there is instead is a chain of explanation, in which each level is accounted for by the one beneath it. This is why a document about the mind opens with the Nernst equation, and why that equation is still doing work by the time the account reaches memory and fear.

One claim recurs so often in what follows that it is worth naming in advance, so that it can be recognised each time it appears. It is that nothing about the mind is given. Seeing feels like opening the eyes and receiving the world; it is a construction performed in stages by cortex, and it fails in strikingly specific ways when particular stages are damaged. The sense of occupying a body in a place feels primitive and unanalysable; it is built, and it can be taken apart. Feeling an emotion feels like the most direct knowledge one has; it is assembled, late, out of a reading of the body's own state. In every case the immediacy is real as an experience and misleading as evidence, because the construction is invisible from the inside. That the seams do not show is a fact about the machinery, not about what the machinery is doing.

The order of the sections follows the lecture series on which they are based, and falls into three movements. The first four sections establish the substrate: the layout of the nervous system, the structure of the neuron, the electrical and chemical events by which neurons signal, and the glial cells without which none of it works. The second movement follows the central nervous system from the top down: the lobes of the cortex, then the thalamus and hypothalamus beneath them, the brainstem, the cerebellum, the cranial nerves, and finally the spinal cord. The cortical sections are where the constructive claim is made repeatedly and in detail, as sensation becomes perception, perception becomes recognition, and recognition becomes a world with a body in it; the thalamus and hypothalamus then gate what reaches awareness and generate the drives that make anything worth doing. The brainstem sets the level of consciousness as distinct from its contents, the cerebellum predicts the consequences of action, and the spinal cord shows how much organised behaviour is completed before awareness is involved at all. The third movement takes up the systems that run across this anatomy rather than sitting in one place within it: the basal ganglia, which select actions; the limbic system, which assigns to events the significance that makes them worth responding to; and the autonomic and enteric nervous systems, through which all of this reaches the body.

Together these sections form the neurological foundation of the study. The parts on psychiatry, psychedelics, and the philosophy of mind are to follow, and they build on that foundation rather than replace it: the serotonin receptor at which the psychedelics act is met in the section on neuron firing, and the picture of the self as a construction, which runs through the cortical sections, is where any later question about what the self finally is has to start.

Within the neurology some things are absent. There is no developmental account and little on the cortical microcircuit, though both are touched where they illuminate something, and psychiatric illness, which has its own part to come, appears there only where it shows how the intact machinery works. Clinical material appears throughout, but as evidence rather than as medicine: a deficit that follows a specific lesion is the closest thing this subject has to a controlled experiment on the human mind, and the argument leans on that evidence heavily.

1 Nervous System Structure

Before we can talk about thoughts, feelings, memories, or the mind itself, we need a map. The nervous system is the organ that does all of this: it senses, decides, remembers, and imagines. Like any complicated machine, it helps enormously to know, at least roughly, where its parts are and what each part is for.

At the highest level the system splits into two halves. The **central nervous system (CNS)** is the brain and the spinal cord: the headquarters, where information is integrated and decisions are made. The **peripheral nervous system (PNS)** is everything else, the sprawling network of nerves that links the CNS to muscles, glands, and sensory organs. Every sensation and every voluntary movement crosses the boundary between these two systems.

The rest of this section walks through the major structures of the CNS and PNS from the top down, establishing where each of the regions discussed later actually sits.

1.1 The Cerebrum

The cerebrum is the familiar part of the brain, the wrinkled, walnut-shaped dome that sits on top of everything else. It is also, by some distance, the largest part of the brain and the part most closely associated with what we usually mean by "the mind." Thinking, reasoning, memory, language, voluntary movement, the experience of being a self, all of these depend on cerebral machinery.

Structurally it is symmetrical. Two hemispheres, left and right, are joined by a thick band of white matter called the **corpus callosum**, which lets them share information. Each hemisphere is divided into four lobes, with a fifth hidden beneath them, and each has a characteristic job:

- **Frontal lobe**: voluntary movement, planning, decision-making, and much of what we call personality.
- **Parietal lobe**: bodily sensations such as touch, temperature, and pain, and spatial awareness.
- **Temporal lobe**: hearing, smell, language comprehension, and the formation of long-term memories.
- **Occipital lobe**: vision.
- **Insula**, hidden in the depths of the lateral sulcus: taste, the internal state of the body, and emotional awareness.

The outer layer of the cerebrum, the **cerebral cortex**, is only a few millimetres thick but is crumpled into deep folds. The folding is not cosmetic: by crumpling the sheet, evolution has packed far more cortical area into the skull than would otherwise fit, and cortical area turns out to matter enormously for cognitive capacity.

1.2 The Cerebellum

Tucked beneath the back of the cerebrum, the cerebellum ("little brain") looks almost like a miniature version of its larger sibling. Despite its size, it contains around four in every five of the brain's neurons, and it is indispensable for smooth, coordinated movement.

Crucially, the cerebellum does not initiate actions. Instead it receives a copy of the motor commands being issued by the cortex together with sensory feedback about what the body is actually doing, and it compares the two. When the intention and the reality diverge, the cerebellum quietly corrects the discrepancy in real time. That is why skilled movements, reaching for a mug, playing a scale on the piano, walking in a straight line, feel effortless.

When the cerebellum is damaged, this silent correction falls apart. The classic picture includes *ataxia* (loss of coordination), tremor, and difficulty maintaining balance. Patients often describe it as feeling perpetually drunk, which is itself a hint: alcohol disrupts cerebellar function, and that is exactly why it impairs coordination.

1.3 The Diencephalon

Between the cerebrum above and the brainstem below sits the **diencephalon**, a compact group of structures wrapped around the third ventricle. It is small relative to the hemispheres that enclose it, and it is easy to overlook on a surface view because almost none of it is visible from outside, but very little reaches the cortex without passing through it or being regulated by it.

It comprises four parts:

- The **thalamus** is by far the largest, a pair of egg-shaped masses of grey matter sitting either side of the midline. It is the great relay station of the brain: with the single exception of smell, every sensory stream (vision, hearing, touch, taste, and the sense of the body's own position) synapses in a specific thalamic nucleus before continuing to the cortex. It is not a passive junction box. The thalamus gates what reaches the cortex according to attention and arousal, and it carries return traffic from the cortex back down as well, taking part in the motor and cognitive loops discussed later.
- The **hypothalamus** lies beneath the thalamus and is, for its size, the most consequential structure in the body. It is the master regulator of **homeostasis**, holding body temperature, fluid balance, hunger, thirst, and the sleep-wake cycle within viable limits. It commands the autonomic nervous system, and through its connection to the **pituitary gland** it also commands the endocrine system, which makes it the point at which the nervous and hormonal systems become one control problem rather than two.
- The **epithalamus** is a small posterior region whose principal component is the **pineal gland**. The pineal secretes **melatonin** on a daily rhythm and is central to the timing of the sleep-wake cycle.
- The **subthalamus** lies below the thalamus and contains the **subthalamic nucleus**, a small structure that plays a disproportionately large role in the control of movement, as the section on the basal ganglia sets out.

The diencephalon is therefore best understood not as one organ but as the junction at which the traffic between cortex, brainstem, and body is routed, filtered, and regulated. Damage here rarely produces a single clean deficit, precisely because so many separate pathways are packed into so small a volume.

1.4 The Brainstem

If the cerebrum is where thought happens and the cerebellum is where movement is refined, the brainstem is where life is sustained. It connects the brain to the spinal cord and is the conduit for almost every signal passing between them. It is also the home of the circuits that drive breathing, set the rhythm of the heart, and maintain consciousness itself.

It is traditionally divided into three levels, each with distinct responsibilities:

- **Midbrain**: visual and auditory reflexes, aspects of motor control, and arousal.
- **Pons**: a relay between the cerebrum and cerebellum, and a participant in sleep and respiration.
- **Medulla oblongata**: the control centre for vital autonomic functions: heart rate, blood pressure, breathing, coughing, swallowing.

Because the brainstem handles these non-negotiable tasks, even small lesions there can be catastrophic. A few millimetres in the wrong place can interrupt consciousness or stop breathing. This is also why brainstem death is used as a clinical definition of death.

1.5 Cranial Nerves

Most nerves reach the body via the spinal cord, but the twelve **cranial nerves** bypass that route and leave the CNS directly, mostly from the brainstem. They handle the close work of the head and neck (vision, eye movement, facial sensation and expression, taste, hearing, balance, and swallowing), and in the case of the vagus nerve they reach deep into the thorax and abdomen to regulate several visceral functions.

Each is conventionally numbered with a Roman numeral, running from front to back in the order the nerves emerge. Each is also classified as *sensory*, *motor*, or *both*, according to the direction of the information it carries:

- **I, Olfactory** (sensory). Carries smell. Like the optic nerve it does not arise from the brainstem, and smell is the one sense that reaches the cortex without first relaying in the thalamus, a peculiarity taken up in the section on the temporal lobe.
- **II, Optic** (sensory). Carries vision from the retina, partially crossing at the optic chiasm so that each hemisphere receives the opposite half of the visual field.
- **III, Oculomotor** (motor). Drives four of the six muscles that move the eye, raises the upper eyelid, and carries the parasympathetic fibres that constrict the pupil and focus the lens.
- **IV, Trochlear** (motor). Supplies a single muscle, the superior oblique, which turns the eye downward and inward. It is the thinnest cranial nerve and the only one to emerge from the back of the brainstem.
- **V, Trigeminal** (both). The great sensory nerve of the face, carrying touch, pain, and temperature from the skin, the cornea, the sinuses, and the front of the tongue through its three divisions, the ophthalmic, maxillary, and mandibular. Its motor fibres drive the muscles of chewing.
- **VI, Abducens** (motor). Supplies the lateral rectus, which turns the eye outward.
- **VII, Facial** (both). Drives the muscles of facial expression, carries taste from the front two-thirds of the tongue, and supplies the glands that produce tears and saliva.
- **VIII, Vestibulocochlear** (sensory). Carries hearing from the cochlea and balance from the vestibular apparatus, in two divisions that travel together but report on quite different things.
- **IX, Glossopharyngeal** (both). Carries taste and sensation from the back of the tongue and the pharynx, including the blood pressure sensors of the carotid sinus and the oxygen sensors of the carotid body, and contributes to swallowing.
- **X, Vagus** (both). By far the most widely distributed, wandering (as its name says) out of the head entirely to supply the heart, lungs, and most of the digestive tract. It is the principal parasympathetic nerve of the body, and it also drives the muscles of the larynx and pharynx that make speech and swallowing possible.
- **XI, Accessory** (motor). Supplies the sternocleidomastoid and trapezius, the muscles that turn the head and shrug the shoulders.
- **XII, Hypoglossal** (motor). Supplies the muscles of the tongue, and so underlies both speech and the manipulation of food in the mouth.

Their arrangement is diagnostically valuable. Because the nerves emerge in an orderly sequence along the brainstem, and because each can be tested individually at the bedside, the pattern of which nerves fail and which are spared localises a brainstem lesion with considerable precision. A deficit combining several adjacent nerves points to a lesion at the level from which they emerge together.

1.6 Ventricles and Cerebrospinal Fluid

Inside the brain is a branching system of cavities called the **ventricles**: two lateral ventricles in the cerebral hemispheres, a third between them, and a fourth between the brainstem and the cerebellum. They are filled with **cerebrospinal fluid (CSF)**, a clear fluid produced mostly by the *choroid plexus* that lines them.

It is tempting to think of CSF as just a liquid cushion, and that is certainly part of its job: the brain floats in it, which greatly reduces its effective weight and protects it against mechanical shock. But CSF also helps maintain a stable chemical environment around neurons and carries metabolic waste away from nervous tissue. It is manufactured in the ventricles, circulates out over the surface of the brain and spinal cord, and is eventually reabsorbed into the venous blood.

When CSF flow is obstructed, the ventricles swell under pressure, a condition called **hydrocephalus**, which we will meet again when we talk about ependymal cells.

1.7 The Spinal Cord

The spinal cord is a bundle of nervous tissue about the thickness of a finger that runs down inside the vertebral column. It does two things at once. First, it is a **highway**: sensory information travels upward to the brain, and motor commands travel downward to the body. Second, and less obviously, it is a **simple decision-making device** in its own right. Reflexes such as withdrawing a hand from a hot surface are mediated by the spinal cord alone, without waiting for the brain to get involved. By the time the pain is consciously registered, the hand has already moved.

The cord is segmented, and each segment gives rise to a pair of **spinal nerves** that serve a particular territory of the body. This segmental organisation is why spinal injuries at different levels produce such different patterns of loss: a cervical lesion affects the arms and everything below, while a lumbar lesion may spare the arms completely.

1.8 Peripheral Nerves and the Autonomic Nervous System

The peripheral nervous system extends from the CNS out into the rest of the body. Functionally, peripheral nerves are either **sensory (afferent)**, carrying information from the body back to the CNS, or **motor (efferent)**, carrying commands from the CNS out to muscles and glands.

Motor control itself splits into two worlds. The **somatic nervous system** governs deliberate action: moving an arm, speaking, walking. The **autonomic nervous system** governs what proceeds without deliberation (heart rate, digestion, pupil size, sweating, and blood pressure), and it does so through two opposing branches:

- The **sympathetic division** prepares the body for action: heart rate rises, pupils dilate, blood is shunted toward muscle. This is the “fight or flight” response.
- The **parasympathetic division** does the opposite: it slows the heart, promotes digestion, and generally encourages the body to conserve and repair. This is “rest and digest.”

A third component, the **enteric nervous system**, is a semi-independent network of neurons in the wall of the gut, which runs digestion largely by itself while taking instructions from the other two.

The division between sympathetic and parasympathetic matters far beyond the textbook. The balance between sympathetic and parasympathetic activity is the physiological signature of emotional arousal, stress, and relaxation, and it is where neurology and psychology start to meet in earnest: a frightened mind produces a sympathetically driven body, and a calm body feeds back into a calmer mind.

1.9 Summary

The nervous system is hierarchical and deeply integrated. The cerebrum handles the most elaborate cognition; the cerebellum refines motor output; the diencephalon relays and gates almost everything passing between them and the body below, and regulates the internal state of the organism; the brainstem keeps the lights on and gives rise to the cranial nerves that serve the head and neck; the spinal cord is both cable and reflex circuit; and peripheral nerves extend the system out into every organ and muscle in the body.

It is worth being clear about what this map does and does not provide. It is an inventory of parts and their locations, and nothing in it yet explains anything mental. Knowing that the occipital lobe is concerned with vision is not knowing how seeing is done, and knowing that the hypothalamus regulates hunger is not knowing why anything is wanted. What the map establishes is only where to look, which is a real thing to have and is the necessary first step. Everything after this section is the attempt to replace a location with a mechanism, and it begins by zooming in on the individual building blocks, the neurons themselves.

2 Neuron Structure

Neurons are the cells that make the nervous system a nervous system. Every tissue in the body signals in one way or another, but neurons have specialised for one job to an extreme degree: they move information, quickly and reliably, over long distances. Everything else about their biology (their shape, their metabolism, their internal traffic system) is bent toward that single purpose.

The most striking consequence of this specialisation is that neurons are **polarised**. They have a clearly defined input side and output side, and the whole architecture of the cell enforces one-way traffic. Understanding that shape is the first step toward understanding how neurons fire and communicate.

A neuron has three main regions, the **soma**, the **dendrites**, and the **axon**, plus a suite of internal machinery adapted to its unusual demands.

2.1 Soma, Dendrites, and the Axon

2.1.1 Soma (Cell Body)

The soma is the neuron's headquarters. It contains the nucleus and most of the cell's organelles, and it is where proteins are manufactured, energy is managed, and the general business of being a living cell is transacted. In a real sense the soma is the only part of the neuron that looks like a generic animal cell; everything else is specialised for signalling.

Its most important computational job is **integration**. The soma, together with the axon hillock that we will meet in a moment, is where the neuron sums the inputs arriving on its dendrites and decides whether to fire. It is the place where "many inputs" becomes "one decision."

2.1.2 Dendrites

Dendrites are the branches that fan out from the soma like the root system of a tree, and they are the neuron's input antennae. A single neuron may have thousands of them, covered in small protrusions called *dendritic spines*, and each spine can host a synapse from another neuron. The

sheer geometry of this arrangement means that one neuron can listen to thousands of others at once.

What arrives at a dendrite is not, at first, an electrical spike. It is a *chemical* signal, neurotransmitter released from the presynaptic cell, that is converted into a small, local change in membrane voltage called a **graded potential**. These graded potentials spread passively toward the soma, and they can either push the cell toward firing or pull it back from the brink. The dendritic tree is, in effect, the neuron's ear and its first stage of processing all at once.

2.1.3 Axon

The axon is the output. It is a single long projection that leaves the soma and carries electrical signals away to whatever the neuron talks to next: another neuron, a muscle cell, or a gland. Axons can be astonishingly long: the motor neurons running from the spinal cord to the foot are over a metre in length in an adult, all in a single cell.

Three regions of the axon are worth naming explicitly:

- The **axon hillock** is the short region where the axon leaves the soma. It has an unusually high density of voltage-gated sodium channels, and it is here that action potentials are typically triggered. The hillock is, in effect, the neuron's "go / no-go" switch.
- The **axon shaft** conducts the action potential along its length.
- The **axon terminals** (or synaptic boutons) are the tiny swellings at the far end, where neurotransmitter is released onto the next cell.

Many axons are wrapped in **myelin**, a fatty insulating sheath produced by glial cells, which dramatically speeds up conduction. The glial cells section takes this up in detail.

2.2 Internal Machinery: Nucleus, Rough ER, and Golgi

Neurons are unusually demanding cells. They need a constant supply of ion channels, receptors, transporters, structural proteins, and neurotransmitter-related machinery, and they need to ship all of this out to parts of the cell that may be a metre away. The internal factory is built accordingly.

2.2.1 Nucleus

The nucleus holds the cell's DNA and controls gene expression. In neurons, that expression has to be finely tuned: the mix of receptors and channels a neuron produces is a large part of what determines its electrical character, which transmitters it responds to, how strongly, and what it releases in turn.

Because most neurons are never replaced, this transcriptional control operates over the whole of a lifetime rather than across cell divisions. Gene expression in a neuron is also unusually responsive to the cell's own activity: patterns of firing feed back on which genes are transcribed, which is one of the routes by which experience leaves a lasting mark on the cell's properties.

2.2.2 Rough Endoplasmic Reticulum (Nissl Substance)

In a neuron, the rough endoplasmic reticulum is so abundant and so densely studded with ribosomes that it stains as a distinct pattern under the microscope. Historically this pattern was named **Nissl substance**. It is the neuron's protein factory: it synthesises the membrane proteins, ion channels, receptors, and secreted proteins that the cell needs in industrial quantities.

The amount of Nissl substance a neuron contains is a reasonable proxy for how metabolically active it is, and its dispersal (*chromatolysis*) is a classic sign of neuronal stress after injury.

2.2.3 Golgi Apparatus

The Golgi apparatus takes proteins fresh from the rough ER, modifies them, sorts them, and packages them into vesicles for delivery. Its sorting role is what matters most in a neuron. A protein leaving the Golgi must be labelled for a destination, and the destinations are unusually far apart: a channel bound for a dendritic membrane, a transporter bound for the axon, and a vesicle protein bound for a terminal a metre away all leave from the same place and must not be confused. Mis-sorting in a cell this asymmetric is not a small error, since a protein delivered to the wrong compartment is both absent where it is needed and disruptive where it arrives.

2.3 Microtubules, Kinesin, and Dynein

Because a neuron's axon can be so long, ordinary diffusion is nowhere near fast enough to move molecules around inside it. Diffusion time scales with the square of distance, so a journey of a metre is not a hundred times slower than one of a centimetre but ten thousand times slower, taking years at the very least. Neurons solve this by building a system of oriented tracks along which motor proteins actively haul their cargo.

2.3.1 Microtubules

The tracks are **microtubules**: rigid, polar tubes of the cytoskeleton that run the length of the axon and into the dendrites. They provide mechanical support and, crucially, a surface that motor proteins can walk along. The polarity matters (microtubules have distinct "plus" and "minus" ends) because it allows the cell to enforce direction on intracellular traffic.

2.3.2 Kinesin and Dynein

The motors themselves are two families of protein, and they differ in the direction they travel:

- **Kinesin** walks toward the plus end, which in an axon means away from the soma and out toward the terminals. This is *anterograde* transport, and it is how freshly made proteins, vesicles, and organelles are delivered to the synapse.
- **Dynein** walks the other way, toward the minus end and back toward the soma. This is *retrograde* transport, and it is how the cell recycles components, clears damaged cargo, and, sometimes, how signals from distant synapses make it back to the nucleus.

This bidirectional traffic is non-negotiable for neuronal life. A synapse at the end of a long axon is essentially a consumer that produces very little for itself; everything it uses has to be manufactured in the soma and delivered by kinesin, and everything it must dispose of has to be carried back by dynein. Disruption of axonal transport is implicated in several neurodegenerative diseases, and the vulnerability is structural: a cell whose furthest working parts are a metre from its only factory has no redundancy if the delivery system fails.

2.4 Vesicles and SNARE Proteins

At the axon terminal, information has to cross from one cell to another, and for most neurons it does so chemically. Neurotransmitter is stored in advance inside small, membrane-bound **synaptic vesicles** that cluster just inside the terminal membrane, primed and ready to release.

Release has to be fast, on the scale of fractions of a millisecond, and it has to be tightly coupled to the arrival of an action potential, or else the signal would be hopelessly noisy. The machinery that accomplishes this is a family of proteins called **SNAREs**.

SNAREs on the vesicle and SNAREs on the presynaptic membrane interlock like a zip, pulling the vesicle tight against the membrane and priming it for fusion. The final trigger is

a rapid influx of calcium through voltage-gated channels that open when the action potential arrives. Calcium binds to a sensor protein, the SNARE complex completes fusion, and the contents of the vesicle are dumped into the **synaptic cleft**, the tiny gap between the two neurons, in a fraction of a millisecond.

This machinery is so central to nervous-system function that several of the most potent biological toxins known target it. Botulinum toxin and tetanus toxin, for example, both cleave specific SNARE proteins; the difference in their clinical effects comes down to which synapses they disable.

2.5 Neurotransmitters

A **neurotransmitter** is simply a chemical messenger released from one neuron that acts on receptors belonging to another. In principle the definition is simple; in practice the list of neurotransmitters is long and the consequences are enormous, because neurotransmitters are where neuroscience starts bleeding directly into psychology and psychiatry.

At the crudest level they fall into two groups:

- **Excitatory** neurotransmitters, the prototypical example being *glutamate*, push the post-synaptic cell toward firing.
- **Inhibitory** neurotransmitters, the prototypical example being *GABA*, pull it away from firing.

Most CNS signalling, by volume, is glutamate and GABA. Two further amino acids belong alongside them: **glycine**, which is the main inhibitory transmitter of the spinal cord and brainstem and works much as GABA does, opening chloride channels; and **aspartate**, which is excitatory and closely related to glutamate. These four do the bulk of fast, point-to-point signalling.

Layered on top of this are the **neuromodulators**, which typically do not simply flip a downstream cell on or off but tune the way whole circuits respond to input. These are the systems most often implicated in mood, attention, reward, and arousal, and they are also the systems most commonly targeted by psychiatric drugs. The best characterised are the **monoamines**, so called because each is built around a single amine group:

- **Dopamine** governs the initiation and selection of movement, and signals reward and motivation. Its motor role is the subject of the basal ganglia section, where the same transmitter is shown to promote or suppress an action depending only on which receptor it reaches.
- **Serotonin** is associated with mood, sleep, and appetite, and is the target of the most widely prescribed class of antidepressant.
- **Noradrenaline** sets the level of arousal and alertness, driving the systems that raise vigilance in response to threat or salience.
- **Adrenaline** is chemically the next step beyond noradrenaline and acts mainly as a circulating hormone, though a small population of brainstem neurons use it as a transmitter in the control of blood pressure and respiration.
- **Histamine** is released from a small hypothalamic population that projects widely across the brain and helps maintain wakefulness. Its role is most obvious in reverse: antihistamines that cross into the brain cause drowsiness.

Acetylcholine sits outside the monoamine family but belongs with the modulators. It does two quite separate jobs. In the periphery it is the transmitter of the neuromuscular junction, where it activates skeletal muscle directly. In the brain it is a modulator, strongly implicated in attention and in the formation of memory.

Beyond these classical transmitters are several groups that break the mould in one way or another:

- The **neuropeptides** are short chains of amino acids, larger and slower-acting than the transmitters above, and typically released alongside them rather than instead of them. They include **substance P**, which carries pain signals into the spinal cord, and the **endorphins** and **enkephalins**, the body's own opioids, which suppress those same signals and act on the receptors that morphine and its relatives exploit.
- The **purines**, chiefly **ATP** and its breakdown product **adenosine**, act as transmitters in their own right. Adenosine accumulates during waking and promotes sleep, which is the mechanism caffeine blocks.
- **Nitric oxide** is a dissolved gas rather than a packaged molecule. It cannot be stored in vesicles, so it is synthesised on demand and simply diffuses out through the membrane to act on neighbouring cells.
- The **endocannabinoids** are lipid messengers, and like nitric oxide they are made when needed rather than stored. Their most striking feature is the direction in which they travel, which the next section takes up.

Whenever we later talk about a psychological phenomenon with a pharmacological correlate (depression treated with SSRIs, Parkinson's disease with L-DOPA, anxiety with benzodiazepines), we will be talking, in the end, about how a drug has tweaked one of these systems.

2.6 Summary

A neuron's shape is not incidental. The soma integrates, the dendrites listen, the axon transmits, and the terminals release chemistry onto the next cell. Inside the cell, a network of ER, Golgi, microtubules, and motor proteins keeps this long, demanding architecture supplied with everything it needs. At the synapse, vesicles wait ready and SNAREs stand by, so that when an action potential arrives, chemical communication can happen in a fraction of a millisecond.

The feature to carry forward is the polarity. A neuron enforces one-way traffic, from dendrite to soma to hillock to terminal, and that constraint is what makes it a computational device rather than merely an excitable one. A cell that conducted equally in all directions could be a signal, but it could not be a step in an argument. Because the traffic runs one way, thousands of inputs can be summed into a single output, that output can become an input elsewhere, and elements of this kind can be chained into circuits in which the order of operations means something. Every capacity discussed in the later sections is built out of that arrangement repeated on a very large scale. With the architecture in place we are ready to ask the next question: how does a neuron actually fire?

3 Neuron Firing

So far we have been saying that neurons "send signals." It is time to be concrete about what that actually means. A neuron does not have tiny wires or its own battery. What it has is a membrane, some ions, and some very cleverly controlled channels through which those ions can move. Everything we call neural activity (every perception, every thought, every movement) is ultimately some pattern of ions crossing membranes.

This section builds the story up from the ground. We begin with the resting state: why an idle neuron has a voltage across its membrane at all. We then see how small, local disturbances (graded potentials) can add up to a threshold crossing, at which point a stereotyped, all-or-nothing **action potential** is born. Finally we follow that action potential down the axon to the synapse, where it triggers the chemical release we just described. The section closes with the drugs that act on these processes, either by imitating a transmitter or by regulating how much of it there is.

3.1 The Resting Membrane Potential

Place a very fine electrode inside an unstimulated neuron and another in the extracellular fluid, and the voltage difference between them is roughly:

$$V_m \approx -70\,\text{mV}$$

The inside of the cell is negative relative to the outside. This is not an accident, and it is not "at rest" in the sense of nothing happening. Maintaining the resting potential is one of the most energy-hungry things a neuron ever does.

Three facts, taken together, explain it.

First, ions are distributed unequally across the membrane. Sodium (Na^+) is much more concentrated outside the cell than inside; potassium (K^+) is much more concentrated inside than outside; chloride (Cl^-) is more concentrated outside.

Second, the membrane is selectively permeable. At rest it is much more permeable to K^+ than to any other ion, because of leak channels that are always open.

Third, an active pump, the **Na^+/K^+ ATPase**, burns ATP to push three sodium ions out and pull two potassium ions in with every cycle. This both maintains the concentration gradients and contributes slightly to the negative interior.

Put these three facts together and the resting membrane potential emerges more or less inevitably. Because the membrane is dominated by potassium permeability, the cell's voltage sits close to the equilibrium potential for potassium.

3.1.1 The Nernst Equation

We can make this quantitative. For any single ion species, there is a particular membrane voltage at which the electrical force on the ion exactly cancels the diffusional force from its concentration gradient. This voltage is the ion's **equilibrium potential**, and it is given by the **Nernst equation**:

$$E_{ion} = \frac{RT}{zF} \ln\!\left(\frac{[ion]_{out}}{[ion]_{in}}\right)$$

Here R is the gas constant, T is temperature in kelvin, F is Faraday's constant, and z is the ion's charge. At body temperature (37°C), plugging in the constants and converting from the natural logarithm to the base-ten logarithm gives the convenient working form:

$$E_{ion} \approx \frac{61}{z} \log_{10}\!\left(\frac{[ion]_{out}}{[ion]_{in}}\right)\ \text{mV}$$

3.1.2 Typical Equilibrium Potentials

For a mammalian neuron under physiological conditions:

- $E_K \approx -90$ mV
- $E_{Na} \approx +60$ mV

- $E_{Cl} \approx -65$ mV

Notice where -70 mV sits in that list. The resting potential is close to E_K because at rest, potassium is effectively the ion whose voice dominates. If a cell were instead to open a large number of sodium channels, the voltage would race toward $+60$ mV instead. This is exactly what happens during an action potential.

3.2 Graded Potentials

Before we reach the action potential, we need a smaller kind of signal: the **graded potential**. Graded potentials are the small, local changes in membrane voltage produced in dendrites and the soma by incoming synaptic input, and they are the raw material out of which the decision to fire is built.

Three properties define them. They are *graded*: their size depends on the size of the stimulus, rather than being all-or-nothing. They *decay* with distance: they are passive spreads of voltage, and they get weaker the further they travel. And they *summate*: if several graded potentials arrive at roughly the same time (temporal summation) or at nearby spots on the dendritic tree (spatial summation), their effects add together.

But where does the voltage change actually come from? A neurotransmitter molecule is not electrically interesting in itself; it carries no current across the membrane and it does not touch the ions on either side. Everything depends on what the transmitter binds to. Postsynaptic receptors come in two architecturally different kinds, and the difference between them accounts for most of the variety in how neurons influence one another.

3.2.1 Ligand-Gated Ion Channels

The first kind is the **ligand-gated ion channel**, also called an **ionotropic** receptor. Here the receptor *is* the channel: a single protein assembly that spans the membrane, encloses a water-filled pore, and carries a binding site for the transmitter on its outer face. When the transmitter (the *ligand*) binds, the protein changes shape, the pore opens, and ions move through it under the forces described in the previous subsection. When the transmitter unbinds, the pore shuts.

Nothing else is involved. There is no intermediate messenger and no chain of enzymes, which is why this arrangement is fast: the delay between transmitter binding and current flowing is a fraction of a millisecond, and the whole event is usually over within a few milliseconds. It is also strictly local, confined to the patch of membrane holding the channel.

What determines whether such a channel excites or inhibits is not the transmitter but the **selectivity of the pore**, that is, which ions it will let through:

- **Glutamate** acts on **AMPA** and **NMDA** receptors, both of which pass cations, chiefly Na^+ inward. The result is depolarisation. This is why glutamate is the workhorse excitatory transmitter.
- **Acetylcholine** acts on **nicotinic** receptors, which likewise pass cations inward, and which are the receptors of the neuromuscular junction.
- **GABA** acts on $\mathbf{GABA_A}$ receptors, which pass Cl^-, and **glycine** acts on receptors that do the same. The result is inhibition.

It is worth stating plainly, because it is a common confusion, that no transmitter is intrinsically excitatory or inhibitory. Glutamate is excitatory because the receptors that happen to bind it happen to pass sodium. Put glutamate on a chloride-selective channel and it would inhibit. The transmitter selects which receptors respond; the receptor decides what that response is.

3.2.2 G Protein-Coupled Receptors

The second kind is the **G protein-coupled receptor (GPCR)**, also called a **metabotropic** receptor, and it works on an entirely different principle. A GPCR is not a channel and cannot pass a current at all. It is a single protein that threads back and forth through the membrane seven times, which is why the family is sometimes called the seven-transmembrane receptors, with its binding site facing outward and its business end facing the cytoplasm. What it does when activated is start a chain of chemistry inside the cell.

The sequence runs as follows.

1. The transmitter binds to the receptor's extracellular face, and the protein changes shape.
2. That shape change is felt on the inside, where the receptor is associated with a **G protein**, a three-part assembly of α, β, and γ subunits. At rest the α subunit holds a molecule of **GDP**.
3. The activated receptor causes the α subunit to swap that GDP for **GTP**. This is the switch being thrown.
4. The GTP-bound α subunit separates from the $\beta\gamma$ pair, and both halves are now free to act.
5. The α subunit acts on an **effector enzyme**, commonly **adenylyl cyclase**, which manufactures a **second messenger**, **cyclic AMP (cAMP)**, that diffuses through the cytoplasm.
6. cAMP activates **protein kinase A (PKA)**, an enzyme that phosphorylates other proteins, including ion channels already sitting in the membrane. Phosphorylation changes how readily those channels open.
7. The α subunit eventually hydrolyses its GTP back to GDP, reassembles with $\beta\gamma$, and the system resets.

Which direction the cascade pushes depends on which G protein the receptor is coupled to. Receptors coupled to $\mathbf{G_s}$ (stimulatory) activate adenylyl cyclase, so cAMP rises and PKA is switched on. Receptors coupled to $\mathbf{G_i}$ (inhibitory) suppress it, so cAMP falls. The same receptor architecture and the same transmitter can therefore produce opposite effects in two different cells, depending only on which G protein sits behind the receptor.

The $\beta\gamma$ pair is not merely left over. It can act on its own, and in one case that matters a great deal it acts directly on ion channels: $\beta\gamma$ released by $\mathrm{GABA_B}$ and by various monoamine receptors binds and opens a class of potassium channels known as **GIRK** channels. Potassium leaves, the cell hyperpolarises, and the membrane voltage has been changed by a metabotropic receptor without any second messenger being involved at all. This is a useful corrective to the tidy picture: metabotropic does not have to mean slow and indirect; it means that the receptor and the channel are separate proteins.

Three consequences of the whole arrangement matter. It is *slow*: a cascade of several enzymatic steps takes hundreds of milliseconds to seconds, against the millisecond of an ionotropic synapse. It is *amplifying*: one bound receptor activates many G proteins, each activated enzyme makes many molecules of second messenger, and so a small amount of transmitter produces a large intracellular effect. And it is *modulatory* rather than commanding: the usual end result is not that the cell fires, but that the channels it already has are tuned to open more or less readily. A GPCR does not so much send a message as change how the cell reads its other messages.

Most transmitters act on both kinds of receptor. Glutamate has metabotropic receptors alongside AMPA and NMDA; GABA has $\mathrm{GABA_B}$ alongside $\mathrm{GABA_A}$; acetylcholine has muscarinic receptors alongside nicotinic. The neuromodulators met in the previous section (dopamine, serotonin, and noradrenaline) act almost entirely through GPCRs, which is precisely why they modulate rather than command.

3.2.3 How These Change the Membrane Voltage

We can now be exact about the step that both mechanisms end in: an open channel, and a voltage that moves. The key idea is that opening a channel does not *set* a voltage. It passes a **current**, and the size and direction of that current depend on how far the membrane currently sits from that ion's equilibrium potential:

$$I_{ion} = g_{ion}\,(V_m - E_{ion})$$

Here g_{ion} is the conductance, essentially how many channels for that ion are open, and the bracketed term $(V_m - E_{ion})$ is the **driving force**. The equation says something simple: an ion flows in whichever direction moves the membrane potential *toward* that ion's equilibrium potential, and it flows harder the further away the membrane currently is. An open channel is a pull toward its own ion's E_{ion}, and the strength of the pull is the distance still to travel.

Read against the equilibrium potentials tabulated earlier, every kind of postsynaptic response follows directly:

- Open a **sodium**-permeable channel at rest. The membrane sits at -70 mV and $E_{Na} \approx +60$ mV, so the driving force is large and inward. Sodium pours in, the membrane depolarises toward $+60$ mV, and the result is an **EPSP**.
- Open a **potassium**-permeable channel, as $\beta\gamma$ does at GIRK channels. Since $E_K \approx -90$ mV lies below rest, potassium leaves, the membrane hyperpolarises toward -90 mV, and the result is an **IPSP**.
- Open a **chloride**-permeable channel, as $\mathrm{GABA_A}$ does. Here $E_{Cl} \approx -65$ mV sits very close to rest, so remarkably little chloride actually moves and the voltage barely changes. The inhibition is real nonetheless: with a large chloride conductance open, the membrane is effectively clamped near -65 mV, and any excitatory current arriving has to fight that clamp. This is **shunting inhibition**, which holds the cell down rather than pushing it down.

This also explains the defining properties listed at the start of the subsection. The response is *graded* because more transmitter binds more receptors, which opens more channels, which is a larger g_{ion} and so a larger current and a bigger deflection; nothing here is all-or-nothing. It *decays* with distance because the charge spreading away from the synapse leaks back out across the membrane as it goes, so less and less of it reaches the axon hillock. And it *summates* because these are currents flowing into a common pool of charge, and currents simply add.

Finally, the two receptor families leave different signatures on the voltage. An ionotropic receptor produces a fast, sharp deflection lasting a few milliseconds, the discrete blip of a single synaptic event. A metabotropic receptor produces a slow shift, taking hundreds of milliseconds to develop and persisting for seconds, which does not so much signal an event as change the terms on which every other input is judged. When we come to the basal ganglia we will find dopamine doing exactly this, setting the gain on a circuit rather than driving it.

With that machinery in place, the two kinds of graded potential can be named:

- **Excitatory postsynaptic potentials (EPSPs)** depolarise the cell, moving it closer to threshold.
- **Inhibitory postsynaptic potentials (IPSPs)** hyperpolarise the cell, moving it away from threshold.

The axon hillock tallies the EPSPs and IPSPs arriving at every moment, and when the net result is a depolarisation above threshold, an action potential is launched.

3.3 The Action Potential

The action potential is a rapid, stereotyped, all-or-nothing spike in membrane voltage that propagates along the axon without decay. It is the only signal a neuron has that can travel the full length of the cell and arrive undiminished.

3.3.1 Threshold

The action potential begins when the membrane at the axon hillock depolarises to about:

$$V_{\text{threshold}} \approx -55\,\text{mV}$$

At this voltage, voltage-gated sodium channels open in large numbers, and what follows is a runaway.

3.3.2 Phases of the Action Potential

The spike unfolds as a short sequence of stages, each dominated by a different set of channels.

1. Resting state (≈ -70 mV). Voltage-gated channels are closed. Leak channels and the Na^+/K^+ pump maintain the resting potential.

2. Depolarisation (up to $\approx +30$ mV). Voltage-gated Na^+ channels open in a positive-feedback avalanche: more sodium in means more depolarisation, which means more channels open. The voltage races upward toward E_{Na}.

3. Repolarisation. Two things happen, slightly delayed. Sodium channels *inactivate* (they physically plug themselves closed with an intrinsic gate), and voltage-gated potassium channels open. Potassium flows out and drags the voltage back down.

4. Hyperpolarisation (≈ -80 to -90 mV). Potassium channels are slow to shut, so K^+ efflux briefly overshoots the resting potential, pulling the cell closer to E_K.

5. Return to rest. Potassium channels close. The Na^+/K^+ pump continues its unending job of restoring the gradients, and the membrane settles back to -70 mV.

The entire spike, from threshold to recovery, takes only a few milliseconds. And because it is triggered by a self-reinforcing avalanche, it is all-or-nothing: a neuron either fires or it does not. There is no such thing as a partial action potential.

3.4 Neurotransmitter Release at the Synapse

When the action potential reaches the axon terminal, it finds yet another class of voltage-gated channels: **calcium channels**. These open, calcium floods in, and, as we saw in the previous section, that calcium is the trigger the SNARE machinery has been waiting for. Vesicles fuse with the presynaptic membrane and dump their neurotransmitter into the synaptic cleft, where it diffuses across and binds to receptors on the next cell.

The amount of neurotransmitter released depends on how much calcium enters, so the strength of the chemical signal is tightly linked to the arrival of the electrical one. In this way a digital, all-or-nothing spike is translated back into a graded chemical signal, which becomes a new graded potential in the next neuron, and the whole cycle repeats.

3.5 Clearing the Signal: Reuptake and Degradation

A signal is only useful if it can also be turned off. Neurotransmitter that is left to linger in the synaptic cleft blurs subsequent signals and, in the case of excitatory transmitters, can actually damage the postsynaptic cell (a phenomenon called *excitotoxicity*). There are three main ways the cleft is cleaned up:

- **Reuptake.** Specialised transporters on the presynaptic neuron (or on nearby glia) pump neurotransmitter back out of the cleft, often to be repackaged for reuse. This is the dominant mechanism for monoamines like serotonin and dopamine.
- **Enzymatic degradation.** At some synapses an enzyme in the cleft breaks the transmitter down. The classic example is *acetylcholinesterase*, which cleaves acetylcholine almost as soon as it is released.
- **Diffusion.** Simple loss of molecules out of the narrow cleft also contributes.

Many psychoactive drugs act here. Selective serotonin reuptake inhibitors (SSRIs) block the reuptake transporter for serotonin, raising its concentration in the cleft. Amphetamines interfere with dopamine reuptake and release. Nerve agents block acetylcholinesterase, producing catastrophic overstimulation. Whenever a drug is said to "increase" or "decrease" a neurotransmitter, it is almost always doing something very concrete and very local at the synapse.

3.6 Refractory Periods

After an action potential, the neuron is not instantly ready to fire another. For a brief interval it refuses absolutely, and for a slightly longer interval it only reluctantly agrees. These intervals are the refractory periods, and they are not flaws but what keeps nervous-system signalling clean.

3.6.1 Absolute Refractory Period

Immediately after a spike, no new action potential can be generated at any price. This is because the voltage-gated sodium channels that just fired are *inactivated*: their inactivation gate is closed, and it will not reopen until the membrane has returned close to rest. No amount of stimulation will fire them. The absolute refractory period effectively sets an upper bound on how fast a neuron can fire.

3.6.2 Relative Refractory Period

During the hyperpolarisation that follows, sodium channels have largely recovered but the cell is further from threshold than usual. Firing is possible, but it takes a stronger-than-normal stimulus to get there, and the deficit shrinks as the membrane drifts back toward rest. The consequence is that the relative refractory period does not block firing so much as price it: the strength of input required to trigger a second spike is a direct measure of how recently the first one occurred. Only an unusually strong input will fire the cell early in this window, so a neuron's firing rate ends up reflecting the intensity of what is driving it.

Together, these two periods do two crucial jobs. They ensure that action potentials propagate **in one direction only**, because the region just behind a travelling spike is refractory and the spike cannot double back; and they cap the maximum firing rate, forcing the nervous system to encode intensity through *rate* and *pattern* rather than amplitude.

3.7 Retrograde Signalling: Endocannabinoids

Everything described so far in this section runs in one direction. The presynaptic terminal releases, the postsynaptic membrane receives, and the polarity of the neuron enforces the arrangement from dendrite to hillock to terminal. There is, however, a class of messenger that runs the other way, from the postsynaptic cell back to the presynaptic terminal that has just signalled to it. The best understood are the **endocannabinoids**, so named because they act on the receptors that the active compounds of cannabis also bind.

Two of them account for most of what is known: **anandamide** and **2-arachidonoylglycerol (2-AG)**. Both are lipids, derived from fatty acid components of the cell membrane itself, and that chemistry dictates almost everything else about how they behave.

3.7.1 Synthesis on Demand

A conventional transmitter is manufactured in advance, loaded into vesicles, and held at the terminal awaiting a calcium signal. Endocannabinoids cannot be handled this way. Being lipids, they would dissolve straight through the wall of any vesicle intended to contain them, so there is nothing to store and nothing to release in the usual sense.

Instead they are synthesised at the moment they are required. When a postsynaptic neuron is strongly depolarised, or when certain of its metabotropic receptors are activated, calcium entering the cell switches on the enzymes that cleave endocannabinoids out of membrane lipids. The molecule is produced on the spot, in the postsynaptic membrane, in response to that cell's own activity.

3.7.2 Travelling Backwards

Having been made in the postsynaptic membrane, the endocannabinoid diffuses the short distance back across the synaptic cleft, in the opposite direction to the traffic described in the rest of this section. On the presynaptic terminal it binds **CB1 receptors**, which are among the most abundant G protein-coupled receptors in the brain.

The consequence is inhibitory. CB1 is coupled to G_i, and its activation suppresses the calcium channels on which vesicle release depends, described earlier as the trigger the SNARE machinery waits for. With calcium entry reduced, fewer vesicles fuse and less transmitter is released. The postsynaptic cell has, in effect, reached back across the synapse and instructed the presynaptic terminal to be quieter.

3.7.3 Why a Backwards Signal Is Useful

The arrangement gives a synapse something it otherwise lacks: a way for the receiving cell to regulate how much it is being sent. Every other mechanism in this section leaves the postsynaptic neuron a passive recipient, able to decide only what to do with the input it is given. Retrograde signalling turns that into a negotiation.

Its clearest use is as a brake against overexcitation. A postsynaptic neuron being driven hard produces endocannabinoids in proportion to that drive, and the resulting suppression of transmitter release turns the input down. The effect can be brief, lasting seconds, or sustained, and in the latter case it becomes one of the mechanisms by which the strength of a synapse is lastingly adjusted, rather than merely a moment-to-moment correction.

This also explains why the cannabinoid system has such diffuse effects when it is engaged from outside. A drug acting at CB1 does not target one circuit; it interferes with a general-purpose feedback mechanism operating at synapses throughout the brain, which is why the consequences reach across appetite, pain, mood, movement, and memory at once.

3.8 Autoreceptors

Endocannabinoids are not the only brake operating on the presynaptic terminal, and the other one comes from a different direction entirely. An **autoreceptor** is a receptor sitting on a neuron's own membrane that binds the very transmitter that neuron releases. The cell is, in effect, listening to itself: what it has just released comes back and binds to it, and the result is a negative feedback loop closed entirely within a single neuron. This distinguishes autoreceptors from **heteroreceptors**, which sit in the same presynaptic position but respond to some other transmitter released by some other cell. It also distinguishes them from the retrograde signalling

just described, where the message originates in the postsynaptic cell. Endocannabinoids are the receiving neuron asking for less; an autoreceptor is the sending neuron regulating itself.

The mechanism divides according to where on the neuron the receptor sits, and the two locations do different jobs.

Terminal autoreceptors sit on the axon terminal alongside the release machinery. When transmitter accumulates in the cleft, these receptors detect it and suppress the voltage-gated calcium channels that vesicle fusion depends on, by exactly the route CB1 uses. Less calcium enters with each arriving action potential, fewer vesicles fuse, and the amount of transmitter released per impulse falls. The neuron continues to fire at the same rate; it simply says less each time it does.

Somatodendritic autoreceptors sit at the other end of the cell, on the soma and dendrites, where they detect transmitter released from the neuron's own dendrites or from neighbouring cells of the same population. Here the effect is on firing itself. These receptors act through the $\beta\gamma$ pathway described earlier, opening GIRK potassium channels; potassium leaves, the membrane hyperpolarises toward E_K, and the cell is carried further from threshold. The neuron now fires less often. Between the two locations, a neuron can throttle its output either by speaking less often or by saying less each time, and it is the somatodendritic population that regulates firing rate proper.

Both kinds are almost always metabotropic, and almost always coupled to $G_{i/o}$. This follows from the job rather than being an accident of which receptors happened to be recruited. A feedback brake needs to be slow enough to reflect accumulated activity rather than a single vesicle, amplified enough that modest transmitter concentrations produce a real effect, and modulatory rather than commanding, since its purpose is to adjust the cell's responsiveness and not to drive it. Those are precisely the three properties of G protein-coupled signalling set out earlier, and they are what an ionotropic receptor could not provide.

The particular receptors involved are familiar ones. Dopamine neurons carry **D2** autoreceptors, noradrenergic neurons carry α**2**, serotonergic neurons carry **5-HT$_{\mathbf{1A}}$** on their cell bodies and **5-HT$_{\mathbf{1B/1D}}$** on their terminals, and GABAergic terminals carry **GABA$_{\mathbf{B}}$**. Each of these subtypes also occurs postsynaptically elsewhere in the brain, doing quite different work. What makes a receptor an autoreceptor is therefore its location, not its identity: the same molecule is a feedback sensor on the cell that releases the transmitter and an ordinary receiver on the cell that listens.

The clinical consequences of this are best seen in the delayed action of the SSRIs. As noted earlier, these drugs block the serotonin reuptake transporter, and that blockade is immediate: within hours of the first dose, serotonin is clearing from the cleft more slowly and its concentration is raised. Yet patients do not feel better for weeks, which is a genuine puzzle if raised serotonin were the whole story. The autoreceptor resolves it. Reuptake is blocked everywhere at once, including around the raphe nuclei where the serotonergic cell bodies live, and the resulting rise in serotonin there activates their somatodendritic 5-HT$_{1A}$ autoreceptors. Those receptors do what autoreceptors do, and the raphe neurons slow their firing. Less serotonin therefore arrives at the distant terminals where the therapeutic effect is thought to arise, and the two changes very nearly cancel: the drug is working perfectly at the molecule it targets while the system as a whole absorbs the change.

What breaks the deadlock is adaptation. Under sustained stimulation the somatodendritic autoreceptors gradually desensitise and reduce in number over a period of two to four weeks. As their restraint weakens, raphe firing returns toward normal, but the reuptake blockade is still in place, so terminal serotonin now genuinely rises for the first time. The clinical improvement tracks this slow adaptation rather than the drug's immediate pharmacology, which is why the delay is measured in weeks and why stopping too early can look like failure when it is only earliness.

The general lesson runs beyond antidepressants. Because an autoreceptor and its postsynaptic

counterpart are frequently the same molecule, a drug directed at one will ordinarily reach the other, and the effect observed in a patient depends on which site dominates and on how each adapts over time. A receptor system is not a fixed target but one that responds to being interfered with.

3.9 Neurotransmitter Mimicry

Almost everything in this section has been described in terms of endogenous molecules: the transmitters the brain makes, the receptors they bind, and the machinery that clears them away. But a receptor cannot tell where the molecule filling it came from. It responds to shape and chemistry, not provenance, and this indifference is the opening that pharmacology exploits. A great many drugs, from the ones a psychiatrist prescribes to the ones sold on a street corner, work for the simple reason that they resemble a neurotransmitter closely enough to be mistaken for it, or resemble it closely enough to sit in its place without doing its job.

It is worth being precise about the vocabulary, because the differences between these cases are exactly the differences between the drugs. A molecule that binds a receptor's normal transmitter site, the **orthosteric** site, and switches the receptor on as the natural transmitter would is an **agonist**. This is mimicry in the strict sense: the drug is a counterfeit key that turns the lock. A molecule that occupies the same site but fails to activate the receptor is an **antagonist**; it turns nothing, and by squatting in the site it keeps the real transmitter out, so its effect is blockade rather than imitation. Between these lies the **partial agonist**, which activates the receptor but only submaximally, so that against a background of high natural signalling it can even look like a brake. And a molecule need not touch the orthosteric site at all: an **allosteric modulator** binds elsewhere on the receptor and changes how the receptor answers its true transmitter, turning the response up (a positive modulator) or down (a negative one) without ever triggering it directly. Only the first of these is mimicry; the others are worth setting beside it precisely because they are so often confused with it.

3.9.1 Serotonin: LSD and the Psychedelics

The canonical example is **LSD**. Its behavioural effects are extraordinary, but its mechanism is mundane in the sense that matters here: the molecule is close enough to serotonin to bind serotonin's receptors, and in particular it is an agonist at the **5-HT$_{2A}$** receptor, a G protein-coupled receptor concentrated on the pyramidal cells of the cortex. **Psilocybin**, the active principle of the relevant mushrooms, does the same thing by way of **psilocin**, the compound it is converted into in the body. Both are partial agonists, and the cortical 5-HT$_{2A}$ receptor is now the agreed common denominator of the classical psychedelics: block it and the experience does not occur.

Note how specific this is. Serotonin acts across a whole family of receptors, among them the 5-HT$_{1A}$ autoreceptor met in the previous subsection, whose job is to quiet the raphe neurons that release serotonin in the first place. LSD's psychedelic action is not a matter of raising serotonin everywhere; it is a matter of pretending to be serotonin at one particular subtype in one particular place. The lesson that receptor identity and location decide the effect, already drawn for autoreceptors, is drawn again here from the other side.

3.9.2 Acetylcholine: Nicotine and Muscarine

Acetylcholine offers the cleanest teaching example, because two natural products carve its two receptor families neatly in half and even supply their names. **Nicotine** is an agonist at the **nicotinic** acetylcholine receptor, the ionotropic, cation-passing receptor of the neuromuscular junction described earlier. Being ionotropic, it acts fast, and because these receptors also sit on the dopamine-releasing neurons of the reward pathway, nicotine's agonism there is a large

part of why it is addictive. **Muscarine**, from the fly agaric mushroom, is an agonist at the other family, the metabotropic **muscarinic** receptors; and **atropine**, from deadly nightshade, is a muscarinic **antagonist**. Here the whole vocabulary is on display in a single transmitter: two agonists selective for two different receptor families, and an antagonist that blocks one of them, which is why atropine dries secretions and races the heart, doing the opposite of what muscarinic activation would.

3.9.3 Opioids: Morphine and Heroin

The body makes its own opioids, the **endorphins** and **enkephalins** met among the neuropeptides, which damp down pain signalling. **Morphine**, and **heroin** once the body converts it into morphine, are agonists at the **μ-opioid receptor** those peptides normally use, a G protein-coupled receptor working through G_i. This is straightforward mimicry of an endogenous transmitter, and its two faces, powerful analgesia and powerful addiction, follow from the same fact: the drug occupies both the pain-suppressing circuits and, as with nicotine, the reward circuitry, more completely and for longer than the body's own opioids ever would.

3.9.4 Cannabinoids: THC and CBD

The cannabis plant furnishes the neatest contrast in the whole section, because its two best-known constituents act in opposite manners. **THC** (tetrahydrocannabinol) is a partial agonist at the **CB1 receptor**, the presynaptic G protein-coupled receptor introduced when endocannabinoids were discussed. It is, quite literally, a mimic of anandamide and 2-AG: it engages the brain's own retrograde-signalling brake from outside, which is exactly why its effects are so diffuse, reaching appetite, mood, memory, and pain at once, as anticipated at the end of the endocannabinoid subsection.

CBD (cannabidiol) is the instructive exception. It is not an appreciable CB1 agonist and does not produce the intoxication THC does; instead it behaves as a **negative allosteric modulator** at CB1, tuning down that receptor's response rather than driving it, and it exerts much of its remaining action elsewhere, including agonism at the **5-HT_{1A}** receptor. Two molecules from the same plant, one an agonist and one an allosteric modulator, thus illustrate in a single example the difference the opening paragraphs laboured to establish.

3.9.5 The Non-Mimics: Caffeine and Alcohol

It is worth ending with two of the most widely consumed psychoactive drugs in the world, precisely because neither is a mimic, and seeing why sharpens the definition. **Caffeine** works chiefly as an **antagonist** at **adenosine** receptors, the **A_1** and **A_{2A}** subtypes. Adenosine accumulates through the waking day and, acting on these receptors, promotes sleepiness; caffeine occupies the receptors without activating them, so the drowsiness signal is blocked. Caffeine does not imitate a transmitter, then. It removes a brake, and alertness is what is left when the brake is gone.

Alcohol is diffuse in a different way, acting not at one clean site but across several, and again by modulation rather than mimicry. Its two principal actions are as a **positive allosteric modulator** of the **$GABA_A$** receptor, the ionotropic chloride channel that is the brain's main inhibitory receptor, potentiating inhibition and producing sedation, and as an **antagonist** at the **NMDA** glutamate receptor, damping excitation. The **benzodiazepines** belong beside alcohol here as the tidier case: they too are positive allosteric modulators of $GABA_A$, binding their own distinct site to make the receptor more responsive to the GABA already present, which is why they calm without directly opening the channel themselves. The general point of this subsection is that "acting at a receptor" covers three quite different things (agonism, antagonism,

and allosteric tuning), and only the first deserves the name of mimicry; the confusion of the three is the confusion of a drug that imitates a signal with one that merely blocks or bends it.

3.10 Neurotransmitter Regulation

The drugs in the previous subsection act by occupying a receptor. A larger group, including most of the medicines prescribed in psychiatry, act one step back, on the machinery that decides how much transmitter is present at the synapse and for how long. Transmission has several such control points: a transmitter must be made, packed into vesicles, released, and then cleared from the cleft by reuptake or breakdown, and a drug that alters any one of these changes the signal without imitating it. This subsection takes the main classes in turn: the antidepressants that block reuptake, the stimulants that block it more powerfully or force release, the antipsychotics that block dopamine receptors outright, and finally the anaesthetics, which suppress signalling so broadly that sensation or consciousness itself is switched off.

Two lessons from earlier in the section keep returning. The first is that a drug's immediate action and its clinical effect are often different things, because the system adapts, as the autoreceptor account of the SSRIs showed. The second is that a drug acts wherever its target is found, and the same target sits in circuits with very different jobs, so that one molecular action can be the treatment in one pathway and a side effect in another.

The first three classes all act on reuptake. The monoamines are cleared from the synapse mainly by transporters on the presynaptic membrane that pull the transmitter back into the terminal to be repackaged, and there is one for each: the **serotonin transporter (SERT)**, the **noradrenaline transporter (NET)**, and the **dopamine transporter (DAT)**. Block a transporter and the transmitter it would have removed stays in the cleft longer and at a higher concentration, so each release has a larger and longer effect on the receptors around it. The antidepressants that follow differ mainly in which of the three transporters they block.

Selectivity is the point of the modern drugs, and their history shows why. The older **tricyclic** antidepressants, such as amitriptyline, block both SERT and NET, but they also block histamine, muscarinic, and α_1 receptors, which causes drowsiness, a dry mouth, constipation, and dizziness on standing, and in overdose they are dangerous to the heart. The drugs that replaced them were designed to act on the transporters and little else.

3.10.1 SSRIs: Fluoxetine (Prozac)

Selective serotonin reuptake inhibitors (SSRIs) block SERT and leave the other transporters and receptors largely alone. **Fluoxetine**, sold as **Prozac** from the late 1980s, was the first to be widely used, and it was followed by sertraline, citalopram, escitalopram, and others that differ mainly in how long they last in the body and how strongly they interact with other drugs. They are the first-line drug treatment for depression, and are also used for anxiety disorders, obsessive-compulsive disorder, and bulimia.

Their mechanism was set out in the subsection on autoreceptors and needs only an outline here. The blockade of SERT is immediate, but the extra serotonin around the raphe nuclei first activates the 5-HT_{1A} autoreceptors there, which slows the serotonergic neurons and cancels much of the gain. Only as those autoreceptors desensitise, over two to four weeks, does serotonin at the distant terminals genuinely rise, and the clinical improvement follows that slower timetable. Even this is probably not the whole story: the later changes also include altered gene expression and the growth of new synaptic connections, and which of these the improvement depends on is still debated.

The side effects follow from the fact that serotonin receptors are found throughout the body. Early nausea comes from 5-HT_3 receptors in the gut and brainstem, which is why it usually fades within the first weeks, and why the 5-HT_3 antagonist ondansetron in Table 1 is an anti-sickness drug. Sexual side effects, emotional blunting, and a brief rise in anxiety at the start of treatment

are also common. The most serious risk comes from combining an SSRI with other drugs that raise serotonin, such as the **monoamine oxidase inhibitors (MAOIs)**, an older class of antidepressant that blocks the enzyme which breaks the monoamines down. Serotonin can then rise far enough to cause **serotonin syndrome**, with agitation, tremor, fever, and muscle twitching.

3.10.2 SNRIs: Venlafaxine and Duloxetine

Serotonin–noradrenaline reuptake inhibitors (SNRIs) block both SERT and NET. The main examples are **venlafaxine** (Effexor) and **duloxetine** (Cymbalta). Venlafaxine shows how selectivity can depend on dose: at low doses it acts mainly on SERT and behaves much like an SSRI, and its effect on NET becomes significant only as the dose rises.

Adding noradrenaline brings two differences. The first is a use beyond mood. Noradrenaline and serotonin both feed the descending pathways from the brainstem that suppress pain signals as they enter the spinal cord, and strengthening those pathways makes SNRIs useful for chronic pain, including the nerve pain of diabetes and fibromyalgia, where SSRIs are less effective. The second is a set of side effects that come from noradrenaline's role in the sympathetic nervous system: sweating, a faster heart rate, and, at higher doses, a rise in blood pressure. Stopping an SNRI abruptly, venlafaxine especially, can also cause unpleasant withdrawal symptoms, because the drug leaves the body quickly.

3.10.3 NDRIs: Bupropion (Wellbutrin)

Bupropion, sold as **Wellbutrin** for depression and as Zyban for stopping smoking, is the main **noradrenaline–dopamine reuptake inhibitor (NDRI)**. It blocks NET and DAT, relatively weakly, and leaves SERT alone, and that difference from the SSRIs shows directly in its profile. It does not cause the sexual side effects or emotional blunting associated with serotonin, and it tends not to cause weight gain, which is why it is often chosen when those side effects matter, or added to an SSRI when they appear. Its action on dopamine is thought to be why it helps with low energy and loss of motivation, and why it helps people stop smoking, easing the craving that follows when nicotine's boost to dopamine is taken away; bupropion also blocks nicotinic receptors directly. Its main risk is that, at high doses, it makes seizures more likely.

Blocking DAT raises an obvious question, since cocaine blocks it too. The answer lies in how far and how fast dopamine rises. Bupropion blocks the transporter weakly and is taken by mouth in a slow-release form, so dopamine rises modestly and gradually, and the drug has little potential for misuse. The same principle separates the medical and recreational uses of the stimulants that follow.

3.10.4 Stimulants: Amphetamine, Methylphenidate, and Cocaine

The **stimulants** raise dopamine and noradrenaline much more strongly, and they do it in two different ways.

Methylphenidate (Ritalin) and **cocaine** are reuptake blockers, like the antidepressants but more potent at DAT. Methylphenidate blocks DAT and NET; cocaine blocks all three monoamine transporters. Cocaine also has a second action, which links it to the end of this subsection: it blocks voltage-gated sodium channels, and in 1884 it became the first local anaesthetic used in surgery, before safer synthetic versions replaced it.

Amphetamine works differently, and the difference matters. It is not a blocker but a substrate: its shape is close enough to dopamine and noradrenaline that the transporters carry it into the terminal. Once inside, it disrupts the storage of transmitter in vesicles, so that dopamine builds up in the cytoplasm, and it makes the transporters run in reverse, pumping

that dopamine out into the cleft. The result is a release of transmitter that does not depend on action potentials at all, which is why amphetamine raises dopamine further than a blocker does.

At low, steady doses taken by mouth, methylphenidate and amphetamine are the main treatments for **attention deficit hyperactivity disorder (ADHD)**, and stimulants are also used to treat narcolepsy. It is often called paradoxical that a stimulant should help hyperactivity, but the effect is not paradoxical. At these doses the important action is in the prefrontal cortex, where extra dopamine and noradrenaline strengthen the circuits that keep attention on a task and hold back impulses (the executive functions taken up in the section on the frontal lobe), and smaller improvements in focus are seen in people without the condition too.

Taken in larger doses, or smoked, snorted, or injected so that they reach the brain within seconds, the same drugs produce euphoria, and they are among the most addictive substances known. What separates the two uses is mainly the size and speed of the dopamine rise in the reward circuitry that the section on the limbic system describes: a slow, modest rise improves attention, while a fast, steep one is felt as a high and is strongly reinforcing. At high doses or with repeated use, amphetamine and cocaine can also produce **psychosis**, with paranoia and hallucinations, and that observation is one of the main clues behind the next class of drug.

3.10.5 Antipsychotics: Blocking Dopamine

If too much dopamine can produce psychosis, blocking dopamine might treat it, and that is broadly how the **antipsychotics** work. The first, **chlorpromazine**, was introduced in the early 1950s, and its mechanism was worked out only later: the drugs that treat psychosis best at the lowest doses are the ones that bind most strongly to the **D2 receptor**, where they act as antagonists. This is the basis of the **dopamine hypothesis** of psychosis, which in its modern form holds that symptoms such as hallucinations and delusions arise from excessive dopamine release in the striatum, lending ordinary perceptions and thoughts a false sense of significance. The drugs treat these *positive* symptoms reasonably well, and do much less for the *negative* symptoms of schizophrenia (withdrawal, flattened emotion, and loss of motivation) or for its effects on thinking.

Blocking D2 receptors is a blunt instrument, because a drug cannot block them in one pathway and spare them in another. They sit in several dopamine systems, and blocking each has a predictable result:

- In the **mesolimbic** pathway to the ventral striatum, blockade is the intended therapeutic effect.
- In the **nigrostriatal** pathway, blockade imitates the loss of dopamine that causes Parkinson's disease, producing stiffness, slowness, and tremor. These are the **extrapyramidal side effects**, and the section on the basal ganglia explains why they take that form. Years of treatment can also cause **tardive dyskinesia**, involuntary movements of the mouth and face that may persist after the drug is stopped.
- In the pathway from the hypothalamus to the pituitary, dopamine normally holds back the hormone **prolactin**, so blocking it raises prolactin, which can cause breast enlargement, milk production, and missed periods.

Brain imaging shows how narrow the working window is: the therapeutic effect generally needs roughly two thirds of the striatum's D2 receptors to be occupied, and movement side effects become likely above about four fifths. This is the clearest case in these notes of a drug acting wherever its target is found: the same molecular action is the treatment in one pathway and a side effect in the next two.

The older **typical** antipsychotics, such as chlorpromazine and **haloperidol**, are strong D2 blockers and often cause movement side effects. The newer **atypical** antipsychotics, such

as **risperidone**, **olanzapine**, **quetiapine**, and **clozapine**, cause them less often, partly because many of them also block the **5-HT$_{2A}$** receptor and partly because some bind D2 more loosely. That 5-HT$_{2A}$ receptor is the one at which the psychedelics act as agonists; the atypical antipsychotics are antagonists at the same receptor, and taking one blunts or blocks a psychedelic experience. The atypicals carry their own costs, chiefly weight gain and a higher risk of diabetes, most of all with olanzapine and clozapine. Clozapine is also the one antipsychotic that often works when others have failed, but it needs regular blood tests because it can dangerously lower the white blood cell count. **Aripiprazole** takes a different approach again. It is a D2 *partial agonist*, the category defined at the start of the mimicry subsection, so it turns dopamine signalling down where dopamine is high while keeping some signalling where it is low.

3.10.6 Anaesthetics: General and Local

Anaesthetics sit at the far end of the scale. Rather than adjusting one transmitter system, they suppress neural signalling broadly enough to remove sensation from part of the body or consciousness from the whole person. They come in two kinds that act in quite different ways.

Local anaesthetics, such as **lidocaine**, block the voltage-gated sodium channels on which the action potential depends. With those channels blocked a nerve cannot fire, and signals from the numbed area never reach the spinal cord. The drug crosses the membrane in an uncharged form and blocks the channel from the inside, and it binds most readily to channels that are open or inactivated, so nerves that are firing hard are blocked first. Thin fibres are also blocked before thick ones, which is why pain and temperature disappear before touch and before the motor fibres, and why an epidural in labour can remove pain while leaving some feeling and movement in the legs.

General anaesthetics act on the brain and produce a reversible state of unconsciousness, together with loss of memory, loss of response to pain, and stillness. Most of them work by strengthening inhibition. **Propofol**, given into a vein, and inhaled anaesthetics such as **sevoflurane** are positive allosteric modulators of the GABA$_A$ receptor, the same action as the benzodiazepines and alcohol in the previous subsection but far stronger; the inhaled agents also open potassium channels that make neurons harder to excite. A second group, which includes **ketamine** and **nitrous oxide**, works mainly by blocking the NMDA glutamate receptor instead. Ketamine produces a distinctive *dissociative* state, in which the patient may keep their eyes open and go on breathing normally while being cut off from their surroundings. At lower doses it has also become a fast-acting treatment for severe depression, working within hours rather than weeks, a finding the section on the limbic system returns to.

How these drugs actually remove consciousness is still not fully understood, which makes them unusually interesting for the larger questions of these notes. Around 1900 it was noticed that an anaesthetic's strength tracks how well it dissolves in oil, which suggested a general effect on the fatty membrane; the targets are now known to be specific proteins, but knowing the receptor does not by itself explain the result. Leading accounts point to a breakdown in communication between the thalamus and the cortex, and between different regions of the cortex, so that the brain remains active but its parts no longer work together as a whole. Anaesthesia is therefore a switch for consciousness that can be turned off and on at will and is used on hundreds of millions of patients a year, whose molecular action is known and whose effect on experience has not yet been explained in terms of that action. Within the neurology itself, it is a reminder of the questions the later parts of these notes take up.

3.11 Summary

The electrical life of a neuron is a story of ions crossing a membrane. The resting potential is set mostly by potassium permeability. Small graded potentials from synaptic input summate at the axon hillock; if they push the membrane past threshold, a self-reinforcing avalanche of sodium

entry produces an action potential, which propagates to the axon terminal. There, calcium influx triggers vesicle fusion, neurotransmitter is released, and the process starts again in the next cell. Refractory periods keep the signal unidirectional and bounded. Layered on top of this forward flow are two presynaptic brakes. Endocannabinoids, synthesised on demand in the postsynaptic membrane, travel back across the cleft to suppress release from the terminal that just fired, giving the receiving cell a say in how much it is sent; autoreceptors let the sending neuron regulate itself, damping release at the terminal and firing rate at the soma according to how much transmitter it has already put out. These same receptors are also where drugs act, mimicking the natural signal as agonists, blocking it as antagonists, or allosterically tuning the response up or down, and the effect a drug produces follows from which of these it does and at which receptor. Other drugs act on the regulation of transmission instead: the antidepressants and many stimulants block the transporters that clear the monoamines from the cleft, amphetamine forces their release, the antipsychotics block dopamine's D2 receptor, and the anaesthetics silence signalling broadly enough to remove sensation or consciousness. The Nernst equation, which at first looks like a piece of physical chemistry, turns out to be quietly responsible for most of it.

One result of this section deserves to be carried further than the others, because it is the first point at which anything resembling *meaning* enters the account. A neurotransmitter has no intrinsic significance. Glutamate is not an excitatory molecule and GABA is not an inhibitory one; glutamate excites because the receptors that bind it happen to pass sodium, and put on a chloride-selective channel it would inhibit. The signal is identical in every case, and what the signal means is decided entirely by the receiver. This is a small fact about pore selectivity, and it is also the seed of something much larger, because it establishes that significance in a nervous system is not carried by a message but assigned by whatever the message reaches. The same principle appears twice more in these notes and does progressively heavier work each time: at the autoreceptor, where a receptor's location rather than its identity determines whether it is a feedback sensor or an ordinary receiver, and in the basal ganglia, where one diffuse broadcast of dopamine promotes movement and suppresses it simultaneously, because the two populations it lands on carry receptors of opposite sign. A system built this way can make one chemical mean many things, which is a large part of how so much is achieved with so short a list of transmitters.

The transmitters and receptor families met across this section are gathered in Table 1 below, extended with their principal receptor subtypes to serve as a single-glance reference.

Table 1: The principal neurotransmitters and the receptor families through which they act, with their synaptic actions and the drugs that bind them. "Type" distinguishes fast ionotropic channels from slower metabotropic G protein-coupled receptors (GPCRs), with the G protein noted where relevant. The receptors drawn on in this section are discussed in the prose above; the remaining subtypes are included to complete the picture. A dash marks a receptor for which no relevant drug is listed. Drugs that act on transporters or ion channels rather than receptors are listed separately in Table 2.

Neurotransmitter	**Receptor**	**Type**	**Action**	**Drug (effect)**
Glutamate	AMPA	Ionotropic	Excitatory	—
	NMDA	Ionotropic	Excitatory	Alcohol, ketamine, nitrous oxide (antagonist)
	Kainate	Ionotropic	Excitatory	—
	mGluR	GPCR	Modulatory	—
GABA	$GABA_A$	Ionotropic	Inhibitory	Alcohol, benzodiazepines, propofol, sevoflurane (positive allosteric modulator)

Table 1 continued from previous page

Neurotransmitter	Receptor	Type	Action	Drug (effect)
	$GABA_B$	GPCR	Inhibitory	— (autoreceptor)
Acetylcholine	Nicotinic	Ionotropic	Excitatory	Nicotine (agonist)
	Muscarinic	GPCR	Modulatory	Muscarine (agonist); atropine (antagonist)
Dopamine	D1	GPCR (G_s)	Excitatory	—
	D2	GPCR (G_i)	Inhibitory	Antipsychotics (antagonist); aripiprazole (partial agonist); autoreceptor
Serotonin (5-HT)	5-HT_{1A}	GPCR	Inhibitory	CBD (agonist); somatodendritic autoreceptor
	5-$HT_{1B/1D}$	GPCR	Inhibitory	— (terminal autoreceptor)
	5-HT_{2A}	GPCR	Excitatory	LSD, psilocin (partial agonist); atypical antipsychotics (antagonist)
	5-HT_3	Ionotropic	Excitatory	Ondansetron (antagonist)
Noradrenaline	α_1	GPCR (G_q)	Excitatory	—
	α_2	GPCR	Inhibitory	— (autoreceptor)
	β_1/β_2	GPCR (G_s)	Modulatory	—
Endocannabinoids (anandamide, 2-AG)	CB1	GPCR (G_i)	Inhibitory	THC (partial agonist); CBD (negative allosteric modulator)
	CB2	GPCR (G_i)	Modulatory	—
Opioid peptides (endorphins, enkephalins)	μ-opioid	GPCR (G_i)	Inhibitory	Morphine, heroin (agonist)
Adenosine	A_1/A_{2A}	GPCR	Modulatory	Caffeine (antagonist)

The drug classes of the regulation subsection, which act on transporters, receptors, and ion channels rather than by imitating a transmitter, are gathered in Table 2.

Table 2: The main classes of drug that regulate neurotransmission, with examples, their molecular target, how they act on it, and their principal uses. Brand names are given in brackets where they are the more familiar name.

Class	Examples	Target	Action	Main uses
SSRI	Fluoxetine (Prozac), sertraline	SERT	Blocks serotonin reuptake	Depression, anxiety, OCD
SNRI	Venlafaxine (Effexor), duloxetine (Cymbalta)	SERT, NET	Blocks serotonin and noradrenaline reuptake	Depression, anxiety, chronic pain
NDRI	Bupropion (Wellbutrin)	NET, DAT	Blocks noradrenaline and dopamine reuptake	Depression, stopping smoking

Table 2 continued from previous page

Class	Examples	Target	Action	Main uses
Stimulant (blocker)	Methylphenidate (Ritalin), cocaine	DAT, NET (cocaine also SERT)	Blocks reuptake	ADHD, narcolepsy (methylphenidate)
Stimulant (releaser)	Amphetamine	DAT, NET, vesicle storage	Reverses the transporters, releasing dopamine and noradrenaline	ADHD, narcolepsy
Typical antipsychotic	Chlorpromazine, haloperidol	D2	Antagonist	Psychosis
Atypical antipsychotic	Risperidone, olanzapine, quetiapine, clozapine	D2, 5-HT_{2A}	Antagonist	Psychosis, bipolar disorder
	Aripiprazole	D2	Partial agonist	Psychosis, bipolar disorder
General anaesthetic	Propofol, sevoflurane	$GABA_A$	Positive allosteric modulator	Surgical anaesthesia
	Ketamine, nitrous oxide	NMDA	Antagonist	Anaesthesia, pain relief; ketamine also depression
Local anaesthetic	Lidocaine	Voltage-gated Na^+ channel	Blocks the channel from inside	Local and regional anaesthesia

4 Glial Cells

Read a textbook written fifty years ago and it might describe neurons as the stars of the nervous system and glial cells as their stagehands: quiet, supportive, and uninteresting. That view is now dead. Modern neuroscience treats glia as active participants: they shape synapses, regulate ions, feed neurons, defend the brain, build the insulation that makes fast signalling possible, and fail or misbehave in many of the diseases we most dread. A surprisingly large portion of what we might loosely call "brain function" is not neuronal at all.

The word "neuroglia" literally means "nerve glue," which was a reasonable guess in the nineteenth century and is a misleading one now. Structural support turns out to be among the least of what these cells do.

This section introduces the main types of glial cell, what each of them does, and where glial failure sits in neurological disease. The theme running through all of it is that a neuron cannot do its job alone: every neuron in the body depends on glia to regulate its environment, recycle its transmitters, insulate its axon, and protect it from damage.

The main glial types, divided by location, are:

- **CNS**: astrocytes, oligodendrocytes, ependymal cells, microglia.
- **PNS**: Schwann cells, satellite cells.

4.1 Astrocytes and Satellite Cells

4.1.1 Astrocytes

Astrocytes are the workhorses of the CNS glial family. Under the microscope they are star-shaped (hence the name), with long processes radiating in every direction. Those processes contact

neurons, synapses, blood vessels, and the surface of the brain, and that connectivity is the key to everything they do. An astrocyte sits at the intersection of almost everything important in nervous tissue.

Their responsibilities divide into seven roles.

1. Structural support. Astrocytes hold the architecture of nervous tissue together. Their processes fill the spaces between neurons, dividing the tissue into domains that barely overlap, so that each astrocyte takes responsibility for a defined territory and its synapses. Nervous tissue has no fibrous connective framework of the kind found elsewhere in the body; this cellular scaffolding is what stands in for it.

2. Ion buffering. Every time a neuron fires, potassium leaves the cell and accumulates briefly in the extracellular space. If this were allowed to build up, nearby neurons would become abnormally depolarised and fire uncontrollably. Astrocytes take up the excess potassium and redistribute it through their own network, a process called **spatial buffering**. Without it, neuronal activity would rapidly become chaotic.

3. Neurotransmitter recycling. Astrocytes are the main clean-up crew for glutamate, the principal excitatory transmitter of the brain. They pull glutamate out of the cleft through dedicated transporters, convert it to glutamine (which is inert and therefore safe), and ship the glutamine back to neurons to be re-converted into glutamate. This loop is called the **glutamate–glutamine cycle**, and it is the main defence against excitotoxicity.

4. Metabolic support. Neurons have extraordinary energy demands and almost no energy storage, so someone has to keep them fed. That someone is the astrocyte. Astrocytes take up glucose from blood vessels, store glycogen, and supply lactate and other substrates to neurons, an arrangement often called the *astrocyte–neuron lactate shuttle*.

5. The blood-brain barrier. The CNS is protected from the rest of the body by the **blood-brain barrier (BBB)**, a highly selective wall that only specific molecules can cross. The BBB is built chiefly by tight junctions between endothelial cells in brain capillaries, but astrocytic end-feet wrap around those vessels and are essential for inducing and maintaining the barrier's properties. Without astrocytes, the BBB does not form properly.

6. Modulation of synapses. The "tripartite synapse" concept treats every synapse as involving three cells, not two: the presynaptic neuron, the postsynaptic neuron, and an astrocytic process sitting alongside them. The astrocyte can detect neurotransmitter release, respond with its own calcium signals, and release *gliotransmitters* that feed back onto nearby neurons. How large a role this plays in moment-to-moment information processing is still an active research question, but the days of astrocytes as inert scaffolding are firmly over.

7. Response to injury. After any significant CNS insult, astrocytes become *reactive*. They enlarge, proliferate, change which genes they express, and in severe cases help form a dense **glial scar**. This response is double-edged: it walls off damage and limits inflammation, but the scar itself is a significant obstacle to the regeneration of neurons, which is one of the reasons CNS injuries heal so poorly.

4.1.2 Satellite Cells

Satellite cells are the PNS cousins of astrocytes. They form a close-fitting envelope around the cell bodies of sensory and autonomic neurons inside peripheral **ganglia**, such as the dorsal

root ganglia. Their functions overlap substantially with those of astrocytes (structural support, control of the local chemical environment, regulation of ion concentrations, and modulation of excitability), but they act on a much smaller, more localised scale.

Interest in satellite cells has grown sharply in recent years because of their role in chronic pain. Changes in satellite cell signalling are thought to contribute to the abnormal excitability of sensory neurons in persistent pain states, which makes them one of the more surprising targets on the emerging map of pain biology.

4.2 Oligodendrocytes and Schwann Cells

4.2.1 Oligodendrocytes

Oligodendrocytes are the myelinating glia of the CNS. Their name means "cells with few branches," a deliberate contrast with the exuberant astrocyte, and their central job is to wrap axons in **myelin**, the lipid-rich insulation that makes fast signalling possible.

A single oligodendrocyte can myelinate segments of several different axons at once, reaching out a process to each and wrapping it repeatedly. This is one of the most important differences between CNS and PNS myelin: a Schwann cell commits itself to a single internode on a single axon, whereas one oligodendrocyte serves many. The arrangement is efficient, but it also means that the loss of a single oligodendrocyte strips myelin from several axons at once.

Their jobs are closely linked. They **produce myelin**, layer upon layer of plasma membrane wrapped tightly around the axon. They also **support axonal health** in ways that go beyond simple insulation: damage to oligodendrocytes tends to cause damage to the axons they serve, even when the axons themselves are initially untouched. And they **organise the axonal membrane** into alternating **internodes** (myelinated) and **nodes of Ranvier** (bare gaps where voltage-gated sodium channels cluster).

The clustering of sodium channels at the nodes is what makes **saltatory conduction** possible. Instead of the action potential regenerating continuously at every point along the axon, it regenerates only at the nodes and effectively jumps from one to the next. The result is faster, more energy efficient, and more reliable.

4.2.2 Schwann Cells

Schwann cells are the PNS counterparts of oligodendrocytes, and they come in two forms.

A myelinating Schwann cell wraps itself repeatedly around a single segment of a single axon, producing one **internode** of myelin. One Schwann cell per internode, one axon at a time. Multiple Schwann cells are therefore needed along the length of any given PNS axon, separated by nodes of Ranvier just as in the CNS.

Not every peripheral axon is myelinated. Small, slowly conducting axons (including many pain fibres and autonomic fibres) are instead embedded in non-myelinating Schwann cells, often multiple axons per cell, in structures called **Remak bundles**. These Schwann cells do not build concentric wrappings, but they still provide mechanical support, metabolic support, and protection.

The single most striking difference between the PNS and the CNS is that peripheral nerves can regenerate and central nerves largely cannot. Schwann cells are a large part of why. After injury to a peripheral nerve, Schwann cells clear away the debris, proliferate, and form longitudinal tracks that actively guide regrowing axons back toward their targets, while secreting growth factors that promote regrowth. A cut peripheral nerve has a real chance of reconnecting; a cut spinal cord generally does not.

4.2.3 Myelin

Myelin is not a cell type in its own right (it is built by oligodendrocytes and Schwann cells), but its electrical role is specific enough to be worth setting out separately.

Physically, myelin is a multilayered sheath of plasma membrane wrapped tightly around an axon, squeezing out almost all of its cytoplasm so that what remains is essentially a stack of lipid bilayers. The stack is heavily lipid-rich, which is why myelinated tissue looks pale and is called *white matter*, and it contains a characteristic set of structural proteins.

There are two electrical tricks at work. First, wrapping an axon in many layers of membrane enormously **increases its resistance** to transverse current flow, which means less current leaks out across the membrane. Second, and equally importantly, it **decreases its capacitance**: a stack of membranes in series stores less charge than a single membrane, so less charge needs to be moved to change the voltage. Both effects mean that a given current spreads further and faster along the inside of the axon.

Instead of regenerating at every point, the action potential now jumps almost passively along the internodes, pausing at the nodes of Ranvier only long enough for sodium channels there to boost it before the next leap. The net result is dramatic: a well-myelinated axon can conduct at roughly 100 m/s, versus perhaps 1 m/s for an unmyelinated one of the same diameter.

When myelin is stripped away, the electrical tricks disappear. Current leaks across the bare membrane, conduction slows, and action potentials may simply fail to propagate. This can cause severe neurological deficits even when the axons underneath are, for the time being, structurally intact, which is precisely what is so frightening about demyelinating diseases. Fix the myelin and the function can return; let the damage accumulate and the underlying axons may eventually die too.

4.3 Ependymal Cells

Ependymal cells form an epithelial-like lining along the walls of the ventricles and the central canal of the spinal cord. They are glial cells of a rather different flavour from astrocytes or oligodendrocytes: their job, essentially, is plumbing.

A specialised subset of ependymal cells, together with associated capillaries and connective tissue, forms the **choroid plexus**, the structure inside the ventricles that actively secretes cerebrospinal fluid. CSF is the fluid we met earlier: it cushions the brain, keeps its chemical environment stable, and removes waste products. Without ependymal cells, there is no CSF.

Beyond production, ependymal cells also help move CSF. Many of them have **cilia** on their apical surface, and the beating of these cilia drives circulation through the ventricular system. The ependymal lining also contributes to the regulation of exchange between CSF and brain tissue, though it is not as tight a barrier as the blood-brain barrier proper.

Clinically, anything that obstructs CSF flow or overwhelms its reabsorption can produce **hydrocephalus**, in which CSF accumulates inside the ventricles. The pressure rises, the ventricles balloon, and, especially in infants, whose skulls are still flexible, the head itself can visibly enlarge. Treatment usually involves surgically diverting CSF with a shunt.

4.4 Microglia

Microglia are the immune system of the CNS. They are unusual among glial cells in that they are not of ectodermal origin: they descend from the same mesodermal lineage as macrophages and other immune cells, and they migrate into the brain during early development. Once installed, they are essentially permanent residents.

In their “resting” state, microglia are not really resting at all. Their processes extend, retract, and extend again, continuously surveying the local tissue for anything out of place: pathogens, debris, abnormally folded proteins, damaged cells, and inappropriate synaptic structures. When they find something, they respond.

4.4.1 What Microglia Do

Immune surveillance. This is the microglial baseline, and it is continuous rather than occasional. The processes described above sweep the surrounding tissue without pause, so that the whole volume of the brain falls under inspection on a timescale of hours despite the cells themselves barely moving. What they are testing for is anything that should not be there: a pathogen, the debris of a dead cell, a protein that has misfolded, a synapse that is no longer appropriate. Surveillance is therefore not a passive readiness to respond but the activity that occupies most of a microglial cell's life, and everything else in this list is what happens on the occasions when it finds something.

Phagocytosis. Once activated, microglia engulf and digest dead cells, debris, pathogens, and protein aggregates. They are the principal professional phagocytes of the CNS.

Inflammatory signalling. Microglia release cytokines and chemokines to coordinate local inflammatory responses. In acute infection or injury this is protective. In chronic or excessive activation it is corrosive, and chronic microglial inflammation is increasingly implicated in neurodegenerative disease.

Synaptic pruning. During development, microglia physically remove unnecessary synapses, helping to refine neural circuits into their mature form. Remarkably, this pruning role continues into adulthood, and disruptions to microglial pruning are now suspected of contributing to several disorders of neurodevelopment.

4.4.2 Activation States

Microglial activation is not a simple switch. A resting microglial cell looks highly branched ("ramified"); an activated one retracts its branches and takes on a more amoeboid shape appropriate to its phagocytic role. But the states in between span a wide spectrum, and depending on context the same cell can be protective or destructive. The current picture is more one of *roles* than of discrete states.

4.5 Comparing CNS and PNS Glia

The two compartments solve the same problems with different cells.

In the CNS: astrocytes maintain homeostasis and support the BBB; oligodendrocytes myelinate; ependymal cells line the ventricles and make CSF; microglia provide immune defence.

In the PNS: Schwann cells myelinate (and non-myelinating Schwann cells envelop small axons); satellite cells support the cell bodies of ganglion neurons.

The parallels are not perfect, but the functional principles are the same (control the environment, insulate the axons, defend the tissue), and each system has specialised cells for each role.

4.6 Summary

Glia are not the supporting cast of the nervous system. Astrocytes and satellite cells regulate the environment in which neurons live, holding ion concentrations, transmitter levels, and extracellular composition within the narrow bounds that electrical signalling requires, and supplying the energy substrates that neurons cannot store for themselves. Oligodendrocytes and Schwann cells produce the myelin without which fast conduction is impossible. Ependymal cells

build and circulate the CSF. Microglia provide the standing immune defence of a compartment that the rest of the body's immune system cannot easily reach, and, with astrocytes, actively shape synapses rather than merely surrounding them. Glia also carry out most of the maintenance and repair of nervous tissue, including the regenerative capacity that distinguishes peripheral nerves from central ones. Neuronal activity is the visible half of the story, but none of it would be possible without this infrastructure, and any serious account of how the brain works, or of how it fails in disease, has to take the glial half seriously.

One glial contribution matters more than the others for what follows, and it is myelin. The figures given above, roughly a hundred metres per second along a myelinated axon against perhaps one along a bare one of the same width, are the difference between a nervous system that can act on the world and one that can only comment on it afterwards. A signal from the foot reaches the brain in something like twenty milliseconds, and a response returns on a similar schedule, which is why balance is possible and why a stumble can be caught. Without that speed the same computations would still be correct and would arrive too late to be of use. Much of what the later sections describe (the binding of vision to touch into a single scene, the interception of a movement before it is completed, the fast appraisal of a threat) presupposes that signals cross the brain in milliseconds rather than seconds. Cognition happens in real time because glia make the wiring fast enough, and that is an achievement of insulation rather than of computation.

5 The Frontal Lobe

We now have a working model of individual neurons, the signals they generate, and the glial tissue that supports them. The remaining sections move up to the systems level, beginning with the region that has, more than any other, been identified with personhood itself.

The **frontal lobe** is the largest of the cerebral lobes and sits just behind the forehead. It is responsible for voluntary movement, the planning and execution of actions, language production, and a cluster of higher cognitive and behavioural functions we collectively call *executive function*: judgment, decision-making, inhibition, and personality. If the occipital lobe is where vision lives and the temporal lobe is where memory and hearing live, the frontal lobe is where thought is translated into action, and where action is held back in the service of longer-term goals.

It is also the region most implicated in the overlap between neurology and psychology, which makes it the natural place to begin the survey of the cortex. Damage to different parts of the frontal lobe produces characteristic and sometimes startling changes in behaviour and personality, changes that force us to confront how thoroughly the "mind" is rooted in very specific pieces of tissue.

5.1 Anatomical Landmarks

The frontal lobe is bounded by three anatomical landmarks:

- The **central sulcus** (Rolandic fissure) runs roughly down the top of the hemisphere and separates the frontal lobe from the parietal lobe. It is the most clinically important sulcus in the brain, because it divides the *motor* cortex in front from the *somatosensory* cortex behind.
- The **lateral sulcus** (Sylvian fissure) runs along the side of the hemisphere and separates the frontal lobe from the temporal lobe below.
- The **longitudinal fissure** runs down the midline and separates the two hemispheres from each other.

Within the frontal lobe, a few gyri (ridges) are worth knowing by name. The **precentral gyrus**, running immediately in front of the central sulcus, contains the primary motor cortex. The **superior, middle, and inferior frontal gyri**, running along the lateral surface, house progressively more abstract motor and cognitive areas toward the front. Broca's area, for example, sits in part of the inferior frontal gyrus.

The rest of this section walks forward from the most concrete function, the movement of muscles, to the most abstract, the governance of conduct and character.

5.2 Primary Motor Cortex (M1)

The **primary motor cortex**, traditionally labelled **M1** and corresponding to Brodmann area 4, occupies the precentral gyrus just in front of the central sulcus. It is the final cortical stage of voluntary movement: the place where the abstract intention to move becomes the concrete commands that drive muscles.

Its output leaves the cortex mainly via the **corticospinal tract** (for the body) and the **corticobulbar tract** (for the face and head), eventually reaching lower motor neurons in the brainstem and spinal cord that directly innervate muscles. Crucially, the corticospinal tract crosses to the opposite side of the body as it descends, so the left M1 controls the right side of the body and vice versa. This **contralateral** organisation is why a stroke in one hemisphere weakens the opposite side.

5.2.1 The Motor Homunculus

M1 is not a featureless sheet. It is organised *somatotopically*: different patches of cortex control different parts of the body, arranged in a roughly orderly map along the gyrus. This map is often drawn as a small distorted human figure, the **motor homunculus**, draped along the cortex.

Walking from medial to lateral along M1:

- The **medial** regions (down in the longitudinal fissure) control the lower limb.
- The **superior lateral** regions control the trunk and upper limb.
- The **lateral** regions control the face, lips, and tongue.

What makes the homunculus so striking visually is that it is wildly disproportionate. The hands and face occupy enormous stretches of cortex, while the trunk and legs get relatively little. The representation is proportional not to body size but to the fineness of motor control that part of the body demands. Hands and lips perform feats of precision on a millisecond timescale; the muscles of the lower back do not.

5.2.2 Microanatomy

M1 contains exceptionally large pyramidal neurons called **Betz cells**, which project long axons into the corticospinal tract described above. These are among the biggest neurons in the entire CNS, and the size is not incidental: a cell body must support an axon reaching from the cortex to the lower spinal cord, over a metre in an adult, and keep the kinesin and dynein traffic described in the section on neuron structure running along the whole of it.

Betz cells are nonetheless only part of the story, accounting for a small fraction of corticospinal fibres; most of the tract arises from smaller pyramidal neurons alongside them.

5.3 Motor Association Cortex: Premotor and Supplementary Motor Areas

Immediately anterior to M1 lies the **motor association cortex** (Brodmann area 6), which is itself divided into two main regions: the **premotor cortex** on the lateral surface and the **supplementary motor area (SMA)** on the medial surface. These are the areas where movements are *planned* rather than executed. A useful way to think about them is that M1 decides how to move, and the motor association cortex decides what to move and in what order.

Their jobs include sequencing complex movements, coordinating the body so that posture and proximal muscles are prepared before the main action begins, and integrating sensory input into motor plans. There is a division of labour between them that is clinically worth remembering: the **supplementary motor area** is particularly important for *internally generated* and *bimanually coordinated* movements (those arising from internal intention rather than external prompting, and those requiring both hands to cooperate), while the **premotor cortex** is more involved in movements *cued by external stimuli*, such as reaching for an object that has just come into view.

Damage to these areas typically does not produce paralysis. It produces **apraxia**: a striking dissociation in which strength and coordination are intact, but the patient cannot perform learned purposeful movements on request. Asked to wave goodbye or mime brushing their teeth, they fumble. They still have the "how" in the low-level sense, but they have lost the "what": the motor plan. It is one of the cleanest demonstrations in neurology that planning and executing a movement live in different places.

5.4 Frontal Eye Field

Tucked into the posterior part of the middle frontal gyrus, and corresponding to Brodmann area 8, is the **frontal eye field (FEF)**. Its job is voluntary eye movement, in particular the rapid, ballistic eye movements called **saccades** that shift gaze from one point to another.

The FEF is the cortical route by which attention is directed through the eyes: when a decision is made to look at something, the FEF is a major node in the network that carries it out. It connects extensively to brainstem gaze centres and to the cranial nerve nuclei that actually drive the eye muscles. Stimulating one frontal eye field causes the eyes to move toward the opposite side; damaging it produces the opposite pattern, with the eyes drifting toward the side of the lesion because the healthy FEF on the other side is now unopposed. This "eyes toward the lesion" picture is a classic sign in the early hours of a frontal stroke.

5.5 Broca's Area

In the **inferior frontal gyrus** of the dominant hemisphere (almost always the left, even in most left-handers) lies **Broca's area**, corresponding to Brodmann areas 44 and 45. It is the part of the brain most closely associated with the *production* of language. It plans the motor sequences of speech, handles much of the grammar and sentence structure, and coordinates the muscles of the tongue, lips, larynx, and diaphragm that actually do the talking.

Damage to Broca's area produces **Broca's aphasia** (also called expressive or non-fluent aphasia). The clinical picture is characteristic and often heartbreaking: speech is slow, effortful, and telegraphic; content words survive while grammatical glue words drop out; and patients know, agonisingly, that what they are saying is not what they mean. Comprehension is relatively preserved, which is what distinguishes Broca's aphasia from the fluent but empty speech of Wernicke's aphasia, whose lesion lies further back in the temporal lobe.

Broca's area is also a reminder that cortical function is **lateralised**. The two hemispheres are not functional mirror images of one another. Language production sits overwhelmingly on the left for most people, while analogous regions on the right handle things like prosody and emotional tone.

5.6 Prefrontal Cortex

Moving still further forward, we arrive at the **prefrontal cortex (PFC)**: the portion of the frontal lobe that lies in front of the motor and premotor areas. This is the most recently evolved part of the frontal lobe and, in humans, proportionally the largest. It is also where the neurology of the frontal lobe bleeds most clearly into psychology.

The PFC does not produce movement directly. It produces the conditions under which movement makes sense. It holds goals in mind over time, compares possible courses of action, suppresses impulses that would be counterproductive, and integrates emotion, memory, and sensation into coherent behaviour. The whole constellation of abilities we call **executive function** lives here: working memory, planning, flexible switching between tasks, inhibition of prepotent responses, and the capacity to act in accordance with long-term goals rather than short-term reward.

5.6.1 Subdivisions

The PFC divides into three functional regions, each with its own characteristic contribution.

Dorsolateral prefrontal cortex (DLPFC). This is the region most associated with classical "cold" executive function: working memory, problem-solving, planning, and flexible control of attention. It is the part of the brain that holds a phone number in mind while it is being dialled, keeps a multi-step instruction in order, or juggles the possibilities in a chess position. DLPFC dysfunction is implicated in attention deficits and is heavily involved in schizophrenia.

Ventromedial prefrontal cortex (VMPFC). This region is more concerned with "hot" cognition: emotion, risk, and reward. It integrates signals from the limbic system into decision-making, which is why damage here can leave formal reasoning intact while devastating the patient's ability to make real-life choices. Patients with VMPFC lesions can pass IQ tests and still ruin their lives through impulsive, poorly weighed decisions.

Orbitofrontal cortex (OFC). Sitting just above the orbits of the eyes, the OFC is closely involved in social behaviour, impulse control, and the moment-to-moment evaluation of rewards and punishments. It is especially important for recognising when something that used to be rewarding has stopped being so, and for holding back socially inappropriate behaviour.

5.6.2 Connections

The PFC is one of the most richly connected regions of the brain. It has strong reciprocal links with the limbic system (amygdala, hippocampus), with sensory and motor cortices, with the thalamus, and with the basal ganglia. This connectivity is what lets it perform its integrative role: executive function is, in a very literal sense, the business of pulling together information from everywhere else.

5.7 Summary

The frontal lobe is where the nervous system turns thought into action, and where action is disciplined in the service of goals. Its organisation reflects a rough gradient. The primary motor cortex at the back of the lobe executes movement; the premotor and supplementary motor areas plan and sequence it; the frontal eye field and Broca's area carry out specialised motor behaviours of gaze and speech; and the prefrontal cortex, sitting in front of all of these, supplies the executive control, judgment, and personality that make any of those actions meaningful. Damage at each level produces a characteristic and clinically revealing deficit, and together these

deficits provide some of the most compelling evidence we have that the mind is, in the end, a function of specific pieces of brain tissue, which is both the starting point of neurology and the doorway through which psychology must pass.

This is also the first section in which the account reaches something a person would recognise as themselves. Judgment, restraint, the weighing of a long-term goal against an immediate temptation, the capacity to be reliably one sort of person rather than another: these are not capacities of neurons or of cortex in general but of particular regions, and they are lost in particular ways when those regions are damaged. A patient with a ventromedial lesion who reasons perfectly and chooses disastrously is difficult to fit into any account in which character sits above and apart from tissue. Everything up to this point has been mechanism with no obvious bearing on personhood. From here on the two are hard to keep separate.

6 The Parietal Lobe

Having mapped the frontal lobe, the cortex of action and intention, we now move backwards across the central sulcus into territory of a different character. The **parietal lobe** is the cortex of *sensation* and *space*: where the body's tactile world is registered, where the position of every limb is tracked moment by moment, and where the brain assembles its sense of where things are, including where the body itself sits within the world.

Where the frontal lobe pushes outward, sending commands to muscles, the parietal lobe pulls inward, taking the river of information arriving from the skin, joints, muscles, and viscera and turning it into a coherent picture of the body and its surroundings. It is also the place where touch meets vision, where felt position meets seen position, and where the multiple sensory streams are bound into the unified spatial scene that we treat, naively, as simply "the world."

6.1 Anatomical Landmarks

The parietal lobe sits behind the frontal lobe and above the temporal lobe, occupying roughly the upper-rear quadrant of each hemisphere. Its boundaries are:

- Anteriorly, the **central sulcus** separates it from the frontal lobe. The first ridge behind the sulcus, the **postcentral gyrus**, mirrors the precentral gyrus in front and contains the primary somatosensory cortex.
- Posteriorly, the **parieto-occipital sulcus**, visible most clearly on the medial surface, marks the boundary with the occipital lobe.
- Inferiorly, the **lateral sulcus** (Sylvian fissure) separates it from the temporal lobe.
- Medially, the **longitudinal fissure** divides it from its counterpart on the other side.

Within the parietal lobe, the most important subdivision is made by the **intraparietal sulcus**, which runs roughly horizontally and divides the surface into a **superior parietal lobule** above and an **inferior parietal lobule** below. The inferior parietal lobule is itself made up of two named gyri: the **supramarginal gyrus**, which curls around the end of the lateral sulcus, and the **angular gyrus**, which sits behind it and curls around the end of the superior temporal sulcus. These are landmarks worth keeping in mind, as they mark the functional subdivisions the rest of this section works through.

Functionally, the parietal lobe walks from the most concrete to the most abstract from front to back. The postcentral gyrus, just behind the central sulcus, registers raw sensation; the cortex behind it interprets and integrates that sensation; and the cortex further back still uses sensation to construct space, attention, and bodily self-awareness.

6.2 Primary Somatosensory Cortex (S1)

The **primary somatosensory cortex**, traditionally labelled **S1** and corresponding to **Brodmann areas 3, 1, and 2**, occupies the postcentral gyrus immediately behind the central sulcus. It is the cortical destination of the body's sensory traffic, the place where information about touch, pressure, vibration, joint position, and pain finally becomes conscious sensation.

The information arriving at S1 has come a long way. **Discriminative touch, vibration, and proprioception** ascend through the **dorsal column–medial lemniscus pathway**; **pain and temperature** ascend through the **spinothalamic tract**. Both pathways relay through the **thalamus**, specifically the **ventral posterolateral (VPL) nucleus** for the body and the **ventral posteromedial (VPM) nucleus** for the face, before fanning out to S1. As with the motor system, the projection is **contralateral**: the right S1 represents the left side of the body and vice versa.

6.2.1 The Sensory Homunculus

S1 is somatotopically organised in just the same way M1 is, and the resulting map is called the **sensory homunculus**. Walking from medial to lateral along the postcentral gyrus:

- The **medial** regions, tucked into the longitudinal fissure, represent the lower limb and genitals.
- The **superior lateral** regions represent the trunk and upper limb.
- The **lateral** regions represent the face, lips, and tongue.

Like its motor counterpart, the sensory homunculus is grotesquely disproportionate. The lips, tongue, fingertips, and face occupy enormous swathes of cortex; the back, trunk, and legs make do with much less. The principle is the same as in M1: cortical real estate scales with the *density of receptors* and the *fineness of discrimination* demanded of that body part, not with its physical size. The fingertips can distinguish points a millimetre or two apart, and they require the cortex to do it.

6.2.2 Microanatomy

The three Brodmann subdivisions of S1 are not redundant. **Area 3** is the most anterior strip, sitting in the wall of the central sulcus, and it is the principal recipient of thalamic input. It is itself split into **3a**, which receives proprioceptive information from muscles and joints, and **3b**, which receives cutaneous information from the skin. Behind these, **area 1** processes mainly cutaneous input as well, and **area 2** integrates cutaneous and proprioceptive signals to support more complex perceptions such as size, shape, and texture. There is, in other words, an early stage of integration happening already within S1 itself.

6.3 Somatosensory Association Cortex

Behind S1, on the **superior parietal lobule**, lies the **somatosensory association cortex**, corresponding to **Brodmann areas 5 and 7**. Where S1 registers raw sensation, the association cortex *interprets* it. It takes the disjointed feel of edges, textures, weights, and contours arriving from S1 and assembles them into the recognition of objects, the sense of one's body shape, and the awareness of where each limb is in space.

A useful way to put it is that S1 registers *that* something is touching the body and *where*, while the association cortex determines *what it is*. Identifying a key in a pocket without looking, distinguishing a coin from a button by feel, judging the heft of a tool well enough to use it: all of this is somatosensory association cortex at work. It also stores tactile memories, so that the

feel of familiar objects becomes immediately recognisable rather than having to be reconstructed from scratch each time.

Damage to this region produces a striking deficit known as **tactile agnosia**, or, when restricted to a hand, **astereognosis**. Strength is intact, primary sensation is intact, and the patient can describe the temperature, texture, and shape of an object placed in the hand, but cannot identify what the object is. The raw materials of perception are present; the act of recognition has been severed from them. As with the motor association cortex's apraxia, this is a clean dissociation between low-level capability and the higher-level integration that makes it useful.

6.4 The Posterior Association Area

Further back still, we reach the point at which cortex stops belonging to any one sense. Everything described so far in this section has been somatosensory: S1 registers touch, and the somatosensory association cortex interprets it, but both are working with a single modality. The **posterior association area** is different in kind. It is the large expanse of cortex where the separate sensory worlds are combined into one, and it is the reason we experience a single scene rather than three parallel reports.

Anatomically it does not respect lobar boundaries, which is precisely the point. It occupies the **parieto-temporo-occipital junction**, the territory where the three posterior lobes meet, and it takes in the **posterior parietal cortex** (the superior and inferior parietal lobules, including the **angular** and **supramarginal gyri** identified earlier) along with the adjacent posterior temporal and lateral occipital cortex. It is discussed here because its centre of mass is parietal, but a boundary drawn around it would cut across the parietal, temporal, and occipital lobes alike.

Its position explains its function. Three great streams of processed information converge on it: **somatosensory** information from the parietal cortex in front, **visual** information from the occipital cortex behind, and **auditory** information from the temporal cortex below, with vestibular input arriving alongside them. Each stream arrives already interpreted by its own association cortex. What the posterior association area adds is *binding*: the felt position of the hand and the seen position of the hand are registered as one hand, and the sound of a voice and the sight of a moving mouth are registered as one person speaking.

The most conspicuous product of that binding is a representation of **space**. The posterior association area constructs where things are, where the body is among them, and how the two relate, holding this in several frames at once, some anchored to the body and some to the world. This is what allows a hand to reach accurately for a cup without being watched, a body to pass through a doorway without striking the frame, a moving object to be tracked while the head turns, and a room to be experienced as stable even though the retinal image lurches with every saccade.

Bound up with space is **attention**. The area is a major node in the brain's attention network, working with the frontal eye field to decide which part of the environment matters next. It is also the destination of the visual **dorsal stream** (the "where" and "how" pathway, which the occipital lobe section describes), and it is here that seeing is converted into the coordinates a movement can actually use, so that a reach or a grasp arrives where the object is rather than where it merely appeared.

Two further contributions are worth naming. The area maintains the **body schema**, the continuously updated model of the body's shape, posture, and extent that makes the body feel like a single owned object rather than a collection of limbs. And through the **angular gyrus**, which sits at the junction proper and adjoins the language cortex of the temporal lobe, it forms the bridge from perception to **symbol**: the step at which a seen mark becomes a letter, a heard pattern becomes a word, and a quantity becomes a number.

Like the frontal lobe, the posterior association area is strongly **lateralised**. The right hemisphere carries the heavier load for spatial and attentional processing, tending to attend to both sides of space, while the left is more involved in the language-related and symbolic operations of the angular gyrus, in calculation, and in the ordered representation of learned actions. This asymmetry is functional rather than absolute, and the two sides work together through the corpus callosum, but it is a real division of labour and it explains why the region's contributions cannot be understood as a single uniform capacity.

6.5 Summary

The parietal lobe is where the body's sensory traffic becomes conscious sensation, where sensation becomes recognition, and where recognition is woven together with vision and movement into the felt sense of a body in space. The primary somatosensory cortex registers touch, proprioception, pain, and temperature in a contralateral, somatotopically organised map; the somatosensory association cortex turns those raw sensations into recognised objects and a coherent body schema; and the posterior association area, straddling the junction of the parietal, temporal, and occipital lobes, binds the separate sensory streams into a single scene and constructs the space, attention, and visuomotor frames within which action becomes possible. Damage at each level produces a characteristic deficit, from contralateral numbness at the first, through the tactile agnosia of the second, to disturbances of space, attention, and the body schema at the third. Taken together, the parietal lobe shows that even something as immediate as "what the world feels like" is the product of layered cortical processing, and that the mind's sense of being embodied in a place is not given but built.

This is the first full instance of the claim flagged at the outset, and it is worth pausing on, because the body schema is about as unpromising a candidate for a construction as could be chosen. Occupying a body does not feel like an achievement. It feels like the background condition against which everything else is experienced, the one thing that could not require assembly. Yet it can be dismantled, and the ways it comes apart are informative: a limb can be felt in the wrong place, or disowned, or duplicated, or experienced as belonging to someone else, and each of these follows damage to a particular part of the machinery described above. What cannot be done is to damage the parietal lobe and leave the sense of embodiment untouched but incorrect in some way the anatomy does not predict. The construction has parts, and the failures respect them.

7 The Temporal Lobe

Dropping below the lateral sulcus, we leave the cortex of touch and space and enter the cortex of *sound*, *smell*, and *meaning*. The **temporal lobe** is where vibrations of air become recognised speech and music, where the volatile chemistry of the world becomes the pull of a familiar smell, and where, in its language regions, the heard word becomes the understood word.

If the frontal lobe turns intention into action and the parietal lobe turns sensation into space, the temporal lobe turns sensation into *recognition* and *meaning*. It is where the auditory and olfactory streams arrive in cortex, where they are interpreted, and where the interpretation of language as comprehensible content first becomes possible. It is also, along the way, the cortex most tightly bound to the limbic system, which is part of why sound and smell are so good at evoking memory and emotion.

7.1 Anatomical Landmarks

The temporal lobe sits below the lateral sulcus and forwards of the occipital lobe, occupying roughly the lower side of each hemisphere. Its principal external landmarks are three roughly parallel gyri running along its lateral surface:

- The **superior temporal gyrus**, immediately below the lateral sulcus, contains the auditory cortices and the receptive language area in the dominant hemisphere.
- The **middle temporal gyrus**, beneath it, contributes to higher-order auditory and visual processing, including aspects of object and motion recognition.
- The **inferior temporal gyrus**, lower still, contributes to high-level visual recognition, particularly of faces and objects.

Two further landmarks matter. Tucked into the floor of the lateral sulcus on the upper surface of the superior temporal gyrus lie **Heschl's gyri**, also called the **transverse temporal gyri**, which house the primary auditory cortex itself. On the medial surface, the temporal lobe curls inward to form the **uncus** and the **parahippocampal gyrus**, structures continuous with the limbic system; the primary and association olfactory cortices live here, alongside the hippocampus, the structure most closely associated with the formation of declarative memory.

As with the other lobes, the temporal lobe walks from concrete to abstract: primary sensory cortex first, association cortex behind it, and language and memory regions further back and inward.

7.2 Primary Auditory Cortex (A1)

The **primary auditory cortex**, traditionally labelled **A1** and corresponding to **Brodmann areas 41 and 42**, occupies Heschl's gyri on the upper surface of the superior temporal gyrus, mostly hidden inside the lateral sulcus. It is the cortical destination of the auditory pathway, where pressure waves in the ear finally become the conscious experience of sound.

The pathway that delivers sound to A1 is one of the longest and most heavily processed in the brain. Hair cells in the **cochlea** convert mechanical vibrations into neural signals carried by the **cochlear division of the vestibulocochlear nerve (CN VIII)**. From there, signals pass through the **cochlear nuclei** of the brainstem, the **superior olivary complex**, the **lateral lemniscus**, the **inferior colliculus** of the midbrain, and the **medial geniculate nucleus (MGN)** of the thalamus before finally fanning out to A1. At several points along this chain, information crosses the midline, with the result that each A1 receives input from *both* ears. This bilateral organisation is why a unilateral lesion of A1 does not produce deafness in either ear; it produces only a subtle deficit in tasks like sound localisation and the perception of complex auditory scenes.

A1 is not a featureless sheet either. It is organised **tonotopically**: different patches of cortex respond to different frequencies, arranged in an orderly map that mirrors the mechanical tuning of the cochlea. Roughly, **high frequencies** are represented at one end of A1 and **low frequencies** at the other, with a smooth gradient between. This is the auditory analogue of the somatotopic organisation of S1 and the retinotopic organisation of the visual cortex: a peripheral sensory surface mapped, in orderly fashion, onto the cortex.

7.3 Auditory Association Cortex

Surrounding A1 on the superior temporal gyrus, and corresponding to most of **Brodmann area 22**, lies the **auditory association cortex**. Where A1 registers raw sound, the association cortex *interprets* it: it recognises the sound as speech, music, a familiar voice, a doorbell, or a piece of meaningless noise. It is also the region that gives auditory experience much of its temporal structure: the perception of rhythm, melody, and the patterning of language at the level of pure sound.

Damage to the auditory association cortex produces **auditory agnosia**, a deficit analogous to the tactile agnosia of the parietal lobe. The patient hears (pure-tone audiometry is normal and sound is consciously perceived), but recognition has been severed from perception. Specific

subtypes are clinically recognised. **Pure word deafness** leaves the patient unable to understand spoken language while still able to hear, read, write, and speak normally; non-speech sounds are recognised, but speech is reduced to noise. **Amusia** leaves the patient unable to recognise melodies that were once familiar, or, in some cases, unable to perceive music as music at all. As with other association-cortex syndromes, what has been lost is not the input but the act of making the input meaningful.

7.4 Primary Olfactory Cortex

Smell is the odd one out among the senses, and the temporal lobe is where its peculiarity is on display. The **primary olfactory cortex** sits on the medial surface of the temporal lobe, around the **uncus**, and comprises the **piriform cortex**, the periamygdaloid cortex, and parts of the amygdala. Its input comes from the **olfactory bulb**, which receives the olfactory nerve (CN I), by way of the **olfactory tract**.

What makes olfaction unusual is that, alone among the major senses, it does not relay through the thalamus before reaching cortex. Olfactory receptor neurons in the nasal epithelium project to the olfactory bulb, the bulb projects to the primary olfactory cortex directly, and only *after* this initial cortical processing does smell information reach the thalamus and the orbitofrontal cortex for further integration. This shortcut is part of why smell has such an immediate, unmediated quality, and why it is so tightly bound to emotion and memory: the primary olfactory cortex is essentially continuous with the limbic system.

7.5 Olfactory Association Cortex

Adjacent to and continuous with the primary olfactory cortex, the **olfactory association cortex** occupies the **entorhinal cortex** (Brodmann area 28) and parts of the parahippocampal gyrus. Its job is to take the raw olfactory signal arriving at primary cortex and turn it into a recognisable, named, emotionally weighted experience: this smell is coffee, that one is petrol, this one is a kitchen not entered in twenty years.

The olfactory association cortex is the most direct anatomical demonstration of why smell is so tied to memory and feeling. Through the entorhinal cortex it has powerful, reciprocal connections with the **hippocampus** (the seat of declarative memory) and the **amygdala** (a key node in emotional processing). A familiar smell can summon a specific memory with a vividness that other senses rarely match, partly because the wiring that delivers it is, in a literal sense, two synapses away from the structures that store autobiographical memory and assign emotional valence.

Damage here produces the recognition counterpart to anosmia: smells are still consciously detected but cannot be identified or attached to memory, an olfactory analogue of the agnosias seen elsewhere. Entorhinal cortex is also one of the very first regions to degenerate in **Alzheimer's disease**, which is part of why a degraded sense of smell can be one of the earliest non-cognitive signs of the illness.

7.6 Wernicke's Area

In the **posterior part of the superior temporal gyrus** of the dominant hemisphere (almost always the left) lies **Wernicke's area**, traditionally identified with the posterior portion of **Brodmann area 22**. It is the part of the brain most closely associated with the *comprehension* of language. If Broca's area, in the inferior frontal gyrus, is where the motor plans for speech are assembled, Wernicke's area is where heard and read words become understood meaning.

Anatomically, Wernicke's area sits where it does for a reason: it is right next door to the auditory association cortex, perfectly placed to interpret the patterned sounds of speech, and it is connected to Broca's area by a long-range white-matter tract called the **arcuate fasciculus**.

The arcuate fasciculus is what allows the comprehended word to be turned, almost without effort, into the spoken word: it ferries information from the back of the language network to the front.

Damage to Wernicke's area produces **Wernicke's aphasia** (also called receptive or fluent aphasia), the mirror image of Broca's. Speech flows easily, with normal rhythm and much of its grammar intact, but it is empty of sense: words are swapped for wrong or invented ones, and comprehension of what others say is poor. Patients are often unaware that anything is wrong, because the system that would check whether speech makes sense is the one that has been damaged.

7.7 Summary

The temporal lobe is where the brain turns sound into meaning, smell into memory, and heard speech into understood language. The primary auditory cortex on Heschl's gyri receives the bilateral output of the auditory pathway and arranges it tonotopically; the auditory association cortex of the superior temporal gyrus turns that raw sound into recognisable speech, music, and familiar voices. The primary olfactory cortex on the uncus, alone among the senses, receives its input directly without a thalamic relay, and the olfactory association cortex of the entorhinal region binds smell to memory and emotion through its links with hippocampus and amygdala. Finally, Wernicke's area in the dominant superior temporal gyrus stands at the centre of language comprehension, paired with Broca's area through the arcuate fasciculus to form the language network. Damage at each of these levels produces a characteristic deficit: cortical deafness after bilateral damage to primary auditory cortex, the auditory and olfactory agnosias of the association cortices, and Wernicke's aphasia. Taken together, the temporal lobe is where the mind acquires much of its capacity for meaning.

The capacity added here is recognition, and the agnosias show it to be a separate operation rather than a strong form of perception. A patient with pure word deafness has intact hearing and intact language, and speech reaches them as noise; the sound arrives and the meaning does not, and the two can be severed because they were never the same thing. Recognition is a further step performed on a signal already fully received. This is the point at which the account acquires the difference between a system that registers what is present and one for which anything is *about* something, and it is a step that primary cortex, however intact, does not take.

8 The Occipital Lobe

We come now to the back of the brain, and to the simplest of the cerebral lobes to describe in terms of what it does. The **occipital lobe** is, almost in its entirety, the cortex of *vision*. Almost everything the brain does with the light that enters the eye, beyond the very earliest processing in the retina and thalamus, begins here.

If the temporal lobe gives sensation its meaning and the parietal lobe gives it its place, the occipital lobe gives the visual world its existence as a coherent scene at all. From the patterned firing of retinal ganglion cells, the occipital lobe reconstructs edges, surfaces, motion, depth, colour, and form, and hands the result on to the rest of the brain for use. Damage to its primary cortex does not produce subtle deficits of recognition or naming; it produces, in its purest form, the inability to see.

8.1 Anatomical Landmarks

The occipital lobe occupies the rear pole of each hemisphere, behind the parietal and temporal lobes. Its boundaries are:

- Anteriorly and superiorly, the **parieto-occipital sulcus**, visible most clearly on the medial surface, separates it from the parietal lobe.

- Anteriorly and inferiorly, there is no sharp external sulcus separating it from the temporal lobe; the boundary is conventional and follows imaginary lines drawn between landmarks such as the preoccipital notch.
- Medially, the **longitudinal fissure** divides it from its counterpart on the other side.

The single most important landmark within the occipital lobe is the **calcarine sulcus** (or calcarine fissure), a deep horizontal groove running across the medial surface of each hemisphere. The cortex lining the upper and lower banks of this sulcus is the **primary visual cortex**, and almost everything else in the occipital lobe is, in one way or another, organised around it.

Functionally, as in the other lobes, there is a gradient from concrete to abstract. The primary visual cortex around the calcarine sulcus does the earliest cortical work; the visual association cortex surrounding it on both medial and lateral surfaces takes that early work and assembles it into recognisable scenes, objects, and motion.

8.2 Primary Visual Cortex (V1)

The **primary visual cortex**, traditionally labelled **V1** (and also called the **striate cortex**), corresponds to **Brodmann area 17** and lies along the banks of the calcarine sulcus on the medial surface of the occipital lobe. It is the cortical destination of the visual pathway, the first place at which the patterned firing of retinal cells, after a long journey, becomes consciously perceived vision.

The pathway that reaches V1 is heavily processed before it arrives. **Retinal ganglion cells** send their axons through the **optic nerve (CN II)** to the **optic chiasm**, where fibres from the nasal halves of each retina cross to the opposite side. After this crossing, each side of the brain carries information about the *contralateral half of the visual field* from *both eyes*. The fibres continue as the **optic tract** to the **lateral geniculate nucleus (LGN)** of the thalamus, which relays them via the **optic radiations** to V1. Within the optic radiations, fibres carrying information from the *upper* visual field loop forward through the temporal lobe (the so-called **Meyer's loop**) before arriving at the lower bank of the calcarine sulcus, while fibres from the *lower* visual field travel a more direct course through the parietal white matter to the upper bank.

8.2.1 Retinotopic Organisation

V1 is, like S1, A1, and M1 before it, organised as a map of its peripheral surface. This is **retinotopic** organisation: each location on the retina projects to a specific location in V1, with neighbouring retinal points represented at neighbouring cortical points. The map has two clinically vital features. First, the **contralateral** representation is strict: the *left* V1 represents the *right* half of visual space, and vice versa. Second, the map is heavily **magnified at the centre**: the macula, the small central region of the retina that does most of our high-acuity vision, occupies a hugely disproportionate share of V1 at the occipital pole. The same principle is at work as in the motor and sensory homunculi: cortical real estate scales with the precision required of that part of the input surface, not its physical size.

A useful corollary is that the upper visual field maps to the lower bank of the calcarine sulcus and the lower visual field to the upper bank, an inversion which, when combined with the left/right contralateral flip, means that V1 represents the visual world upside-down and mirror-reversed.

8.2.2 Microanatomy

V1 has two anatomical features worth knowing by name. It is called the *striate* cortex because of a prominent white horizontal band visible to the naked eye in cross-section, the **stria of**

Gennari, which is a particularly heavy concentration of myelinated fibres carrying input from the LGN. V1 also receives that input in two parallel streams, originating in the **magnocellular** and **parvocellular** layers of the LGN: magnocellular input carries information about motion and contrast at the expense of fine detail and colour, while parvocellular input carries information about colour and fine detail at the expense of temporal resolution. These two streams are the seeds of the visual association cortex's later division of labour.

8.3 Visual Association Cortex

Surrounding V1, and occupying most of the rest of the occipital lobe, is the **visual association cortex**, corresponding to **Brodmann areas 18 and 19** (often referred to as **V2**, **V3**, and beyond). Where V1 registers the basic features of the visual world (edges, orientations, contrasts, and motion direction), the association cortex *interprets* that early activity and assembles it into the world we actually experience: recognisable objects, faces, scenes, motion, and depth.

A great deal of what we know about visual association cortex is captured by the idea that it splits, beyond the immediate surroundings of V1, into **two streams**, each running into a different lobe. The **ventral stream** (sometimes called the "what" pathway) runs forward from the occipital lobe into the inferior temporal cortex; it specialises in object and face recognition, colour, and form. The **dorsal stream** (the "where" or "how" pathway) runs upward into the posterior parietal cortex; it specialises in motion, depth, spatial relationships, and the visuomotor transformations that convert a seen object into a reach toward it. The two streams are not completely independent, and the simple "what versus where" summary is something of an oversimplification, but it captures a real anatomical and functional division.

Damage to the visual association cortex produces a family of **visual agnosias**: the patient sees, in the sense that V1 still works and the visual field is intact, but cannot make sense of what is seen. **Visual object agnosia** is the inability to recognise objects by sight, despite being able to recognise them by touch or sound. **Prosopagnosia**, classically associated with damage to the **fusiform face area** on the inferior temporal surface where the ventral stream ends, is the strikingly specific inability to recognise faces, sometimes including one's own in a mirror, while general visual recognition remains relatively preserved. **Akinetopsia** is the inability to perceive motion, with the visual world experienced as a series of still photographs; it follows damage to area V5 (also called MT) on the lateral occipital surface, a key node in the dorsal stream's motion processing. **Achromatopsia**, cerebral colour blindness, follows damage to colour-specialised regions on the ventral surface of the occipital lobe.

These syndromes, taken together, make the same point that the agnosias of the parietal and temporal lobes made: vision, like touch and hearing, is not a single function but a stack of functions, and damage at the right level produces a startlingly specific dissociation. A patient with prosopagnosia cannot recognise their spouse's face but can recognise their voice; a patient with akinetopsia can describe a stationary cup but cannot pour from it because they cannot see the rising liquid as moving. The fact that such dissociations exist at all is one of the strongest arguments for the modular, layered architecture of cortical processing.

8.4 Summary

The occipital lobe is where light becomes vision. The primary visual cortex around the calcarine sulcus receives the heavily processed output of the retina and lateral geniculate nucleus, lays it out as a contralateral, retinotopic, and macula-magnified map, and performs the earliest cortical analysis of edges, orientation, contrast, and motion. The visual association cortex surrounding it interprets that early work and splits into a ventral "what" stream that runs into the temporal lobe to recognise objects and faces, and a dorsal "where" stream that runs into the parietal lobe to track motion, space, and the visual control of action. Damage at each level produces a characteristic deficit: visual field defects from V1 lesions, cortical blindness from bilateral V1

damage, and the cleanly dissociable agnosias (prosopagnosia, akinetopsia, achromatopsia, and object agnosia) from damage to specific patches of association cortex. The occipital lobe is the clearest demonstration we have that even something as immediate as "seeing the world" is not given by the eye but constructed, in stages, by cortex.

That is the same claim the parietal lobe made about the body, now made about the visual world, and the agnosias are what make it more than an assertion. If seeing were a single capacity, damage could only make it better or worse. Instead it comes apart along seams: a patient who sees faces as faces but cannot say whose, a patient for whom a poured drink is a series of stills rather than a moving stream, a patient who sees shape and depth in a world drained of colour. Each of these is a stage of the construction failing while the others continue, and their existence is difficult to account for on any view in which vision is delivered rather than assembled. The seams are invisible in a working brain, which is precisely why they have to be found in a broken one.

9 The Insula

Having walked the four classical lobes (frontal, parietal, temporal, and occipital), we now turn to a fifth, often forgotten one. The **insula**, or **insular cortex**, is the cortex of *the body's interior*. It is where the brain registers the state of the viscera, the feel of taste on the tongue, the slow tides of hunger and thirst, the racing of the heart, and, more abstractly, the felt quality of emotion itself. Because it has no exposed surface, it is the lobe most often left off the standard surface tour, despite carrying functions that reach from taste and visceral sensation to emotion and self-awareness.

If the occipital lobe constructs the visual world and the temporal lobe gives it auditory and linguistic meaning, the insula constructs something stranger and more intimate: the felt sense of one's own body. Interoception, the perception of internal states, is the insula's signature function. It is the part of cortex where the workings of the gut, the lungs, and the heart are turned into the conscious feelings that we recognise, after the fact, as hunger, breathlessness, anxiety, or warmth.

9.1 Anatomical Landmarks

Unlike the other lobes, the insula has no surface visible from outside the brain. It sits hidden in the depths of the **lateral sulcus** (Sylvian fissure), buried beneath the surrounding cortex of the frontal, parietal, and temporal lobes. The flaps of overlying cortex that cover it are collectively called the **operculum** (Latin for "little lid"), and they are conventionally divided into a **frontal operculum** above and in front, a **parietal operculum** above and behind, and a **temporal operculum** below. Pulling these apart in dissection or imaging reveals the insula like the meat of a clam.

Within the insula, a deep groove called the **central sulcus of the insula** runs obliquely across its surface and divides it into two regions:

- The **anterior insula**, made up of three short gyri, sits forward of the sulcus and is more closely tied to emotion, taste, and subjective feeling.
- The **posterior insula**, made up of two long gyri, sits behind the sulcus and is more closely tied to interoceptive and somatosensory processing of the body's internal state.

This anterior/posterior split runs through almost everything we know about the insula's function. Information tends to enter through the posterior insula, where bodily signals arrive in something like a raw form, and to be progressively re-represented and integrated as it moves forward, finally emerging in the anterior insula as the conscious, emotionally weighted feeling of being in a body.

9.2 Primary Gustatory Cortex

The insula's most concretely sensory role is as the cortical home of **taste**. The **primary gustatory cortex** occupies the anterior insula along with the adjacent **frontal operculum**, corresponding to **Brodmann area 43** and parts of the surrounding insular cortex. It is the cortical destination of the gustatory pathway, the place where signals from the taste buds finally become the conscious experience of sweet, salty, sour, bitter, and umami.

The pathway that delivers taste to the insula draws on three cranial nerves. Taste afferents from the anterior two-thirds of the tongue travel in the **facial nerve (CN VII)**, those from the posterior third in the **glossopharyngeal nerve (CN IX)**, and those from the epiglottis and pharynx in the **vagus nerve (CN X)**. All three converge in the **nucleus of the solitary tract** in the medulla, then ascend, with a relay in the **ventral posteromedial nucleus (VPM) of the thalamus**, to the gustatory cortex. Unusually, the projection is largely *ipsilateral*: each side of the tongue projects mainly to the same side of the cortex, in contrast to the strict contralateral organisation of the somatosensory system.

Damage to the gustatory cortex produces **ageusia** (loss of taste) or, more commonly, blunted and distorted taste, often noticed first as a loss of pleasure in food. As with smell, much of what we casually call "taste" is in fact flavour (the integration of taste with smell), which is why insular damage and olfactory damage can both produce strikingly similar complaints from patients about food.

9.3 Interoception and Visceral Sensation

Beyond taste, the larger and more distinctive role of the insula is as the cortical home of **interoception**: the perception of the body's internal state. Where S1 in the parietal lobe maps the surface of the body, the posterior and mid-insula map its interior: the heart, lungs, gut, bladder, blood vessels, and the slow chemical milieu of hunger, thirst, temperature, and breathlessness.

The pathway that delivers interoceptive information is anatomically distinct from the classical somatosensory pathways. Visceral and small-fibre afferents, including the C fibres that carry slow, burning pain and warmth, ascend in the **spinothalamic** and related tracts, relay through specialised thalamic nuclei (notably parts of the **ventral medial nucleus, posterior division (VMpo)**), and project to the posterior insula. The result is a topographic map of the body's interior: a kind of visceral homunculus. Awareness of one's own heartbeat, of the urge to breathe, of fullness in the stomach, of needing to urinate: all of these depend on insular cortex.

A theme of modern cognitive neuroscience is that the posterior insula's relatively raw representation of the body is progressively **re-represented** as it is passed forward through the mid-insula to the anterior insula. The anterior insula is where the bodily state stops being a mere physiological reading and becomes a *feeling*: the hot, restless, queasy, anxious quality that distinguishes one internal state from another. The anterior insula's right side in particular has been argued to host a unified "sentient self" representation, the felt sense of being alive in this body at this moment, although that strong claim remains debated.

9.4 The Anterior Insula: Emotion, Disgust, and Awareness

Whatever its precise role in selfhood, the anterior insula is firmly implicated in a cluster of emotional and social functions that go well beyond simple visceral sensation. It is one of the most consistently activated regions in neuroimaging studies of **emotion**, and especially of negative emotions such as **disgust**, **anxiety**, **empathy for pain**, and the **craving** that drives addiction. Damage to the anterior insula can blunt the experience of disgust to the point that patients no longer recognise disgusted facial expressions in others, a deficit that mirrors the way amygdala damage blunts the experience of fear.

The anterior insula is also a major node in the brain's **salience network**, working with the anterior cingulate cortex to detect and flag events that matter, whether for survival, social interaction, or goal-directed behaviour. This is part of why the insula keeps appearing in studies of conditions as varied as anxiety disorders, depression, addiction, eating disorders, and chronic pain: in each, the basic machinery for noticing and weighing internal and emotional states appears to be tuned wrongly.

9.5 Autonomic and Vestibular Roles

The insula does not merely *perceive* the body's interior; it also helps regulate it. It has strong reciprocal connections with autonomic centres in the brainstem and hypothalamus, and stimulation of different parts of the insula can produce changes in heart rate, blood pressure, and respiration. Damage to the insula, particularly the right insula, has been linked to **autonomic instability** after stroke, including arrhythmias and sudden cardiac complications, a clinically important reason to recognise insular involvement on imaging.

A separate role worth mentioning is in **vestibular processing**: the posterior insula and the adjacent parietal operculum host parts of what is sometimes called the *parieto-insular vestibular cortex*, which contributes to the conscious sense of balance and bodily orientation in space.

9.6 Summary

The insula is the cortex of the body's interior, the felt self, and the emotional weighting of experience. Hidden beneath the opercula of the frontal, parietal, and temporal lobes, and divided into a posterior region tied to interoception and somatosensory processing of the body's internal state, and an anterior region tied to taste, emotion, and subjective feeling, it stands at the crossroads of sensation, autonomic regulation, and affect. The primary gustatory cortex turns the chemistry of food into the experience of taste; the posterior and mid-insula construct an interoceptive map of the viscera; and the anterior insula re-represents that bodily state as feeling, contributing to disgust, empathy, craving, and the salience of events that matter. Damage produces ageusia, blunted disgust and empathy, autonomic instability, and altered awareness of one's own bodily state. The insula is, more than any other cortex, where the body becomes the self.

That last sentence is worth unpacking, because this section is where the account first arrives at self-awareness proper, and it arrives by a route that is not obvious. The preceding lobes constructed a world: a visual scene, a heard and recognised environment, a space with a body positioned in it. All of that is a model of what is outside. The insula does the same work on what is inside. Interoceptive afferents deliver the state of the heart, lungs, gut, and vasculature to the posterior insula in something close to raw form, and that reading is then re-represented forward until, in the anterior insula, it has become a felt quality rather than a measurement. The operation is the one performed by every association cortex met so far, applied to an unusual input. What comes out of it is not a percept of an object but a percept of the organism doing the perceiving.

This suggests a way of understanding what a self is that requires nothing beyond the machinery already described. Rather than a further thing that possesses the body and observes the world, the felt self can be read as another construction, built from the same materials as the visual scene and by the same sort of layered re-representation, differing only in that its subject matter is the body's own condition. That reading is not a settled result; the strong version of it, on which the right anterior insula hosts a unified representation of the sentient self, was flagged above as contested and remains so. But something like it is the reason this section sits where it does. Up to this point the notes have been assembling a model of the world. From here they are also assembling the thing that model appears to belong to, and the emotional and motivational sections that follow depend on there being an organism whose states can matter to it.

10 The Thalamus

With the cortex described, we now go beneath it, into the diencephalon. The **thalamus** is a pair of egg-shaped masses of grey matter, one either side of the third ventricle, and it accounts for roughly four fifths of the diencephalon by volume. It sits almost exactly at the centre of the brain, and its position is not incidental: nearly everything travelling between the cortex and the rest of the nervous system passes through it.

The standard description is that the thalamus is the brain's relay station, and as a first approximation that is right. With the single exception of smell, every sensory stream synapses in a specific thalamic nucleus before it is allowed to reach the cortex. But a relay station in the ordinary sense is a passive thing, and the thalamus is not passive. Only a minority of the synapses on a thalamic relay neuron come from the ascending pathway it is supposedly relaying; the majority come from the cortex above, the brainstem below, and the thalamus's own inhibitory shell. What arrives at the cortex is therefore not a copy of what arrived at the thalamus but a version of it that has been weighted by attention, by arousal, and by what the cortex was already expecting. The thalamus decides what gets through.

It is also more than a sensory structure. The same anatomy that carries sensation upward carries the return legs of the motor and limbic loops that later sections describe: the basal ganglia and the cerebellum both reach the motor cortex through it, and the memory circuit running out of the hippocampus passes through it too. The three subsections that follow take those three roles in turn, and the organising claim of the section is that they are the same mechanism applied to different traffic.

10.1 Structure and Components

Each thalamus lies with its medial surface forming the upper wall of the **third ventricle** and its lateral surface pressed against the **posterior limb of the internal capsule**, the great sheet of white matter through which its fibres leave. In most people the two halves are joined across the ventricle by the **interthalamic adhesion**, a bridge of grey matter which, despite appearances, carries little traffic between the sides. Posteriorly the thalamus swells into the **pulvinar**, which overhangs the midbrain, and tucked beneath the pulvinar sit two small bumps that have already appeared in these notes as the last stops on the visual and auditory pathways: the **lateral** and **medial geniculate nuclei**.

Internally the thalamus is divided by a sheet of white matter, the **internal medullary lamina**, which runs front to back through the structure and splits at its anterior end into a Y. That Y is the whole of the organising scheme. The two arms enclose the **anterior group**; the space medial to the stem holds the **medial group**; and everything lateral to it forms the large **lateral group**, itself divided into a dorsal tier of association nuclei and a ventral tier of relay nuclei. Three further populations sit outside this scheme: the **intralaminar nuclei**, embedded within the lamina itself, the **midline nuclei** against the ventricle, and the **reticular nucleus**, a thin shell of cells wrapped around the lateral surface within the **external medullary lamina**.

The reticular nucleus deserves separating from the rest, because it is the exception that explains the system. Every other thalamic nucleus projects to the cortex. The reticular nucleus does not: it is **GABAergic**, and its targets are the other thalamic nuclei. Corticothalamic and thalamocortical fibres both pass through it on their way, giving off collaterals as they go, so it is informed of everything crossing in either direction and inhibits the relay nuclei accordingly. It is, in other words, the gate itself, and it is the structural reason the thalamus can be selective about what it forwards.

It is useful to sort the remaining nuclei into three functional classes rather than to memorise them by location:

- **Relay** (or specific) nuclei receive a defined ascending pathway and project to a defined

cortical area. The sensory relays (VPL, VPM, LGN, MGN) and the motor relays (VA, VL) are all of this kind, and their maps are preserved: neighbouring cells represent neighbouring parts of the body, the retina, or the cochlea.

- **Association** nuclei receive most of their input from the cortex rather than from an ascending tract, and project back to association cortex. The **pulvinar**, the **mediodorsal nucleus**, and the **lateral dorsal** and **lateral posterior** nuclei belong here. They serve the integrative regions of cortex rather than the primary ones.
- **Nonspecific** nuclei (the intralaminar and midline groups) project diffusely to wide areas of cortex and to the striatum. They carry arousal and attentional signals from the brainstem reticular formation rather than any particular modality, and they are the route by which the level of consciousness is set.

The fibres leaving the thalamus are gathered into four **thalamic radiations** which fan through the internal capsule to reach the frontal, parietal, occipital, and temporal cortices. Because these radiations are packed so tightly, a small lesion here can produce a combination of deficits that would require a much larger cortical lesion to reproduce.

10.2 Limbic Involvement

The limbic role of the thalamus is carried by the anterior and medial groups, and it is a role in memory and in the emotional colouring of experience rather than in sensation.

The **anterior nuclear group** is the thalamic member of the **Papez circuit**, a loop running out of the hippocampus and back to it that the sections on the hypothalamus and the limbic system set out in full. It receives the **mammillothalamic tract** from the mammillary bodies, which are themselves fed by the fornix from the hippocampus, and it projects onward to the **cingulate cortex**, from which the loop returns to the hippocampus. The circuit is no longer read as the seat of emotion in the way Papez intended, but as a memory pathway it holds up well, and the clinical evidence is unambiguous: damage to the mammillary bodies and the anterior and mediodorsal thalamus, as occurs in the thiamine deficiency of **Korsakoff's syndrome**, produces a dense anterograde amnesia while leaving perception and intelligence largely intact. A structure usually described as a relay for sensation turns out to be a necessary component of laying down new episodic memory.

The **mediodorsal nucleus (MD)** is the largest of the medial group and is the thalamic partner of the **prefrontal cortex**. Its connections with the prefrontal cortex are reciprocal and extensive, so much so that the extent of the prefrontal cortex was once defined by where MD projects. It also receives from the **amygdala**, from the olfactory cortex, and from the hypothalamus. Two consequences follow. The first is that MD is the point at which the amygdala's assessment of emotional significance reaches the executive machinery of the frontal lobe, which is what allows an emotionally salient event to capture deliberation rather than merely producing a bodily response. The second concerns olfaction: smell is the sense that bypasses the thalamus on its way to cortex, but it does not avoid the thalamus altogether. It reaches MD after its first cortical stop, and from MD reaches the orbitofrontal cortex, which is where conscious discrimination of odours occurs. The thalamic step is postponed rather than absent.

Two smaller contributions complete the picture. The **midline nuclei** connect with the hippocampus, amygdala, and hypothalamus, and are involved in arousal and in visceral aspects of emotion. The **intralaminar nuclei** receive the spinothalamic fibres that do not terminate in VPL and project to the **anterior cingulate** and **insula**. This is the anatomical basis of a distinction met in the insula section: the sensory-discriminative component of pain (its location and intensity) travels the lateral route through VPL to S1, while the affective component (the part that makes pain unpleasant rather than merely detected) travels the medial route through

the intralaminar nuclei to the cingulate and insula. The two can be separated by lesions, and patients in whom the medial route is interrupted may report that the pain is still there but no longer bothers them.

10.3 Sensory Involvement

The sensory role is the one the thalamus is best known for, and the arrangement is systematic: each modality has its own nucleus, each nucleus projects to its own primary cortex, and each preserves the map of the receptor surface along the way.

- **Ventral posterolateral (VPL)** receives the dorsal column–medial lemniscus and spinothalamic pathways from the **body** and projects to S1, somatotopically arranged.
- **Ventral posteromedial (VPM)** does the same for the **face**, receiving the trigeminal pathways. Its most medial part, **VPMpc**, receives **taste** from the solitary nucleus and projects to the gustatory cortex of the insula.
- **Lateral geniculate nucleus (LGN)** receives the optic tract and projects through the optic radiations to V1, retinotopically arranged and layered so that input from the two eyes is kept separate until cortex.
- **Medial geniculate nucleus (MGN)** receives the auditory pathway from the inferior colliculus and projects to A1, tonotopically arranged.
- **VMpo** and neighbouring nuclei relay the interoceptive and small-fibre afferents to the posterior insula, as described in the section on the insula.
- **Vestibular** information relays through the ventral posterior region to the parietoinsular vestibular cortex, the one major sensory stream without a nucleus named after it.

Smell remains the sole exception, a difference usually put down to evolutionary history: the olfactory system is among the oldest of the sensory systems, and its direct route to cortex appears to predate the thalamic arrangement the others share.

The more interesting question is what the relay actually does, since a pathway that simply copied its input to cortex would not need a synapse in the middle. Two mechanisms answer it. The first is the inhibitory gating already described: the reticular nucleus, informed by cortical feedback about what is currently relevant, suppresses relay cells carrying information that is not. Attention, at this level, is a matter of which thalamic neurons are permitted to fire. The second is that relay neurons have two modes of firing. In the **tonic mode**, which prevails during waking, they fire in a graded way that faithfully transmits the detail of their input. In the **burst mode**, which prevails during drowsiness and slow-wave sleep, they fire in stereotyped bursts that carry little information about the stimulus. The transition between them is controlled by the brainstem arousal systems acting on the membrane potential of the relay cell, and it is why sounds that would be vivid to a waking listener can pass through a sleeper's thalamus without being registered. The rhythmic activity of the reticular nucleus during burst mode generates the **sleep spindles** visible on the EEG.

The association nuclei of the dorsal tier work on sensory information after the primary cortices have had it. The **pulvinar**, the largest nucleus in the human thalamus, is connected chiefly with the visual, parietal, and temporal association areas and is heavily implicated in visual attention and in the selection of targets for eye movements; lesions of the pulvinar impair the ability to attend to one object among several without impairing the ability to see any of them. The **lateral posterior** and **lateral dorsal** nuclei serve the parietal and cingulate cortices in a similar integrative fashion.

The clinical consequences follow directly from the anatomy. Infarction of the VPL and VPM territory produces a contralateral loss of all sensory modalities on the body and face together, a pattern that no cortical lesion of comparable size can produce. In a proportion of these patients, the numbness is followed weeks later by **thalamic pain syndrome** (**Déjerine–Roussy syndrome**), in which the deafferented side becomes the source of severe spontaneous burning pain, and in which ordinary touch may be experienced as painful. Pain, on this evidence, is not merely reported by the thalamus but is shaped there.

10.4 Motor Involvement

The thalamus issues no motor commands. It has no descending projection to the spinal cord and no access to a motor neuron; everything it contributes to movement it contributes by talking to the motor cortex. Its importance is that it is the point at which the two great subcortical motor systems, the basal ganglia and the cerebellum, hand their output back to the cortex. Both are loops, and the thalamus is the return leg of both.

The relevant nuclei are the **ventral anterior (VA)** and **ventral lateral (VL)**, which sit in the ventral tier just in front of VPL. They receive from two sources that remain largely segregated within them:

- The **basal ganglia** reach VA and the anterior part of VL by the GABAergic output of the **GPi** and **SNr**. As the section on the basal ganglia sets out, this input is tonically active and inhibitory, so the thalamic contribution to movement is made by *having its brake released* rather than by being driven. When the direct pathway silences the GPi, VA/VL is disinhibited and excites the premotor and supplementary motor areas on glutamate; when the indirect pathway strengthens GPi firing, the brake is pressed harder and the movement is suppressed. The thalamus is the structure on which the whole arithmetic of the basal ganglia is performed.
- The **cerebellum** reaches the posterior part of VL by an excitatory route. Purkinje output leaves through the **dentate** and other deep nuclei, ascends in the **superior cerebellar peduncle**, decussates in the midbrain, and terminates in VL, which projects on to the primary motor cortex. This is the anatomical reason the cerebellum's corrections, which arise from comparing intended with actual movement, are applied by adjusting the next cortical command rather than by acting on the muscle directly.

Two points are worth drawing out. The first is that the two inputs are opposite in sign and largely separate in territory, which is why disorders of the basal ganglia and of the cerebellum produce quite different movement disorders despite converging on the same nucleus. The second is that both loops are closed: the cortex projects down into the striatum and the pons, and receives back through the thalamus, so the motor cortex is in continuous conversation with both systems rather than issuing commands into them.

There is a clinical corollary that has become therapeutically important. Because VL is the funnel through which both systems act, interrupting it interrupts the abnormal signal as well as the normal one. The **ventral intermediate nucleus (VIM)**, a subdivision of VL carrying the cerebellar input, is the standard target for **deep brain stimulation** in essential tremor, and stimulation there can abolish a tremor that no cortical or cerebellar intervention would reach. Lesions of the motor thalamus more generally produce ataxia, tremor, and, when the lesion is larger, the peculiar syndrome of a limb that is neither weak nor uncoordinated but is simply not used.

10.5 Summary

The thalamus is a paired mass of grey matter at the centre of the brain through which almost all traffic between the cortex and the rest of the nervous system passes. Its internal medullary lamina

divides it into anterior, medial, and lateral groups, with the intralaminar, midline, and reticular populations sitting outside that scheme; of these the reticular nucleus is the key structure, being the one thalamic nucleus that inhibits the others rather than projecting to cortex, and therefore the gate through which selection is exercised. Its limbic role runs through the anterior group, the thalamic member of the Papez circuit and a structure whose damage produces amnesia, and through the mediodorsal nucleus, which links the amygdala and the olfactory cortex to the prefrontal cortex; the intralaminar nuclei carry the affective component of pain to the cingulate and insula. Its sensory role is the familiar one of a modality-specific relay (VPL and VPM for body and face, LGN for vision, MGN for hearing, VPMpc for taste, with smell alone excepted), but the relay is an active one: cortical feedback and reticular inhibition determine what is forwarded, and the switch between tonic and burst firing determines whether the cortex is being informed about the world at all. Its motor role is to close the loops of the basal ganglia and the cerebellum, receiving inhibition from the GPi and SNr into VA and anterior VL and excitation from the dentate into posterior VL, and returning both to the motor cortices. The nuclei and their connections are gathered in Table 3. The unifying idea is that the thalamus is not a junction box but a gate, and that consciousness of a sensation, the recollection of an event, and the release of a movement all depend on something being permitted through it.

Two of the results here bear directly on the larger argument, and both concern awareness rather than sensation. The first is that attention, described in earlier sections as a capacity of cortex, has a mechanism at this level: the reticular nucleus, informed by cortical feedback about what currently matters, inhibits the relay cells carrying what does not. Attending to something is not a matter of examining an internal image more closely. It is a matter of which thalamic neurons are permitted to fire, and the material that is not selected does not arrive in a diminished form but largely fails to arrive at all. This is why the peripheral detail of an absorbing moment is not merely forgotten afterwards; much of it was never forwarded in the first place.

The second is the distinction between tonic and burst firing, which is as close as these notes come to a physiological switch for awareness itself. The same relay cell, receiving the same input, either transmits its detail faithfully or emits stereotyped bursts carrying almost nothing about the stimulus, and which of these it does is set by the brainstem arousal systems acting on its membrane potential. The sensory pathway does not shut down during sleep; it continues to carry signals that no longer inform the cortex of anything. Whether there is experience of a stimulus therefore turns not on whether the stimulus was received, nor on whether cortex is intact, but on the mode in which an intervening cell happens to be firing.

Table 3: The principal thalamic nuclei, grouped as the internal medullary lamina divides them, with their major input, their cortical target, and the role each plays. The reticular nucleus is the sole entry with no cortical projection, which is what makes it the gate rather than a relay.

Group	Nucleus	Main input	Main output	Role
Anterior	Anterior nuclei	Mammillothalamic tract	Cingulate cortex	Limbic (memory)
Medial	Mediodorsal (MD)	Amygdala, olfactory cortex, hypothalamus	Prefrontal cortex	Limbic (emotion, executive)
Lateral, dorsal tier	Lateral dorsal (LD)	Hippocampal, cingulate	Cingulate, parietal	Limbic association
	Lateral posterior (LP)	Parietal cortex	Parietal association	Sensory integration

Table 3 continued from previous page

Group	Nucleus	Main input	Main output	Role
	Pulvinar	Visual, parietal, temporal cortex	Visual and temporal association	Visual attention
Lateral, ventral tier	Ventral anterior (VA)	GPi/SNr (basal ganglia)	Premotor, SMA	Motor (gating)
	Ventral lateral (VL)	GPi/SNr; dentate (cerebellum)	Primary motor cortex	Motor (execution, coordination)
	Ventral posterolateral (VPL)	Medial lemniscus, spinothalamic (body)	S1	Somatic sensation
	Ventral posteromedial (VPM)	Trigeminal (face); solitary nucleus (taste, via VPMpc)	S1; gustatory insula	Facial sensation, taste
	Medial geniculate (MGN)	Inferior colliculus	A1	Hearing
	Lateral geniculate (LGN)	Optic tract	V1	Vision
Intralaminar	Centromedian, parafascicular	Reticular formation, spinothalamic	Striatum, diffuse cortex	Arousal, affective pain
Midline	Paraventricular, others	Hypothalamus, brainstem	Hippocampus, amygdala	Visceral, emotional
Reticular	Reticular nucleus	Thalamocortical and corticothalamic collaterals	Other thalamic nuclei (GABA)	Gating, sleep spindles

11 The Hypothalamus

The hypothalamus weighs about four grams, roughly the size of an almond, and is by that measure one of the smallest structures given a section of its own in these notes. It is also the one whose failure the body would notice fastest. It holds temperature, fluid balance, energy intake, and the daily cycle of sleep and waking within the narrow ranges on which every other tissue depends, and it does so continuously and without conscious involvement.

What makes it worth studying as a system rather than as a list of functions is its position. The hypothalamus sits at the junction of three control systems that otherwise have no way of reaching one another: the limbic system, which determines what matters emotionally; the endocrine system, which acts on the body slowly and chemically; and the autonomic nervous system, which acts on it quickly and electrically. None of the three can act on the body in the others' currency. The hypothalamus is what converts between them, and the subsections that follow are best read as five views of that single job.

11.1 Structure and Components

The hypothalamus lies beneath the thalamus, from which it takes its name, forming the floor and the lower parts of the lateral walls of the **third ventricle**. Its boundaries are easier to state than most: the **optic chiasm** marks its front edge, the **mammillary bodies** its back edge, and below it narrows into the **infundibulum**, the stalk by which it hangs the pituitary gland beneath the brain.

Internally it is not a uniform mass but a collection of **nuclei**, each a cluster of cell bodies

with its own job. The naming is easier to hold on to once its logic is clear. Nuclei are grouped in two directions at once: from front to back into **anterior**, **tuberal**, and **posterior** regions, and from the midline outward into **periventricular**, **medial**, and **lateral** zones. A given nucleus is located by both, so that the supraoptic nucleus, for instance, is anterior and lateral.

The nuclei that matter most for what follows are these:

- The **supraoptic** and **paraventricular nuclei** contain large neurons that manufacture **vasopressin** and **oxytocin** and send their axons down the stalk into the posterior pituitary. They are the cells that make the hypothalamus an endocrine organ in its own right.
- The **suprachiasmatic nucleus**, sitting just above the optic chiasm as its name records, is the body's master clock. It receives a direct projection from the retina and uses it to keep an internal daily rhythm aligned with the external light cycle.
- The **arcuate nucleus** lies close to the median eminence and produces several of the releasing hormones that govern the anterior pituitary. It also carries receptors for circulating signals of energy state, which places it at the centre of appetite regulation.
- The **ventromedial nucleus** is associated with satiety (the signal to stop eating), and the **lateral hypothalamic area** with the opposite (the drive to seek food), along with a broader role in arousal and motivated behaviour.
- The **preoptic area**, at the front, holds the thermosensitive neurons that monitor blood temperature and the populations involved in initiating sleep.
- The **mammillary bodies**, two small round swellings on the underside at the back, are not endocrine or autonomic at all. They belong to a memory circuit, and they reappear in the subsections on limbic involvement and pathways.

11.2 Limbic Involvement

The **limbic system** is the set of structures concerned with emotion, motivation, and memory, and the hypothalamus is best understood as its output stage. Structures such as the amygdala and hippocampus determine what a situation means; they cannot, by themselves, do anything about it. The hypothalamus is where meaning is converted into bodily change, and this is the anatomical reason emotion is felt physically rather than merely registered.

The classical description of this circuitry is the **Papez circuit**, proposed as a loop connecting emotion with memory. It runs: hippocampus, then out along the **fornix**, a prominent arching bundle of white matter, to the **mammillary bodies**; from there along the **mammillothalamic tract** to the **anterior nucleus of the thalamus**; onward to the **cingulate cortex** lying above the corpus callosum; and from the cingulate back to the hippocampus, closing the loop. The circuit as Papez conceived it is no longer taken to be the seat of emotion in the way he intended, but the connections themselves are real and the loop remains the standard way of describing how these structures relate.

The amygdala reaches the hypothalamus by two routes of its own, considered in more detail among the pathways below. Its influence is what allows a stimulus judged threatening to produce, within a second or so, the entire pattern of bodily changes associated with fear, without any deliberate decision being taken. Feeding, defensive behaviour, aggression, and sexual behaviour all depend on this limbic traffic reaching a structure that can act on the body.

11.3 Endocrine Involvement

The hypothalamus governs the **pituitary gland**, and it does so by two mechanisms that are entirely separate anatomically. Distinguishing them is the key to the whole subsection, because

the two lobes of the pituitary are not variations on one arrangement but genuinely different kinds of structure.

The **posterior pituitary** is not really a gland. It is nervous tissue, an extension of the hypothalamus itself. The large **magnocellular** neurons of the supraoptic and paraventricular nuclei manufacture **oxytocin** and **vasopressin**, package them, and transport them down axons running through the stalk. The axon terminals sit in the posterior lobe among a bed of capillaries, and when the neuron fires, the hormone is released directly into the bloodstream. The mechanism is precisely the one described earlier for transmitter release, differing only in where the vesicle contents end up: the blood rather than a synaptic cleft. This is **neurosecretion**, and the neuron is itself the secretory cell.

The **anterior pituitary** is a true gland, made of glandular rather than nervous tissue, and no hypothalamic axon reaches it. The connection is instead vascular. Smaller **parvocellular** neurons release **releasing** and **inhibiting hormones** into a specialised set of vessels at the **median eminence**, the **hypophyseal portal system**, which carries them the short distance down to the anterior lobe. There they instruct the glandular cells to release, or to withhold, the hormones that act on targets throughout the body:

- **CRH** drives release of **ACTH**, which acts on the adrenal cortex to release **cortisol**. This is the axis that matters most in these notes, and it reappears in the limbic system's subsections on memory and emotion.
- The remaining axes follow the same pattern with different targets: **TRH** drives **TSH** and the thyroid, **GnRH** drives **LH** and **FSH** and the gonads, and **GHRH** drives **growth hormone**, opposed by **somatostatin**.
- **Dopamine**, met earlier as a neurotransmitter, acts here as an inhibiting hormone, holding **prolactin** release in check. The same molecule doing an entirely different job in a different setting is the pattern noted when transmitters were introduced.

Two features of this arrangement are worth drawing out. The first is the economy of the portal system: by delivering hormones straight to their target through a private circulation, the hypothalamus achieves high local concentrations using quantities far too small to have any effect if released into the general bloodstream. The second is that none of these axes runs one way. The hormones produced by the peripheral glands circulate back to the hypothalamus and the pituitary and suppress the signals that called for them. Each axis is therefore a closed negative feedback loop, of the same general form as the feedback mechanisms met at the synapse, operating over hours rather than milliseconds.

11.4 Autonomic Involvement

The hypothalamus has been called the head ganglion of the autonomic nervous system, and the description is apt provided it is read as a statement about hierarchy. The hypothalamus does not innervate the heart, the gut, or the blood vessels itself. It issues instructions to the brainstem and spinal cord nuclei that do, chiefly through descending projections to the autonomic centres of the medulla and to the **intermediolateral cell column**, the strip of preganglionic sympathetic neurons running down the thoracic and upper lumbar cord.

The two autonomic divisions introduced in the first section are represented unevenly across the structure. Anterior and medial regions bias output toward the **parasympathetic** division, and posterior and lateral regions toward the **sympathetic**. This is a tendency rather than a map, and stimulation experiments produce mixed patterns rather than clean ones, but it is a useful first approximation.

What the hypothalamus contributes that the brainstem cannot is the **set point**. Autonomic reflexes at brainstem level are corrective but have no independent notion of what the correct

value should be; the hypothalamus supplies it, and can change it. Two examples show the whole arrangement in one structure:

- In **thermoregulation**, thermosensitive neurons in the preoptic area measure the temperature of the blood passing through and compare it against a set point. Deviation upward recruits sweating and cutaneous vasodilation; deviation downward recruits vasoconstriction, shivering, and, over a longer period, endocrine changes that raise metabolic rate. The sensing and the effector command sit within millimetres of one another.
- In **osmoregulation**, osmoreceptive neurons detect the concentration of the blood. A rise triggers vasopressin release from the posterior pituitary, which instructs the kidney to retain water, and simultaneously generates thirst, which is a behavioural rather than an autonomic response. A single measurement thus drives an endocrine, an autonomic, and a behavioural limb at once.

That last point generalises. Because the hypothalamus commands all three, its responses are typically coordinated across them rather than confined to one, which is why hunger, cold, and fear each produce a bodily state rather than an isolated adjustment.

11.5 Pathways

The hypothalamus is a small structure with a large number of connections, and several of its major tracts have been named individually because they are visible to the naked eye in dissection.

The **fornix** is the principal route from the **hippocampus**. It is a large arched bundle, one of the most conspicuous white matter tracts in the brain, which sweeps from the hippocampus in the temporal lobe up and forward beneath the corpus callosum before descending to terminate mainly in the mammillary bodies. It is the hippocampus's major output, and it is the reason a structure associated with memory has a direct line into one associated with drive. From the mammillary bodies the signal continues along the **mammillothalamic tract** to the anterior thalamus, as traced in the Papez circuit above.

The **amygdala** connects by two routes that differ in shape rather than in purpose. The **stria terminalis** is long and thin and takes an indirect arcing course that roughly parallels the fornix, following the curve of the lateral ventricle. The **ventral amygdalofugal pathway** takes the direct route underneath, a short and diffuse projection into the medial hypothalamus. Both carry the amygdala's assessment of emotional significance to the structure that can act on it, and their targets substantially overlap.

Connections with the **prefrontal cortex** are reciprocal, and arise mainly from its medial and orbitofrontal regions, the parts already described as most concerned with emotion, value, and social behaviour. Their significance is regulatory. The hypothalamus generates drives that are, by themselves, indifferent to circumstance, and prefrontal input is what allows context, social consequence, and long-term goals to modulate them. Hunger does not compel eating in every setting, and the anatomical basis of that restraint runs along this connection.

Two further pathways are worth naming. The **medial forebrain bundle** runs longitudinally through the lateral hypothalamic area, linking the brainstem below with the basal forebrain and cortex above, and carries much of the traffic associated with reward and motivated behaviour. The **retinohypothalamic tract** is a small, direct projection from the retina to the suprachiasmatic nucleus; it is the anatomical means by which the master clock is kept aligned to the light cycle, and it is entirely separate from the visual pathway described in the occipital lobe section, carrying information about ambient light rather than about images.

11.6 Summary

The hypothalamus is a structure of a few grams occupying the floor of the third ventricle, and its importance lies in its position at the junction of three systems that cannot otherwise communicate.

Its nuclei, grouped from front to back and from the midline outward, each take responsibility for a different regulated variable: the supraoptic and paraventricular for fluid balance and neurosecretion, the suprachiasmatic for circadian timing, the arcuate, ventromedial, and lateral areas for energy intake, the preoptic for temperature and sleep, and the mammillary bodies for the memory circuitry they share with the hippocampus. As the output stage of the limbic system, it converts emotional significance arriving from the amygdala and hippocampus into bodily change. As an endocrine organ, it commands the pituitary by two separate mechanisms, sending axons directly into the posterior lobe to release oxytocin and vasopressin into the blood, and reaching the anterior lobe indirectly through the hypophyseal portal system with releasing and inhibiting hormones that govern the peripheral endocrine axes. As the head of the autonomic hierarchy, it supplies the set points that brainstem reflexes lack, and coordinates endocrine, autonomic, and behavioural responses to a single measurement. Its named tracts (the fornix, the stria terminalis and ventral amygdalofugal pathway, the mammillothalamic tract, the medial forebrain bundle, and the retinohypothalamic tract) are the anatomical form of those relationships. For a structure so small, the hypothalamus is where a remarkable amount of what the body does without being asked is actually decided.

The cortex constructs a world and the thalamus decides how much of it reaches awareness, but neither gives the organism any reason to act on what it finds. This section supplies that reason. A system that merely represented the world and could move would have no reason to do anything, and the hypothalamus is where reasons come from: it holds a set of variables that must stay within limits, detects departures from them, and generates the states that push behaviour toward correcting them. Hunger, thirst, cold, and the drive to sleep are not perceptions of the body but demands made upon the organism, and they are constructed here from measurements taken here. That the same structure supplies both the set point and the command is what allows a deviation to become a motive rather than merely a reading. Wanting, in its simplest and most physiological form, starts as a regulated variable drifting out of range.

12 The Midbrain

Beneath the thalamus and hypothalamus the brain narrows into the **brainstem**, a stalk of tissue not much larger than a thumb, which joins the forebrain to the spinal cord and carries the cerebellum on its back. The first section described it as the place where life is sustained and consciousness maintained. The next three sections take its three levels in turn from the top down, the midbrain, the pons, and the medulla, and this one also sets out the plan they share.

That plan is worth having first, because the brainstem is hard to learn as a list and easy to learn as a pattern. Four kinds of thing are packed into it at every level. There are the **long tracts**, the motor fibres descending from the cortex to the spinal cord and the sensory fibres ascending the other way, for which the brainstem is simply the road. There are the **cranial nerve nuclei**, the clusters of cell bodies that send out or receive ten of the twelve cranial nerves. There are the connections with the **cerebellum**, which is attached to the back of the brainstem by three pairs of stalks, one pair at each level. And there is the **reticular formation**, a core of loosely arranged neurons running the whole length of the stem, in and around which sit the small nuclei that manufacture the brain's modulatory transmitters. A cross-section is organised the same way at every level too. Behind lies the **tegmentum**, the "covering", which is the old core containing the nuclei and the reticular formation; in front lies a **base** that carries the great descending motor fibres. The midbrain alone adds a substantial roof behind the tegmentum.

The **midbrain** is the shortest of the three levels, about two centimetres from top to bottom, and for the purposes of these notes it is the most important. It holds the circuitry that turns the eyes and head towards an event before the event has been identified, the cells that supply the dopamine of the striatum and the cortex, a column of grey matter that organises defensive behaviour and can switch off pain, and the upper end of the system that decides whether a

person is awake at all.

12.1 Structure and Components

Seen from the front, the midbrain presents two thick columns of white matter, the **cerebral peduncles**, which diverge as they climb towards the hemispheres. The hollow between them is the **interpeduncular fossa**, and the oculomotor nerves emerge from its walls. Seen from behind, it presents four rounded swellings, an upper pair called the **superior colliculi** and a lower pair called the **inferior colliculi**, the name meaning "little hills". The trochlear nerve leaves just below the lower pair, which is what makes it the one cranial nerve to emerge from the back of the brainstem. The pineal gland rests in the groove between the superior colliculi, a detail that becomes clinically important below.

Through the centre runs the **cerebral aqueduct**, a channel only a millimetre or two across that connects the third ventricle above to the fourth below. It is the narrowest point in the whole circulation of cerebrospinal fluid, and therefore the place where that circulation is most easily blocked, which is one of the common causes of the hydrocephalus described in the section on glial cells.

The aqueduct is also the landmark by which the interior is divided, into three regions from back to front:

- The **tectum**, or roof, is everything behind the aqueduct. It consists of the four colliculi.
- The **tegmentum** lies in front of the aqueduct and is the core of the midbrain. It contains the nuclei of the oculomotor and trochlear nerves, the **red nucleus**, the **ventral tegmental area**, the midbrain part of the reticular formation, and, wrapped around the aqueduct itself, a sleeve of cells called the **periaqueductal grey**.
- The **base** is formed by the **crus cerebri** on each side, the fibre mass that makes up the visible peduncle. Between the base and the tegmentum lies the dark band of the **substantia nigra**.

The crus cerebri deserves a figure, because it corrects a natural assumption. Each crus carries around twenty million fibres descending from the cortex, and it would be reasonable to suppose that most of them are corticospinal fibres on their way to command the muscles. Only about one million are. The great majority are **corticopontine** fibres, which end in the pons and are relayed from there into the cerebellum. Measured by the number of fibres devoted to it, the cortex's largest descending conversation is not with the spinal cord but with the cerebellum, a point taken up in the section on the pons.

Anatomists describe the midbrain at two levels, named for the colliculus through which a cross-section passes. A section through the superior colliculi contains the oculomotor nucleus and the red nucleus. A section through the inferior colliculi contains the trochlear nucleus and, in the centre of the tegmentum, the **decussation of the superior cerebellar peduncles**, the point at which the cerebellar output traced in the section on the thalamus crosses the midline on its way to the ventral lateral nucleus.

12.2 The Tectum

The **superior colliculus** is a visual structure, and it is the piece of the visual system that the section on the occipital lobe left out. About a tenth of the fibres in the optic nerve do not go to the lateral geniculate nucleus at all; they come here. In fish, amphibians, reptiles, and birds the equivalent structure, the optic tectum, is the principal visual centre of the brain, and the mammalian arrangement, in which cortex has taken over the analysis of the visual scene, is the

later development. What the colliculus retains is a narrower and more urgent job: not working out what something is, but turning towards it.

Its layers show how. The superficial layers hold a retinotopic map of the opposite half of visual space, fed by the retina and by the visual cortex. The deep layers hold a motor map laid out in the same coordinates, in which activity at a given point produces a saccade of a particular direction and size, so that a stimulus registered at one location in the upper map lies directly above the command that would bring the eyes to it. The deep layers also receive auditory and somatosensory input, arranged in maps that are in register with the visual one. A flash, a sound, and a touch arising from the same point in space therefore converge on the same collicular neurons, and those neurons respond more strongly to the combination than to any single cue. Output runs to the brainstem gaze centres that move the eyes and, through the **tectospinal tract**, to the neck muscles that turn the head. The frontal eye field, described in the section on the frontal lobe as the cortical route by which gaze is directed, does much of its work by projecting here.

The colliculus also projects upward, to the pulvinar of the thalamus and from there to the visual association cortex and the amygdala, by a route that bypasses the primary visual cortex entirely. This route is the usual explanation for one of the most instructive findings in neurology. In 1974 Weiskrantz and his colleagues described a patient, D.B., in whom part of the right primary visual cortex had been removed and who was, by his own account and on ordinary testing, blind in the corresponding part of the visual field. Asked to guess where a light had appeared in the blind region, he pointed to it accurately, and he could discriminate the orientation and movement of stimuli he insisted he could not see. Weiskrantz called the phenomenon **blindsight**. It has since been found in other patients, one of whom, with both primary visual cortices destroyed, walked the length of a cluttered corridor without touching the obstacles placed in it. How much of this residual capacity runs through the colliculus and how much through a direct projection from the lateral geniculate nucleus to association cortex is still debated, but the result itself is secure: visual information can guide accurate behaviour in a person who has no visual experience of it. The same route is the leading candidate, at least for vision, for the fast and crude pathway to the amygdala that the section on the limbic system describes, although how much it contributes in primates is disputed.

The **inferior colliculus** belongs to hearing, and unlike its neighbour it lies on the main line. Almost every ascending auditory fibre synapses here, arriving in the lateral lemniscus from the cochlear nuclei and the superior olive, and its output passes to the medial geniculate nucleus and so to the auditory cortex. It is the point at which the separate analyses carried out lower in the brainstem (the differences in timing and loudness between the two ears that indicate where a sound is, and the spectral cues that indicate what it is) are brought together, and it passes the location of a sound across to the superior colliculus so that the eyes and head can be turned towards it.

Just in front of the superior colliculus lies the **pretectal area**, which runs the **pupillary light reflex**. Light-sensitive ganglion cells in the retina project here, the pretectum projects to the **Edinger–Westphal nucleus** on both sides, and the parasympathetic fibres of that nucleus travel in the oculomotor nerve to constrict the pupil. Because the projection is bilateral, light shone into one eye constricts both pupils. The circuit never leaves the midbrain and does not involve the cortex, so it works in a person who is deeply unconscious, and this is why a torch shone into the eye is among the first tests applied to a patient in coma: it interrogates the midbrain directly. The ganglion cells that feed it are largely the same population that supplies the retinohypothalamic tract described in the section on the hypothalamus, cells that report the level of ambient light and contribute little to the seen image.

The tectum's position makes it vulnerable from above. A tumour of the pineal gland presses down on it and produces **Parinaud's syndrome**: the patient cannot look upward, because the centre for vertical gaze lies immediately beneath, and the pupils respond poorly to light while

still constricting when the eyes converge on a near object.

12.3 The Tegmentum

The **oculomotor nucleus** lies in the midline of the tegmentum at the superior collicular level, immediately in front of the aqueduct. It drives four of the six muscles that move each eye and the muscle that raises the eyelid, and the Edinger–Westphal nucleus sits against it. The **trochlear nucleus** lies in the same position one level down and supplies a single muscle, the superior oblique. Its fibres do something no other cranial nerve does: they run backwards, cross the midline completely, and leave from the back of the brainstem, so that each trochlear nucleus controls the eye on the opposite side. Both nuclei are joined to the abducens nucleus in the pons, and to the vestibular nuclei, by the **medial longitudinal fasciculus**, the fibre bundle that keeps the two eyes moving together.

The fibres of the oculomotor nerve pass forward through the tegmentum and the base to reach the interpeduncular fossa, and that course makes the midbrain a place where lesions can be located with unusual confidence. A small infarct in the base catches the oculomotor fibres together with the corticospinal fibres of the crus. The result, **Weber's syndrome**, is a drooping lid and an eye turned down and out with a dilated pupil on the side of the lesion, combined with weakness of the opposite arm and leg, because the corticospinal fibres have not yet crossed. If the lesion lies a little further back, in the tegmentum, the weakness is replaced by tremor and incoordination of the opposite limbs, from damage to the red nucleus and the cerebellar fibres passing through it; this is **Benedikt's syndrome**. A cranial nerve sign on one side with a limb sign on the other is the signature of a brainstem lesion, and the nerve involved gives the level.

The **red nucleus** is a large round nucleus in the centre of the tegmentum, pink in fresh tissue because of its iron content and rich blood supply. It receives from the motor cortex and from the cerebellum. In most mammals it is the origin of a major motor pathway, the **rubrospinal tract**, which crosses at once and descends to drive the flexor muscles of the limbs; in humans that tract is small, its work having been largely taken over by the corticospinal tract. What remains prominent in the human red nucleus is a projection down to the inferior olive of the medulla, which makes it a station on a loop running from the cerebellum to the red nucleus to the olive and back to the cerebellum, taken up in the section on the cerebellum.

The **substantia nigra** is taken up at length in the section on the basal ganglia, and what belongs here is its location and its colour. It is the dark band between tegmentum and base, visible to the naked eye, and the darkness is **neuromelanin**, a pigment that accumulates in the dopaminergic neurons of the pars compacta as a by-product of dopamine synthesis. In Parkinson's disease the band is visibly pale. Medial to it, in the floor of the tegmentum beside the interpeduncular fossa, lies the **ventral tegmental area (VTA)**, the origin of the mesolimbic pathway met among the antipsychotics and of a companion pathway to the prefrontal cortex, both of which the section on the limbic system takes up. The two cell groups are continuous with one another and form a single sheet of dopaminergic neurons across the ventral midbrain, the lateral part projecting mainly to the dorsal striatum and concerned with movement, the medial part projecting mainly to the ventral striatum and prefrontal cortex and concerned with value. Together they amount to something like half a million neurons, a very small population on which the release of every action and the pursuit of every goal depend, which is part of why the loss of these particular cells is so disabling.

The **periaqueductal grey (PAG)** is the sleeve of grey matter around the aqueduct, and it is the structure through which, as the section on the limbic system sets out, the amygdala's judgment of a threat becomes defensive behaviour. It receives from the central nucleus of the amygdala, from the hypothalamus, and from the medial prefrontal and anterior cingulate cortex, and it organises what those structures decide into coordinated patterns of defence. It is arranged in columns running along the aqueduct, and the columns do different things. Stimulating the lateral columns in an animal produces the active responses: confrontation or flight, with a raised

heart rate and blood pressure. Stimulating the ventrolateral column produces the passive ones: freezing and immobility, with a slowed heart and a fall in blood pressure. Each is a complete package of movement, autonomic change, and altered responsiveness, selected as a unit.

The PAG is also the head of the brain's own system for suppressing pain. In 1969 Reynolds showed that a rat could undergo abdominal surgery without anaesthetic, and without apparent distress, while the PAG was being stimulated electrically. The mechanism is a descending one. The PAG projects to the raphe nuclei of the medulla, these project down to the dorsal horn of the spinal cord, and there they inhibit the transmission of pain signals at the first synapse, before the signal has entered the brain at all. The PAG is dense with opioid receptors and with neurons that release enkephalin, and it is one of the principal sites at which morphine acts. This is the descending system that the SNRIs were described as strengthening, and it is the source of the analgesia that lets an injury sustained in an emergency go unnoticed. It can also be engaged by expectation alone. The pain relief produced by a placebo is reduced by naloxone, a drug that blocks opioid receptors, which shows that at least part of it is carried by this same opioid system. A belief about a treatment, held in the cortex, is converted here into a reduction of the signal leaving the spinal cord.

12.4 The Reticular Formation and Arousal

The **reticular formation** is the core of the brainstem tegmentum, running from the medulla to the midbrain. It takes its name from its appearance, a net (*reticulum*) of scattered neurons threaded through with fibres running in every direction, without the sharp boundaries that mark off a named nucleus. Its lower parts, described in the next two sections, generate the rhythm of breathing, regulate the heart and blood vessels, and set the background tone of the muscles. Its upper part, in the midbrain and the top of the pons, does something else: it keeps the forebrain awake.

The evidence for that came in two steps. In the 1930s Bremer compared two transections in cats. A cut at the bottom of the medulla, separating the whole brain from the spinal cord, left an animal that slept and woke in normal alternation. A cut through the midbrain, between the superior and inferior colliculi, left a forebrain that still received vision and smell and yet remained permanently in a state resembling sleep. Whatever sustained wakefulness therefore lay in the brainstem between the two cuts. Bremer took it to be the sensory traffic ascending through that region. In 1949 Moruzzi and Magoun showed otherwise: electrical stimulation of the reticular formation itself converted the slow waves of a sleeping cortex into the fast, low-voltage activity of an alert one, and lesions confined to the reticular formation, sparing the sensory pathways, produced lasting unresponsiveness. Wakefulness is not the passive consequence of sensory input arriving. It is actively imposed on the forebrain from below.

The system that imposes it is now called the **ascending arousal system**, and it reaches the cortex by two routes.

- The first route runs through the **thalamus**. Cholinergic neurons in two nuclei at the junction of the midbrain and pons, the **pedunculopontine** and **laterodorsal tegmental nuclei**, project to the thalamic relay nuclei, the intralaminar nuclei, and the reticular nucleus. Their effect is to depolarise the relay cells, and this is the hand on the switch described in the section on the thalamus: it is what moves those cells from the burst mode, in which they tell the cortex almost nothing, to the tonic mode, in which they transmit faithfully.

- The second route bypasses the thalamus. It runs forward through the lateral hypothalamus and the basal forebrain and reaches the cortex directly, and it is made up of the small modulatory nuclei, each of which broadcasts a single transmitter across most of the forebrain: the **locus coeruleus** of the pons, which supplies noradrenaline; the **raphe**

nuclei along the midline of the brainstem, which supply serotonin; the dopaminergic cells of the ventral midbrain; and, joining them from the hypothalamus, the histamine neurons of the **tuberomammillary nucleus** and the **orexin** neurons of the lateral hypothalamic area.

These are the neuromodulators introduced in the section on neuron structure, and it is worth registering how few cells are involved. The locus coeruleus contains a few tens of thousands of neurons and the raphe nuclei a few hundred thousand, and yet their axons branch so widely that between them they reach essentially every part of the cortex. The serotonin that the SSRIs conserve, and that the psychedelics imitate at the 5-HT_{2A} receptor, is released from the terminals of raphe neurons, and it is on the cell bodies of these same neurons that the autoreceptors responsible for the delayed action of the SSRIs are found. These nuclei fire steadily during waking and slow during ordinary sleep, and the noradrenergic, serotonergic, and histaminergic groups fall almost silent in the dreaming phase of sleep, which the next section takes up. They are gathered in Table 4.

Sleep is produced by switching this system off. Neurons in the preoptic area of the hypothalamus, noted in the section on that structure as involved in initiating sleep, are GABAergic and project to the arousal nuclei, and when they fire the arousal nuclei are inhibited together. The arousal nuclei in turn inhibit the preoptic neurons, so that each side suppresses the other and the arrangement tends to sit firmly in one state or the other rather than drifting between them. The orexin neurons stabilise the waking side, and when they are lost, as they are in **narcolepsy**, the switch becomes unstable and the patient falls abruptly into sleep during the day. Several drugs met in the section on neuron firing act on this machinery: the stimulants raise the noradrenaline and dopamine that these nuclei release, which is why they are used in narcolepsy; antihistamines that enter the brain block the tuberomammillary signal and cause drowsiness; and caffeine removes the adenosine brake that accumulates against the system through the waking day.

No single one of these nuclei is indispensable. Destroying any one of them in an animal changes the amount of wakefulness surprisingly little, and more recent work points to glutamatergic neurons in the same region of the upper pons as carrying much of the burden. The system is redundant, as might be expected of something so important. What is not redundant is the region. Bilateral damage to the paramedian tegmentum of the upper pons and midbrain, even a lesion a few millimetres across, produces **coma**: the eyes are closed, the patient cannot be roused, and the cortex, although structurally undamaged, shows no sign of awareness. Damage to one cerebral hemisphere, however large, does not do this unless swelling presses down on the brainstem.

12.5 Summary

The midbrain is the top of the brainstem, two centimetres of tissue pierced by the cerebral aqueduct and divided by it into a roof, a core, and a base. The roof is the tectum: the superior colliculus holds aligned visual, auditory, and somatosensory maps over a motor map and uses them to turn the eyes and head towards an event, and it supports the residual visual capacity of blindsight; the inferior colliculus is the obligatory midbrain station of the auditory pathway; and the pretectal area closes the pupillary light reflex. The core is the tegmentum, which holds the oculomotor and trochlear nuclei, the red nucleus, the dopaminergic cells of the substantia nigra and ventral tegmental area, and the periaqueductal grey, which organises defensive behaviour into active and passive patterns and heads the descending opioid system that suppresses pain at the spinal cord. The base is the crus cerebri, most of whose fibres are bound not for the spinal cord but for the pons and cerebellum. Running through the tegmentum is the upper reticular formation, the source of an ascending arousal system that reaches the cortex both through the thalamus, where cholinergic input sets the relay cells to their transmitting mode, and around it, through the diffuse projections of the noradrenergic, serotonergic, dopaminergic, histaminergic,

and orexin neurons. Lesions here are localised by the pairing of a cranial nerve sign on one side with a limb sign on the other, as in Weber's and Benedikt's syndromes, and bilateral damage to the paramedian tegmentum produces coma.

The sections on the cortex were about the *contents* of experience: what is seen, heard, felt, and recognised, and how each is assembled. This section adds a distinction that those sections could not make, between the contents of consciousness and its *level*. The two are supplied by different machinery and they fail separately. A cortical lesion removes a content and leaves the person awake: there is no colour, or no recognition of faces, or no sense of where a limb is, and the patient is otherwise present. A lesion of the upper brainstem tegmentum removes no content in particular and abolishes all of them at once, with the cortex that constructs them intact. The cortex, on this evidence, does not keep itself awake. It is held in a state in which it can do its constructive work by a few small groups of cells beneath it, and when that support is withdrawn the machinery described in the earlier sections is all still there and nothing is experienced. The section on the thalamus located a switch for awareness in the firing mode of the relay cell; this section has found what operates it.

Blindsight makes the complementary point from the other side. There the level of consciousness is normal and a content is missing, and yet the processing that would ordinarily accompany that content continues and guides behaviour accurately. Seeing in the sense of using visual information to act, and seeing in the sense of having visual experience, turn out to be separable, and the second is not required for the first. Like the agnosias, this is a seam that only damage reveals. It also sets a caution that applies to everything in these notes: that a system responds appropriately to a stimulus is not, by itself, evidence that the stimulus was experienced.

Table 4: The arousal and modulatory systems of the brainstem and hypothalamus. Each is a small group of cells that broadcasts one transmitter over a wide territory. The last row is the system that opposes the rest and produces sleep.

Transmitter	**Nucleus**	**Location**	**Main projections**	**Role**
Dopamine	Substantia nigra pars compacta	Midbrain	Dorsal striatum (nigrostriatal)	Gain on the release of movement
	Ventral tegmental area	Midbrain	Nucleus accumbens (mesolimbic); prefrontal cortex (mesocortical)	Reward prediction error; motivation
Serotonin	Dorsal and median raphe	Midbrain and upper pons	Cortex, thalamus, striatum, limbic structures	Arousal state and mood; silent in dreaming sleep
	Raphe magnus and neighbours	Medulla	Dorsal horn and other levels of the spinal cord	Descending control of pain
Noradrenaline	Locus coeruleus	Upper pons	Cortex, thalamus, hippocampus, cerebellum, spinal cord	Vigilance; response to salient events
Acetylcholine	Pedunculopontine and laterodorsal tegmental nuclei	Midbrain–pons junction	Thalamus (relay, intralaminar, and reticular nuclei)	Sets thalamic relay cells to tonic mode; active in waking and dreaming sleep
	Basal forebrain (nucleus basalis, medial septum)	Forebrain	Cortex; hippocampus	Attention; hippocampal theta rhythm

Table 4 continued from previous page

Transmitter	Nucleus	Location	Main projections	Role
Histamine	Tuberomammillary nucleus	Posterior hypothalamus	Cortex and thalamus, widely	Wakefulness
Orexin	Lateral hypothalamic area	Hypothalamus	The arousal nuclei above	Stabilises waking; lost in narcolepsy
GABA	Ventrolateral preoptic area	Anterior hypothalamus	The arousal nuclei above	Inhibits them to produce sleep

13 The Pons

Below the midbrain the brainstem swells forward into a broad rounded bulge, the **pons**. The name is Latin for "bridge", and it was given for the look of the thing: seen from the front, a thick band of fibres appears to arch across the brainstem from one half of the cerebellum to the other. The appearance is misleading about where those fibres come from, as will become clear, but the name is apt in another way. The pons is where the cerebral cortex is joined to the cerebellum, and most of its bulk exists for that purpose.

It is about two and a half centimetres long and, following the plan set out in the previous section, it divides into a base in front and a tegmentum behind. The base is the bridge. The tegmentum holds the nuclei of four cranial nerves, the principal source of the brain's noradrenaline, the cells that generate dreaming sleep, and a group of neurons that shapes the rhythm of breathing. It is also the level at which wakefulness and the capacity to move can be separated from one another more completely than anywhere else in the nervous system.

13.1 Structure and Components

The front of the pons is a convex mass, the **basilar pons**, marked by a shallow midline groove in which the basilar artery lies. At each side the base narrows into a thick stalk, the **middle cerebellar peduncle**, which curves backwards into the cerebellum. The **trigeminal nerve** emerges through the side of the pons about halfway down, and its point of exit is the conventional boundary between pons and peduncle. Three more nerves emerge in the groove between the pons and the medulla below: the **abducens** near the midline, and the **facial** and **vestibulocochlear** together at the outer corner, in the recess between pons, medulla, and cerebellum known as the cerebellopontine angle.

Behind, the pons has no roof of its own. Its back surface forms the upper half of the floor of the **fourth ventricle**, a diamond-shaped cavity roofed by the cerebellum, into which the aqueduct opens from above.

Internally the two parts could hardly be more different. The **base** is made of fibres running in two directions with grey matter scattered between them. The longitudinal fibres are the descending corticospinal and corticobulbar tracts, which enter from the crus cerebri as a compact mass and are broken up here into separate bundles. The transverse fibres cross from one side to the other and are the ones visible on the surface. The grey matter between them is the **pontine nuclei**. The **tegmentum** continues the core of the midbrain downward. It contains the reticular formation, the nuclei of the trigeminal, abducens, facial, and vestibulocochlear nerves, the ascending sensory tracts on their way to the thalamus, the medial longitudinal fasciculus, and two groups of cells that the last subsection takes up, the **locus coeruleus** and the **parabrachial nuclei**.

The division into base and tegmentum matters more here than at any other level, and two lesions a few millimetres apart show why. A lesion that destroys the base on both sides, most often from a clot in the basilar artery, cuts every corticospinal and corticobulbar fibre at once.

The patient cannot move the limbs, the face, or the tongue, and cannot speak or swallow. The tegmentum is spared, and with it the arousal system and the sensory pathways, so the patient is fully awake, hears and sees and feels normally, and understands everything that is said. All that usually remains under voluntary control is blinking and the vertical movement of the eyes, because the nuclei for these lie in the midbrain, above the damage. This is the **locked-in syndrome**. A lesion of the same size placed slightly further back, in the tegmentum, spares the motor fibres and destroys the upper reticular formation, and the result, as the previous section described, is coma. The first patient is conscious and cannot show it; the second has every motor pathway intact and no one to use them.

13.2 The Bridge to the Cerebellum

The previous section noted that of some twenty million fibres descending in each crus cerebri, only about a million are bound for the spinal cord. Most of the rest end here. These **corticopontine** fibres arise from a large part of the cortex, from motor and sensory areas and from the association areas of the frontal and parietal lobes, and they synapse on the pontine nuclei of the same side. The axons of the pontine neurons then cross the midline, gather into the middle cerebellar peduncle of the opposite side, and enter the cerebellum. These crossing axons are the transverse bands seen on the surface, and they are why the pons looks like a bridge between the two halves of the cerebellum. The real connection is from each cerebral hemisphere to the opposite cerebellar hemisphere, with the pons as the relay.

The middle cerebellar peduncle is the largest of the three stalks attaching the cerebellum to the brainstem, and it is almost entirely an input pathway. What it carries is, in effect, a copy of what the cortex is currently doing: the plan for a movement that is about to be made and, from the association areas, the state of operations that are not motor at all. This is the input the first section had in mind when it described the cerebellum as receiving a copy of the motor commands issued by the cortex. The cerebellum's reply returns by a different stalk, the superior cerebellar peduncle, which crosses in the midbrain and reaches the cortex by way of the thalamus, as the section on the thalamus set out. The whole arrangement is a loop from cortex to pons to cerebellum to thalamus and back to cortex, and it is among the largest fibre systems in the brain.

Because the base contains both this relay and the corticospinal fibres, a small stroke on one side of it produces a characteristic pair of signs in the opposite arm and leg: weakness, from the corticospinal fibres, and clumsiness out of proportion to the weakness, from the interrupted traffic to the cerebellum.

13.3 Cranial Nerve Nuclei

Four cranial nerves have their nuclei in the pontine tegmentum, and three points about them are worth more than a list.

The first concerns the **trigeminal nerve**, which carries sensation from the face. It has not one sensory nucleus but three, forming a column that runs the whole length of the brainstem, and they are divided by the kind of sensation rather than by the part of the face. The **principal sensory nucleus**, in the pons, receives discriminative touch. The **spinal trigeminal nucleus** extends from the pons down through the medulla into the top of the spinal cord and receives pain and temperature. The **mesencephalic nucleus** extends up into the midbrain and receives the sense of position from the muscles of the jaw. The same separation of touch from pain and temperature that the spinal cord makes for the body is thus made for the face, in different nuclei at different levels, so that a lesion can abolish one kind of sensation over the face and leave the other intact. A fourth nucleus, the motor nucleus, drives the muscles of chewing.

The second concerns the eyes. The **abducens nucleus** drives the lateral rectus, the muscle that turns the eye outward, and beside it in the reticular formation lies the centre for horizontal

gaze. To look to the left, the left eye must turn outward and the right eye inward at exactly the same moment, and the two muscles involved are supplied from nuclei several centimetres apart, the abducens in the pons and the oculomotor in the midbrain. The solution is that the abducens nucleus contains, besides its motor neurons, a second population whose axons cross the midline and ascend in the medial longitudinal fasciculus to the oculomotor neurons supplying the opposite eye. One command to one nucleus therefore moves both eyes. The fasciculus is heavily myelinated and is a favoured target of multiple sclerosis, and when it is interrupted the result is exact: on looking to one side, the eye that should turn inward fails to follow.

The third concerns hearing and balance. The **cochlear** and **vestibular nuclei** lie at the junction of the pons and medulla, where the eighth nerve enters. Just above the cochlear nuclei, the **superior olivary complex** is the first place at which signals from the two ears are compared, and the comparison is extraordinarily fine: a sound from one side reaches the nearer ear a fraction of a millisecond before the farther one, and these neurons detect differences of a few tens of microseconds and use them to compute where the sound came from. The result ascends in the lateral lemniscus to the inferior colliculus. The vestibular nuclei, for their part, drive the **vestibulo-ocular reflex**: when the head turns, they command an equal and opposite rotation of the eyes through the medial longitudinal fasciculus, within about a hundredth of a second, so that the image on the retina stays still. It is one of the fastest reflexes in the body, and it is the reason print remains readable while the head is moving.

The **facial nucleus** completes the set. Its fibres take a peculiar course, looping backwards around the abducens nucleus before leaving the brainstem, which is why a single small lesion in the back of the pons can paralyse outward movement of an eye and the muscles of the face on the same side together.

13.4 Arousal, Sleep, and Breathing

The **locus coeruleus**, the “blue spot”, is a thin column of pigmented neurons in the upper pontine tegmentum, just beneath the floor of the fourth ventricle. It is the source of nearly all the noradrenaline in the forebrain. Its axons reach the whole cortex, the thalamus, the hippocampus, the amygdala, the cerebellum, and the spinal cord, and a single one of its neurons may branch to several of these at once. Its cells fire slowly and steadily during quiet waking, more slowly in sleep, and in brief bursts when something unexpected or significant occurs. What it delivers is not a message about the event but a change in how its targets respond: under noradrenaline, circuits answer their own inputs more strongly. It is driven by the central nucleus of the amygdala, which is the route by which fear raises vigilance, and by the orexin neurons of the hypothalamus.

The locus coeruleus also provides one of the clearest illustrations of the principle, stated in the section on neuron firing, that a receptor system adapts to being interfered with. Its neurons carry μ-opioid receptors, and opioids inhibit them. Under continued opioid use the cells compensate and restore their firing in spite of the drug. When the drug is withdrawn the compensation is left unopposed, the locus coeruleus fires far above its normal rate, and much of the physical misery of opioid withdrawal (the sweating, the racing heart, the agitation) is the result. Drugs that activate the α_2 autoreceptor quiet the nucleus and relieve those symptoms, which is the autoreceptor of the earlier section put to clinical use.

Sleep is not a single state. Through the night the brain alternates between slow-wave sleep, in which the thalamic relay cells are in burst mode and the activity of the cortex is slow and synchronised, and a second state, described by Aserinsky and Kleitman in 1953 and named for its most visible sign, **rapid eye movement (REM) sleep**. In REM sleep the electrical activity of the cortex is fast and of low voltage and is hard to tell apart from that of waking, the eyes dart beneath closed lids, the muscles of the body are paralysed, and a sleeper woken at this point will usually report a vivid dream. Jouvet showed around 1960, by a series of transections in cats, that the machinery for it lies in the pons: with the brainstem cut above the pons, the signs

of REM sleep continue to appear below the cut on schedule, and with the pontine tegmentum damaged they do not appear at all.

The generator is a small group of neurons in the dorsal tegmentum just below the locus coeruleus, working together with the cholinergic nuclei at the junction with the midbrain. They are restrained during waking by, among other inputs, the noradrenergic and serotonergic neurons, and REM sleep begins only once the locus coeruleus and the raphe have fallen silent, which is also why antidepressants that raise noradrenaline and serotonin suppress it. Once active, the generator acts in two directions. Upward, the cholinergic neurons set the thalamic relay cells to their transmitting mode and the cortex is activated. Downward, a projection to inhibitory neurons in the medulla and spinal cord releases glycine and GABA onto the motor neurons, which are hyperpolarised and cannot fire. Only the muscles of the eyes and the diaphragm are spared.

The result is a brain in a remarkable configuration. The cortex is as active as in waking and the thalamus is in the mode in which it transmits. But very little is arriving from the senses, the motor output is blocked at the spinal cord, and the cortex is running without noradrenaline or serotonin. Imaging adds that the amygdala and other limbic structures are highly active and the dorsolateral prefrontal cortex unusually quiet, which fits the character of dreams, emotionally charged and accepted without criticism, though that correspondence is an interpretation rather than a demonstration. Dreaming and REM sleep should also not be simply equated: dreams are reported on waking from other stages too, if less often and less vividly.

What the paralysis is for was shown by removing it. In 1965 Jouvet made small lesions in the dorsal pons of cats that destroyed the downward projection and left the rest of the mechanism intact. When these animals entered REM sleep they stood up, stalked, pounced, and fought, all without waking and without responding to anything around them. The same thing occurs in humans as **REM sleep behaviour disorder**, in which sleepers shout, punch, and leap from the bed, and on being woken describe a dream that matches what they were doing. Most people with the disorder go on, often many years later, to develop Parkinson's disease or a related condition, apparently because the same degenerative process reaches these pontine cells before it reaches the substantia nigra.

Two further groups of cells in the dorsal pons concern the body rather than the forebrain. The **parabrachial nuclei** surround the superior cerebellar peduncle in the outer corner of the upper tegmentum. They are a major relay for information from the body's interior: taste, visceral sensation, and pain arrive from the medulla and spinal cord and are passed on to the hypothalamus, the amygdala, and the thalamic nuclei that feed the insula. Part of the same complex forms the **pontine respiratory group**, which does not generate the rhythm of breathing, since the medulla does that, but shapes it, timing the switch from breathing in to breathing out and fitting the pattern to speech, swallowing, exertion, and emotion. It is the structure through which, as the section on the limbic system describes, the amygdala alters the pattern of breathing in fear. A neighbouring group of cells does a comparable job for the bladder, coordinating contraction of its wall with relaxation of its sphincter, and is held in check by the prefrontal cortex until circumstances are suitable: an everyday instance of the cortical restraint of an automatic programme.

13.5 Summary

The pons is the middle level of the brainstem, divided into a base in front and a tegmentum behind, with the fourth ventricle at its back. The base is a relay: corticopontine fibres from much of the cerebral cortex synapse on the pontine nuclei, whose axons cross and enter the opposite half of the cerebellum through the middle cerebellar peduncle, forming the input limb of a loop that returns to the cortex through the thalamus. The corticospinal fibres pass through the base on their way down, which is why bilateral destruction of it produces the locked-in syndrome. The tegmentum holds the nuclei of the trigeminal, abducens, facial, and vestibulocochlear nerves:

the trigeminal sensory column sorts sensation from the face by kind, the abducens nucleus and the neighbouring gaze centre move both eyes together through the medial longitudinal fasciculus, the superior olive compares the two ears to locate a sound, and the vestibular nuclei hold the eyes steady against movement of the head. It also holds the locus coeruleus, the source of the forebrain's noradrenaline; the generator of REM sleep, which activates the cortex from below while paralysing the body; and the parabrachial nuclei, which relay the state of the viscera upward and shape the rhythm of breathing.

The pons supplies the clearest demonstration in neurology that being conscious and showing it are different things. Every means we have of knowing that another person is aware passes through their motor output: what they say, what they do, where they look. In the locked-in syndrome that output is cut at the base of the pons with the awareness above it left whole, and the patient is easily taken to be in coma. In one survey of such patients the correct diagnosis took on average more than two months to reach, and in over half of the cases it was a relative rather than a clinician who first noticed that the eyes were answering. The opposite dissociation is found a few millimetres away. Jouvet's cats, and the human sleepers who resemble them, carry out complex and apparently purposeful behaviour while asleep and unresponsive. Together with blindsight in the previous section, these cases establish that behaviour is neither a necessary sign of awareness nor a sufficient one, which is an uncomfortable result for a science that has little else to go on, and one that the later parts of these notes will need.

Dreaming bears on the claim that runs through these notes from the other direction. In a dream the cortex produces a visual scene, a body located in a space, other people, speech, and feeling, and it does so with almost nothing coming in from the senses. The construction described in the sections on the cortex evidently does not need a world in order to run; it needs only to be switched on, and the pons is what switches it on. On one reading of this, which should be marked as a reading, waking perception is the same generative process held to account by the senses, and the difference between seeing a room and dreaming one lies less in the machinery that builds the scene than in whether anything outside is constraining it. Whatever is made of that, the dream is the plainest evidence available to anyone that the world as experienced is something the brain makes.

14 The Medulla

The lowest level of the brainstem is the **medulla oblongata**, about three centimetres of tissue tapering from the pons above to the spinal cord below, which it joins at the foramen magnum, the opening in the base of the skull. Nothing else in the nervous system combines so little bulk with so little margin for error. Every fibre connecting the brain to the body below the head passes through it, the two largest of those fibre systems cross the midline within it, and in its reticular formation lie the neurons that generate each breath and hold up the blood pressure from one heartbeat to the next.

The section follows the same order as the last two: the layout, the pathways that pass through, the cranial nerve nuclei, and then the functions that have given this region its reputation as the part of the brain that cannot be done without.

14.1 Structure and Components

On the front of the medulla, either side of the midline, run two long ridges, the **pyramids**. Each contains the million or so corticospinal fibres that have come down through the crus cerebri and the base of the pons, here gathered together again into a compact bundle. At the lower end of the medulla the two ridges can be seen to interlace across the midline, and this is the **decussation of the pyramids**. Lateral to each pyramid is an oval swelling, the **olive**, raised by the **inferior olivary nucleus** beneath it. The hypoglossal nerve emerges in the groove

between pyramid and olive, and the glossopharyngeal, vagus, and accessory nerves emerge in a line behind the olive.

The back of the medulla differs between its upper and lower halves. The upper half is open: it forms the lower part of the floor of the fourth ventricle. In the lower half the ventricle narrows to the fine central canal that continues into the spinal cord, and the back surface closes over it. On that closed surface are two pairs of small elevations, the **gracile** and **cuneate tubercles**, which mark the nuclei of the same names. At the outer edge of the upper medulla the **inferior cerebellar peduncle** climbs backwards into the cerebellum.

Inside, the plan of the brainstem holds. The base is the pyramids. Everything behind them is tegmentum: the reticular formation in the core, the nuclei of the cranial nerves beneath the floor of the ventricle, the folded grey ribbon of the inferior olive towards the front, and the ascending sensory tracts, with the medial lemniscus stacked beside the midline and the spinothalamic tract further out to the side.

14.2 Pathways Through the Medulla

The sections on the frontal and parietal lobes stated that each hemisphere moves and feels the opposite half of the body. The medulla is where most of that crossing takes place, and it happens separately for each system.

- The **corticospinal fibres** cross in the decussation of the pyramids. Between eighty-five and ninety per cent of them pass to the opposite side and descend the cord as the lateral corticospinal tract; the remainder continue uncrossed and supply mainly the muscles of the trunk.
- The fibres carrying **discriminative touch, vibration, and the sense of position** ascend the spinal cord in the dorsal columns on the same side as the part of the body they serve. They end in the **gracile nucleus**, for the leg and lower trunk, and the **cuneate nucleus**, for the arm and upper trunk. The neurons of those nuclei send their axons across the midline, where they turn upward as the **medial lemniscus** and run to the ventral posterolateral nucleus of the thalamus.
- The fibres carrying **pain and temperature** have already crossed, within the spinal cord and close to the level at which they entered it. They pass through the lateral part of the medulla as the spinothalamic tract without interruption.

The consequence is a rule that the section on the spinal cord puts to use. Above the medulla, everything relates to the opposite side of the body. Below it, movement and fine touch are carried on the same side as the body part concerned, and pain and temperature on the opposite side. Why the vertebrate nervous system should be crossed at all is not settled; several explanations have been proposed and none commands general agreement.

The **inferior olivary nucleus** is the one large nucleus of the medulla that is neither a cranial nerve nucleus nor a relay on a pathway to the cortex. It works for the cerebellum. It receives from the red nucleus above, from the motor cortex, and from the spinal cord, and its axons cross the midline and enter the opposite half of the cerebellum through the inferior cerebellar peduncle. There they end as **climbing fibres**, each of which winds around the dendrites of a single Purkinje cell, and the olive is the only source of such fibres in the brain. Its neurons fire slowly, about once a second, and they are joined to their neighbours by gap junctions so that groups of them tend to fire together. What that sparse and synchronised signal is for is the central question of the section on the cerebellum.

The inferior cerebellar peduncle carries, besides the olivary fibres, the tracts that report the position of the limbs from the spinal cord and the input from the vestibular nuclei. The three cerebellar stalks can therefore be given rough characters, one to each level of the brainstem: the

inferior, at the medulla, brings in the actual state of the body; the middle, at the pons, brings in the intentions of the cortex; and the superior, at the midbrain, takes the cerebellum's output away.

14.3 Cranial Nerve Nuclei

The **hypoglossal nucleus** lies beside the midline beneath the floor of the ventricle and supplies the muscles of the tongue. When it or its nerve is damaged, the tongue, on being put out, swings towards the weak side, pushed there by the intact muscle of the other.

The glossopharyngeal and vagus nerves are best taken together, because they share their nuclei, and the nuclei are divided by function rather than by nerve:

- The **nucleus ambiguus** is motor. It drives the muscles of the pharynx, the larynx, and the soft palate, and so underlies swallowing and the voice. It also contains most of the vagal neurons that slow the heart.
- The **dorsal motor nucleus of the vagus** is parasympathetic. It sends the preganglionic fibres that travel in the vagus to the lungs and to most of the length of the gut.
- The **nucleus of the solitary tract**, or solitary nucleus, is sensory, and it receives everything these nerves bring in from the interior of the body. Its upper part receives taste, from the facial nerve as well as from the other two. Its lower part receives the pressure sensors of the great arteries, the chemical sensors that monitor oxygen in the blood, and the stretch receptors of the lungs, heart, and gut.

The solitary nucleus is the point of entry for the interoceptive signals whose later fate was described in the section on the insula. It passes them to the reflex circuits of the medulla itself, which are the subject of the next subsection, and upward, largely by way of the parabrachial nuclei of the pons, to the hypothalamus, the amygdala, and the thalamic nuclei that project to the insula. What the insula eventually represents as the felt state of the body begins as a pattern of activity here.

Beside the solitary nucleus, at the lower tip of the fourth ventricle, lies the **area postrema**, one of the few places in the brain that has no blood-brain barrier. It samples the blood directly for substances that should not be there, and when it finds them it triggers vomiting. Its neurons carry 5-HT_3 and D2 receptors, which accounts for several facts noted in the section on neuron firing: ondansetron, a 5-HT_3 antagonist, is a treatment for sickness; drugs that stimulate dopamine receptors cause nausea; and several of the D2-blocking antipsychotics are also used to stop it.

Two strokes test this map and confirm it in detail. Occlusion of the artery supplying the lateral medulla produces the **lateral medullary syndrome**, described by Wallenberg in 1895, and each of its signs can be assigned to a structure named above. Pain and temperature are lost over the face on the side of the lesion, from the spinal trigeminal nucleus, and over the body on the opposite side, from the spinothalamic tract. The voice is hoarse and swallowing is impaired, from the nucleus ambiguus. There is vertigo and nausea, from the vestibular nuclei, and clumsiness of the limbs on the side of the lesion, from the inferior cerebellar peduncle. Strength and the sense of touch are normal, because the pyramid and the medial lemniscus lie near the midline and have a different blood supply. The **medial medullary syndrome** is the exact complement: weakness and loss of touch and position sense on the opposite side of the body, with a tongue that deviates towards the lesion, and with pain and temperature spared. Neither pattern could arise from a lesion anywhere else in the nervous system.

14.4 The Vital Centres

The older literature speaks of the cardiovascular and respiratory "centres" of the medulla. They are better thought of as networks, spread through the reticular formation, chiefly in its lower outer part. What they share is that they run continuously, that they are corrected from moment to moment by sensory feedback arriving through the solitary nucleus, and that none of their work is ordinarily experienced.

Blood pressure is held up by a group of neurons in the **rostral ventrolateral medulla (RVLM)**. They fire tonically, and their axons descend to the sympathetic neurons of the spinal cord, which keep the blood vessels partly constricted at all times. If the spinal cord is cut at its upper end, this drive is lost and the pressure falls at once to a level that barely sustains life. Around that tonic drive is wrapped a reflex. A rise in pressure stretches the walls of the carotid arteries and the aorta, and sensors there signal the solitary nucleus through the glossopharyngeal and vagus nerves. The solitary nucleus excites a group of inhibitory neurons in the **caudal ventrolateral medulla (CVLM)**, which suppress the tonic drive:

$$\text{pressure sensors} \xrightarrow{+} \text{solitary nucleus} \xrightarrow{+} \text{CVLM} \xrightarrow{-} \text{RVLM} \xrightarrow{+} \text{sympathetic outflow}$$

At the same time the solitary nucleus excites the vagal neurons of the nucleus ambiguus, and the heart slows. A fall in pressure does the reverse on both limbs. This is the **baroreflex**, and it acts within a heartbeat or two. It is the reason that standing up, which shifts half a litre of blood towards the legs, does not ordinarily cause a faint. As the section on the hypothalamus noted, the reflex corrects deviations from a set point without determining what the set point should be; in exercise or in fear the hypothalamus and amygdala reset it upward, and the pressure is then defended at the new level.

Breathing has to be generated, not merely regulated, because unlike the heart the respiratory muscles have no rhythm of their own. The generator is a small cluster of neurons in the ventrolateral medulla known as the **pre-Bötzinger complex**. In 1991 Smith, Feldman, and their colleagues showed that a slice of medulla half a millimetre thick, containing this cluster and little else, continues to produce a respiratory rhythm in a dish. Its output is passed to neighbouring groups of neurons that drive the motor neurons of the diaphragm, in the third to fifth cervical segments of the spinal cord, and those of the muscles of the chest wall. The rate and depth are set chiefly by carbon dioxide rather than by oxygen: sensors on the surface of the medulla respond to the acidity that carbon dioxide produces in the surrounding fluid, and it is the rise in carbon dioxide, not the fall in oxygen, that makes holding the breath intolerable. The pre-Bötzinger neurons carry μ-opioid receptors, and their inhibition is a principal part of the mechanism by which an overdose of morphine or heroin kills. The antagonist naloxone displaces the drug from the receptor and restores breathing within minutes.

Breathing is also the one vital function with two controllers. The motor cortex can drive the respiratory muscles directly through the corticospinal tract, bypassing the medullary generator, which is how the breath is held, shaped into speech, or deliberately slowed. Each controller can fail alone. In **congenital central hypoventilation syndrome** the automatic system is defective: in the typical case affected people breathe adequately while awake and stop, or nearly stop, when they fall asleep, so that they must be ventilated every night of their lives. In the locked-in syndrome of the previous section the opposite holds: the patient breathes steadily and automatically and cannot take a breath on request.

The medulla also stores a set of complete motor programmes that are released by a trigger and then run to completion. **Swallowing** is the most elaborate. It involves more than twenty pairs of muscles contracting in a fixed order over about a second, with breathing suspended while it happens, and it is organised by a network in and around the solitary nucleus and the nucleus ambiguus. Coughing, sneezing, gagging, and vomiting are of the same kind. Like the defensive patterns of the periaqueductal grey, each is a whole coordinated act held below the

level of the forebrain, which the forebrain can sometimes start or postpone and cannot perform in any other way.

Finally, the medulla contains the relay of the descending pain system introduced in the section on the midbrain. The **raphe magnus** and the reticular formation beside it receive from the periaqueductal grey and project to the dorsal horn of the spinal cord. This relay is not only a brake. It contains cells that enhance the transmission of pain as well as cells that suppress it, so that the descending system sets the gain on incoming pain in both directions, and states such as fear, attention, and expectation can make an injury hurt more as well as less.

The dependence of life on this small region is the basis of a definition of death. In the United Kingdom death may be diagnosed in a patient whose heart is still beating when the brainstem has irreversibly ceased to function, and the bedside tests used in most other countries are in practice tests of the same structure. The reasoning follows the anatomy of the last three sections. The upper brainstem holds the arousal system, without which there is no capacity for consciousness; the medulla holds the respiratory generator, without which there is no capacity to breathe. The diagnosis is made by showing that the reflexes of every level are absent and that no breath is attempted when carbon dioxide is allowed to rise. The reflexes used for the purpose are set out in the section on the cranial nerves.

14.5 Summary

The medulla is the lowest and smallest level of the brainstem and the one with the least room for damage. Its base is the pyramids, which carry the corticospinal fibres and in whose decussation most of them cross; its tegmentum contains the dorsal column nuclei, where the pathway for fine touch relays and crosses to form the medial lemniscus, and the inferior olive, the sole source of the climbing fibres of the cerebellum. Its cranial nerve nuclei are divided by function: the hypoglossal nucleus for the tongue, the nucleus ambiguus for the throat and for slowing the heart, the dorsal motor nucleus for the vagal supply to the viscera, and the solitary nucleus for everything that returns from the body's interior, with the area postrema beside it sampling the blood. The lateral and medial medullary syndromes divide these structures between them and confirm the map sign by sign. In the reticular formation lie the networks that hold up the blood pressure and correct it through the baroreflex, the pre-Bötzinger complex that generates the rhythm of breathing, the stored programmes for swallowing, coughing, and vomiting, and the raphe relay through which the brain turns pain up or down. The three levels of the brainstem are compared in Table 5.

Everything described in this section goes on without being experienced. The pressure in the arteries is measured and corrected with every beat of the heart, the carbon dioxide in the blood is measured and converted into a rate of breathing, and the blood is sampled for poisons, and none of it enters awareness unless it fails. That is not a defect in the account but the design of the thing. The functions on which life depends from second to second have been placed where they do not require attention, cannot be forgotten, and continue through sleep, anaesthesia, and coma. The arrangement of the nervous system from top to bottom follows this rule quite closely: the more immediately a function is needed, the lower it is housed and the less it has to do with the cortex. The mind is among the things this machinery keeps alive, and it is not consulted.

Breathing is the instructive exception, because it is the only one of these rhythms that has been given a second, voluntary controller, and the two can be told apart by the conditions in which each fails alone. The same muscles are moved by a generator that knows nothing of the person and by a cortex that can count the breaths. There is no other point at which deliberate control reaches so directly into the automatic regulation of the body, which is presumably why practices that aim to change the state of the body by an act of attention so often begin with the breath. What that reach amounts to physiologically is taken up in the section on the autonomic nervous system.

Table 5: The three levels of the brainstem compared. Each holds a set of cranial nerve nuclei, a set of other nuclei, and fibre systems passing through, and each produces a characteristic picture when damaged.

Level	Cranial nerve nuclei	Other nuclei	Fibre systems	Effect of damage
Midbrain	III, IV	Superior and inferior colliculi; red nucleus; substantia nigra; ventral tegmental area; periaqueductal grey; upper reticular formation	Crus cerebri; decussation of the superior cerebellar peduncles; medial lemniscus	Weber's and Benedikt's syndromes; loss of upward gaze; coma after bilateral tegmental damage
Pons	V, VI, VII, VIII	Pontine nuclei; locus coeruleus; parabrachial nuclei; superior olivary complex; horizontal gaze centre; REM sleep generator	Middle cerebellar peduncle; corticospinal fibres; medial longitudinal fasciculus; lateral lemniscus	Locked-in syndrome after bilateral damage to the base; horizontal gaze palsy; REM sleep behaviour disorder
Medulla	IX, X, XII	Inferior olive; gracile and cuneate nuclei; solitary nucleus; nucleus ambiguus; area postrema; cardiovascular and respiratory networks; raphe magnus	Pyramids and their decussation; origin of the medial lemniscus; spinothalamic tract; inferior cerebellar peduncle	Lateral and medial medullary syndromes; failure of breathing and circulation

15 The Cerebellum

Behind the pons and medulla, filling the back of the skull beneath the occipital lobes, lies the **cerebellum**. The last three sections met it only through its attachments, the three pairs of stalks that join it to the brainstem, and the first section gave it a single job, that of comparing what was intended with what is actually happening. This section sets out how it does that job, and argues that the job is larger than the correction of movement.

Two numbers frame the problem. The cerebellum makes up about a tenth of the brain by volume and contains about four fifths of its neurons, some seventy billion of them, most of which are the smallest neurons in the brain. Its cortex is folded far more tightly than that of the cerebrum and would, if flattened, cover an area not much smaller than the cerebral cortex itself. A structure built on that scale is plainly doing something that requires an enormous number of elements. Yet a person can lose most of it, or be born without it, and remain awake, aware, and recognisably themselves. Both facts need explaining, and each sheds light on the other.

15.1 Structure and Components

The cerebellum occupies the **posterior cranial fossa**. A tough sheet of dura, the **tentorium**, separates it from the cerebrum above, and in front it forms the roof of the fourth ventricle. It has a narrow strip in the midline, the **vermis** ("worm"), and a large **hemisphere** on either side. Its surface is thrown into fine parallel folds called **folia**, much narrower than the gyri of the cerebrum, and deeper fissures divide it into three lobes: an **anterior lobe**, a much larger **posterior lobe**, and a small **flocculonodular lobe** tucked underneath against the brainstem, which is the oldest part in evolutionary terms.

Like the cerebrum it has a cortex of grey matter over a core of white matter, with nuclei

buried in the core. There are four **deep cerebellar nuclei** on each side. From the midline outward they are the **fastigial**, the **globose** and **emboliform** (which work as one and are together called the **interposed nuclei**), and the **dentate**, much the largest. The rule to hold on to is that the cerebellar cortex sends almost nothing out of the cerebellum directly. Whatever it computes leaves through the deep nuclei.

The three stalks, or **peduncles**, were characterised at the end of the previous section. The **inferior** brings in the state of the body, from the spinal cord and the vestibular system, together with the climbing fibres from the inferior olive. The **middle** brings in the activity of the cerebral cortex, relayed through the pontine nuclei. The **superior** carries most of the output, to the red nucleus and the thalamus.

The division that explains function is not the one into lobes but one into three zones, defined by where each receives its input and sends its output:

- The **vestibulocerebellum** is the flocculonodular lobe. It receives from the vestibular apparatus and projects back to the vestibular nuclei, and it is concerned with balance and with keeping the eyes steady while the head moves.
- The **spinocerebellum** is the vermis together with the adjoining strip of each hemisphere. It receives the tracts from the spinal cord that report the position and movement of the trunk and limbs. The vermis projects through the fastigial nucleus to the brainstem pathways that control posture and gait; the adjoining strip projects through the interposed nuclei to the red nucleus and the motor cortex and corrects movements of the limbs while they are in progress.
- The **cerebrocerebellum** is the large lateral part of each hemisphere, and in humans it is by far the largest of the three. It receives from the cerebral cortex by way of the pons and projects through the dentate nucleus and the thalamus back to the cerebral cortex. It is concerned with the planning and timing of movement and, as a later subsection describes, with more than movement.

15.2 The Cerebellar Circuit

The cerebral cortex varies from region to region, enough that Brodmann could divide it into some fifty areas by its appearance under the microscope. The cerebellar cortex does not vary. The same circuit, in the same three layers, built from the same handful of cell types, is repeated without alteration across the whole sheet. One part differs from another only in what it is connected to. Whatever the cerebellum does, then, it does the same thing everywhere, to different inputs.

The circuit has two inputs and one output, and it can be traced in a few steps.

1. **Mossy fibres** are the main input. They arrive from the pontine nuclei, the spinal cord, and the vestibular nuclei, and they excite the **granule cells**, the tiny and enormously numerous neurons that account for most of the cerebellum's cell count.
2. The axon of each granule cell rises to the outermost layer of the cortex and divides into a T, forming a **parallel fibre** that runs along the folium for several millimetres.
3. The parallel fibres pass through the dendrites of the **Purkinje cells**. These are large neurons whose dendritic trees are among the most elaborate in the nervous system and are completely flat, like a fan, set at right angles to the direction of the parallel fibres. Each parallel fibre therefore crosses hundreds of Purkinje cells, and each Purkinje cell is crossed by something of the order of a hundred thousand parallel fibres, every one of which makes a weak excitatory synapse on it.

4. **Climbing fibres** are the second input. They come only from the inferior olive, and each Purkinje cell receives exactly one. The fibre winds through the cell's dendritic tree making hundreds of synapses on it, so that a single impulse in a climbing fibre produces a massive response in its Purkinje cell. It is the most powerful excitatory connection in the brain, and it is used sparingly, about once a second.

5. The axon of the Purkinje cell is the only output of the cerebellar cortex, and it is inhibitory. It releases GABA onto the neurons of the deep nuclei.

The deep nuclei are not waiting for instructions. The mossy and climbing fibres give off branches to them on the way in, so the nuclei are excited directly by the same signals that go on to the cortex, and they project that excitation out of the cerebellum:

$$\text{mossy fibres} \xrightarrow{+} \text{granule cells} \xrightarrow{+} \text{Purkinje cells} \xrightarrow{-} \text{deep nuclei} \xrightarrow{+} \text{thalamus and brainstem}$$

The cortex is thus a side loop laid over a direct path, and what it contributes is a precisely timed and patterned inhibition. The cerebellar cortex shapes its output by subtraction. The same device, an output stage that works by inhibiting something already active, appears again in the basal ganglia, as a later section sets out.

The geometry suggests what the arrangement is for. A Purkinje cell with a hundred thousand inputs, each reporting some small aspect of what the body and the cortex are currently doing, is in a position to recognise an enormous number of different situations. The granule cells, by recombining a modest number of mossy fibre signals into a far larger number of parallel fibre signals, spread those situations out so that they can be told apart. What the cell needs in addition is some way of learning which situations call for it to fire, and that is what the climbing fibre supplies.

15.3 Error and Learning

The proposal that the climbing fibre is a teacher was made on theoretical grounds by Marr in 1969 and Albus in 1971, before there was evidence for it. Their argument was that a cell with a hundred thousand weak inputs and one overwhelming input is built like a device for supervised learning: the weak inputs describe the situation, and the strong input reports that something went wrong. The evidence came in 1982, when Ito showed that a parallel fibre synapse that is active at the same moment as the climbing fibre is weakened, and stays weakened. This is **long-term depression**, and it has the same logic of coincidence as the long-term potentiation that the section on the limbic system describes in the hippocampus, with the sign reversed. The pattern of parallel fibre activity that was present when the error occurred loses some of its power to drive the Purkinje cell, and the next time that situation arises the cerebellar output is different. It is now clear that this is not the only site at which the cerebellum changes with experience, and the simple account is known to be incomplete, but its central idea has held: the inferior olive reports errors, and the cerebellar cortex adjusts itself to make them smaller.

Three experiments show the mechanism at work. The first uses the vestibulo-ocular reflex met in the section on the pons, which turns the eyes against every movement of the head. If a person wears spectacles that magnify the visual world, the reflex is suddenly too weak and the image slips across the retina with each turn of the head. Over hours to days the reflex recalibrates until the image is steady again, and the recalibration fails if the flocculus is damaged. The second is **eyeblink conditioning**. A tone is repeatedly followed, after a fixed interval, by a puff of air to the eye, and the subject comes to blink at the tone, with the blink timed to close the lid just before the puff arrives. A small lesion of the interposed nucleus abolishes the learned blink permanently and leaves the reflex blink to the puff itself untouched. Patients with dense amnesia from damage to the hippocampus, by contrast, acquire the conditioned blink

normally while being unable to recall the training sessions, so this is a form of memory with a separate anatomy from the kind that can be reported. The third is adaptation to prisms. A person throwing at a target while wearing spectacles that shift the visual field sideways misses at first, corrects within a few dozen throws, and on removing the spectacles misses in the opposite direction. Patients with cerebellar damage do not adapt.

Why should the brain need such a system? The answer lies in timing. Sensory feedback is slow: a signal from the limbs takes tens of milliseconds to reach the brain and visual feedback takes longer still, well over a tenth of a second by the time it can influence a movement. Many movements are over, or past the point of correction, before any report of them can return. A system that waited to see how each movement was going would always be acting on old news. The solution is to predict. The cerebellum receives a copy of each motor command as it is issued, through the pontine relay, and from long experience of what such commands produce it computes what the movement *will* feel like before the feedback arrives. This is called a **forward model**. The prediction can be used at once to steer the movement, and when the real feedback comes in it is compared with what was predicted. Any discrepancy is the error, and the error is what the olive reports.

A familiar fact follows from this. It is nearly impossible to tickle oneself. The sensation produced by one's own moving hand has been predicted, and what has been predicted is turned down; a touch delivered by someone else has not, and arrives at full strength. Blakemore, Wolpert, and Frith confirmed this in the late 1990s with a device that let subjects stroke their own palm through a robot arm. When the robot reproduced the subject's movement exactly, the touch felt weak. When a delay of a fraction of a second or a change of direction was inserted between the movement and the touch, it felt progressively more intense and more ticklish, and activity in the cerebellum tracked the mismatch. The same attenuation explains why a blow one delivers always seems lighter than the one received in return.

What is at stake in this is larger than tickling. The prediction is part of how the nervous system sorts sensations into those it caused and those that happened to it. A sensation that matches the forward model is tagged, in effect, as one's own doing. There is evidence that this sorting is disturbed in some people with schizophrenia, who attenuate self-produced touch less than others do, and it has been proposed that the experience of one's own actions or thoughts as produced by an outside agency reflects a failure of the same prediction. That proposal has experimental support and is not settled.

It is also worth holding on to the form of the olive's signal, an error with a direction, because it differs from the dopamine signal that the sections on the basal ganglia and the limbic system describe. That signal reports only that an outcome was better or worse than expected, which says that something should change without saying what. The cerebellum is told how it was wrong.

15.4 Pathways

The input pathways divide along the lines of the three zones. Vestibular fibres enter the flocculonodular lobe. The **spinocerebellar tracts** ascend the cord without crossing and end in the vermis and the adjoining strip, bringing the position sense of the limbs and, in one of the tracts, a report of the activity of the spinal motor circuits themselves. None of this information reaches awareness; conscious position sense travels separately to the thalamus and the parietal cortex. The corticopontine projection, crossing in the pons, supplies the lateral hemispheres. Climbing fibres from the opposite inferior olive go to all three.

The principal output can be followed as a loop, and it was traced from the other end in the section on the thalamus. Fibres from the dentate and interposed nuclei leave in the superior cerebellar peduncle, cross the midline in the tegmentum of the midbrain, and end in the ventral lateral nucleus of the thalamus, which projects to the motor cortex. Some end on the way in the red nucleus. Added to the input side, this closes the circuit:

cortex → pontine nuclei → cerebellar cortex → dentate nucleus → thalamus → cortex

The loop crosses the midline twice, once in the pons on the way in and once in the midbrain on the way out, so each cerebellar hemisphere is partnered with the opposite cerebral hemisphere. Since the cerebral hemisphere in turn controls the opposite side of the body, the cerebellar hemisphere ends up serving the body on its own side. This is the reason for a rule of some clinical value: the signs of cerebellar damage appear on the same side as the lesion.

Those signs are distinctive, and what is absent from them is as informative as what is present. There is no paralysis and no loss of sensation. Strength is normal. What is lost is the fit between a movement and its purpose. The gait is wide-based and staggering, a condition called **ataxia**. A hand reaching for a target overshoots or stops short (**dysmetria**) and develops a tremor that grows worse as the target is approached, the **intention tremor**, which is the reverse of the resting tremor of Parkinson's disease. Rapid alternating movements cannot be performed, a movement involving several joints is broken into its separate parts and carried out one joint at a time, speech is slurred and irregular in rhythm, and the eyes oscillate. Damage to the vermis affects mainly stance and gait, which is why long-term heavy drinking, which damages the front of the vermis, produces an unsteady walk; damage to a hemisphere affects mainly the limbs on that side. A patient with damage to one cerebellar hemisphere, reported by Holmes in 1939, put the matter exactly. The movements of his sound arm, he said, were done without thinking, whereas he had to think out each movement of the affected one.

15.5 Beyond Movement

The lateral hemispheres and the dentate nucleus are much larger in humans and the other apes than in other mammals, relative to the rest of the brain, and the enlargement is out of proportion to any difference in the precision of movement. Tracing studies in monkeys have shown where the additional output goes. A substantial part of the dentate nucleus projects, through the thalamus, not to the motor cortex but to the prefrontal and posterior parietal cortex, and those same areas project back to it through the pons, forming closed loops of exactly the kind described above. Studies of connectivity in the living human brain suggest that most of the cerebellar cortex is coupled to the association areas of the cerebrum and only a minority of it to the motor and somatosensory areas.

The clinical counterpart was described by Schmahmann and Sherman in 1998 as the **cerebellar cognitive affective syndrome**. Patients with damage to the posterior lobe show difficulties with planning, with shifting from one task to another, and with verbal fluency, impaired spatial reasoning, and changes in personality that range from a flattening of emotion to disinhibited and inappropriate behaviour. Damage to the vermis is particularly associated with the emotional changes.

The attractive interpretation follows from the uniformity of the circuit. If the cerebellar cortex performs the same operation everywhere, and if what that operation does for the motor cortex is to predict the consequences of its commands and smooth out its errors, then the parts connected to the prefrontal cortex presumably do the same for thought: they make its operations quick, accurate, and automatic. On this view a cerebellar patient is clumsy in thinking for the same reason as in reaching. That is a hypothesis, and it should be held as one. The cognitive deficits are real but they are considerably milder than the motor ones and than those that follow damage to the cortex itself, and how to interpret the activity seen in the cerebellum during mental tasks is still argued over.

15.6 Summary

The cerebellum sits behind the brainstem, joined to it by three peduncles, and consists of a tightly folded cortex over four pairs of deep nuclei through which nearly all its output passes. It divides into the vestibulocerebellum for balance and the stabilising of gaze, the spinocerebellum for posture, gait, and the correction of limb movements in progress, and the cerebrocerebellum for the planning and timing of movement and for its loops with association cortex. Its circuit is the same throughout. Mossy fibres excite granule cells, whose parallel fibres converge in vast numbers on the Purkinje cells; each Purkinje cell also receives one climbing fibre from the inferior olive; and the Purkinje cell inhibits the deep nuclei, which are otherwise excited directly by the incoming fibres. The climbing fibre carries an error signal, and synapses active when it fires are weakened, so that the circuit learns to reduce the error. What it learns is a forward model: a prediction, computed from a copy of the motor command, of what the movement will feel like, which allows movements to be steered without waiting for feedback and allows self-produced sensations to be recognised and turned down. The loop through the pons and thalamus crosses twice, so cerebellar signs are on the side of the lesion, and they consist of ataxia, dysmetria, and intention tremor with strength and sensation intact. The lateral hemispheres are connected to the prefrontal and parietal association areas, and their damage produces a milder syndrome of cognitive and emotional change. The three divisions are set out in Table 6.

The capacity this section adds is prediction. The sections on the cortex described a brain that receives the world and constructs a scene from it, and that description left out the fact that the brain is fast only in parts: by the time a sensation has been assembled, the event that caused it is over. A nervous system that could only react would be perpetually behind. The cerebellum is the clearest case in these notes of a structure whose business is to run ahead, computing what is about to be sensed and passing only the surprises upward. Holmes's patient describes what is left when it is gone: each movement has to be thought out. The effortlessness of ordinary action, which the first section mentioned in passing, is therefore not the absence of computation but the sign of a very large computation that never reaches awareness, and the same may be true of the effortlessness of fluent thought.

The other result concerns awareness more directly, and it is a constraint rather than an explanation. The cerebellum contains most of the neurons in the brain. People who lose large parts of it are clumsy and slurred in speech and are in no doubt about who and where they are. A small number of people are known to have been born with no cerebellum at all, among them a woman described in 2014 whose condition was discovered on a scan at the age of twenty-four; she had been unsteady on her feet since childhood, spoke indistinctly, and had a mild intellectual impairment, and there was no question of her being anything other than fully conscious. Whatever it is about the brain's activity that goes with being aware, then, it is not the sheer number of neurons involved, nor the number of synapses, nor the sophistication of the computation, since the cerebellum has all of these in abundance and contributes, as far as can be told, nothing to experience that its loss would remove. What it lacks, in comparison with the cortex and thalamus, is a particular kind of organisation. Its circuit runs in one direction, from input to output, in a great many parallel strips that have little to say to one another, whereas the thalamus and cortex are bound together by connections running in both directions at every level. Some theorists take that difference to be the important one. Whether they are right is not settled, but the observation stands on its own, and any account of which neural processes are accompanied by awareness has to accommodate it.

Table 6: The three functional divisions of the cerebellum, with the spinocerebellum split into its two zones. The circuit within each is the same; they differ only in their inputs and in the deep nucleus through which each sends its output.

Division	Region	Main input	Output	Role	Effect of damage
Vestibulo-cerebellum	Flocculonodular lobe	Vestibular nerve and nuclei	Directly to the vestibular nuclei	Balance; steady gaze	Unsteadiness; oscillating eyes
Spino-cerebellum	Vermis	Spinal cord (trunk); vestibular nuclei	Fastigial nucleus to vestibular and reticular nuclei	Posture and gait	Wide-based, staggering gait
	Strip beside the vermis	Spinal cord (limbs)	Interposed nuclei to red nucleus and thalamus	Correcting limb movements in progress	Limb ataxia; intention tremor
Cerebro-cerebellum	Lateral hemispheres	Cerebral cortex, through the pontine nuclei	Dentate nucleus to thalamus, then motor and association cortex	Planning and timing; loops with association cortex	Dysmetria; movements broken into parts; cognitive and emotional change

16 The Cranial Nerves

The first section listed the twelve cranial nerves and said what each one does, and the sections on the brainstem have since placed most of their nuclei. This section does not repeat either. It supplies the logic by which the nerves and their nuclei are arranged, takes the nerves in groups according to what they serve, and shows how they are used at the bedside to examine the brainstem one level at a time. The detail for each nerve is gathered in Table 7 at the end.

16.1 Organisation

The nerves are numbered in the order in which they attach to the brain, from front to back, and the levels are easily remembered. The first two attach to the forebrain. The third and fourth belong to the midbrain, the fifth to the pons, the sixth, seventh, and eighth to the groove between pons and medulla, and the ninth, tenth, and twelfth to the medulla. The eleventh is the odd one: it arises mainly from the upper segments of the spinal cord, climbs into the skull through the foramen magnum, and leaves again with the ninth and tenth.

The first two are not nerves in the same sense as the others. The **olfactory nerve** is a bundle of short axons from receptor cells in the lining of the nose to the olfactory bulb. The **optic nerve** is not a peripheral nerve at all but a tract of the brain, an outgrowth of the diencephalon that carries the axons of the retinal ganglion cells. It is wrapped in the meninges, and its axons are myelinated by oligodendrocytes rather than by Schwann cells. Two consequences follow from what the section on glial cells established: it is attacked by multiple sclerosis, a disease of central myelin that spares peripheral nerves, and like the rest of the central nervous system it does not regenerate when cut.

For the remaining ten, the arrangement of the nuclei follows from a single rule. In the spinal cord, as the next section describes, motor neurons lie in front and sensory neurons behind. In the brainstem the opening of the fourth ventricle spreads that arrangement sideways, so that the

motor nuclei lie near the midline and the **sensory nuclei lie to the side**. Within each group the nuclei form interrupted columns running the length of the brainstem, one column to each kind of function:

- Nearest the midline is the **somatic motor** column, supplying the muscles that move the eyes and the tongue: the oculomotor, trochlear, abducens, and hypoglossal nuclei.
- Next is the **branchial motor** column, supplying the muscles of chewing, facial expression, swallowing, and the voice, which develop from the gill arches of the embryo: the trigeminal motor nucleus, the facial nucleus, and the nucleus ambiguus.
- Outermost of the motor columns is the **parasympathetic** column: the Edinger–Westphal nucleus for the pupil, the salivatory nuclei for the glands of the head, and the dorsal motor nucleus of the vagus for the organs of the chest and abdomen.
- On the sensory side, the **solitary nucleus** receives taste and sensation from the internal organs, the long **trigeminal column** receives sensation from the skin and lining of the head, and the **cochlear** and **vestibular nuclei**, furthest out, receive hearing and balance.

The point to take from this is that a nucleus belongs to a function and not to a nerve. The nucleus ambiguus sends its fibres out through two nerves, the solitary nucleus receives from three, and the trigeminal column receives the pain fibres that arrive in four. Conversely a single nerve may draw on several nuclei: the facial nerve carries motor fibres from one, parasympathetic fibres from a second, and delivers taste to a third. A cranial nerve is a cable, defined by the opening in the skull through which it passes. The organisation is in the nuclei.

16.2 The Nerves by Function

Smell and sight. The receptor cells of the **olfactory nerve** are unusual in being replaced throughout life, each surviving a month or two, and their fine axons pass through a perforated plate of bone in the roof of the nose, where a blow to the head can shear them and abolish the sense of smell. Each **optic nerve** carries about a million axons. At the optic chiasm the fibres from the inner half of each retina cross, as the section on the occipital lobe described, and this makes the pattern of visual loss a precise guide to the site of a lesion. Damage in front of the chiasm blinds one eye. Damage at the chiasm, typically from a tumour of the pituitary gland pressing up from below, removes the outer half of the visual field of both eyes. Damage behind it removes the same half of the field in both.

Moving the eyes. The **oculomotor**, **trochlear**, and **abducens nerves** between them drive the six muscles of each eye. The task is demanding, since the two eyes must be aimed at the same point to within a fraction of a degree or the result is double vision, and it is achieved by the yoking of the three nuclei through the medial longitudinal fasciculus described in the section on the pons. The nuclei themselves are the common end point for several systems that want to move the eyes for different reasons: the frontal eye field and the superior colliculus for voluntary and reflexive shifts of gaze, the vestibular nuclei for holding the gaze steady against movement of the head, and the cerebellum for keeping all of this calibrated. The oculomotor nerve also carries the parasympathetic fibres that constrict the pupil, and they lie on its surface, where they are the first to suffer if the nerve is compressed. A pupil that has become large and unresponsive in an unconscious patient is for that reason one of the most urgent signs in medicine: it usually means that a swollen hemisphere is being forced downward against the nerve and the midbrain.

The face. The **trigeminal nerve** supplies sensation to the face through its three divisions and drives the muscles of chewing; the **facial nerve** drives the muscles of expression. Two observations about the facial nerve bear on more than anatomy. The first is that the part of the facial nucleus supplying the forehead receives its instructions from both cerebral hemispheres, and the part supplying the lower face only from the opposite one. A stroke in one hemisphere therefore weakens the lower half of the face and spares the forehead, whereas damage to the nerve itself paralyses the whole of one side. The second is that the nucleus is reached by two separate descending routes. A patient whose motor cortex or its fibres have been damaged may be unable to raise one corner of the mouth when asked to smile, and yet smile symmetrically when something amuses them. The reverse pattern is also seen, usually with lesions deeper in the hemisphere or in the medial frontal lobe: the smile to command is normal and the spontaneous one is lopsided. The deliberate smile and the felt one are produced by different pathways converging on the same motor neurons, and either can be lost alone.

Hearing and balance. The **vestibulocochlear nerve** is two nerves travelling together. Its cochlear division carries about thirty thousand fibres from the hair cells of the inner ear. Its vestibular division reports rotation of the head from the semicircular canals and tilt and linear acceleration from two smaller organs beside them. The two sides normally fire in balance, and vertigo, the false sensation of turning, is what results when disease on one side makes them disagree.

Throat and viscera. The **glossopharyngeal** and **vagus nerves** work the throat between them, the first mainly bringing sensation in and the second mainly sending motor commands out, so that the gag reflex has the one as its incoming limb and the other as its outgoing limb. The vagus then leaves the head altogether. It is usually thought of as a motor nerve, the main channel of parasympathetic control, but four fifths or more of its fibres are sensory, carrying the state of the heart, lungs, and gut to the solitary nucleus. It is more an organ of report than of command, a point the sections on the autonomic and enteric nervous systems depend on.

Neck and tongue. The **accessory nerve** supplies two muscles that turn the head and lift the shoulder, and the **hypoglossal** supplies the tongue. Their interest is chiefly in localisation, the tongue in particular pointing towards the side of a lesion in the medulla or the nerve.

16.3 Brainstem Reflexes

Because each cranial nerve reflex runs in at one nerve and out at another through a known level of the brainstem, a short series of them tests the whole structure from top to bottom, and none requires the cooperation of the patient.

- The **pupillary light reflex** runs in through the optic nerve and out through the oculomotor. It tests the midbrain.
- The **corneal reflex**, a blink when the surface of the eye is touched, runs in through the trigeminal and out through the facial. It tests the pons.
- The **vestibulo-ocular reflex** is tested by turning the head, or by running cold water into the ear, and watching for the compensating movement of the eyes. It runs in through the vestibular nerve and out through the three nerves to the eye muscles, and so tests the pons and the midbrain together with the fasciculus that links them.
- The **gag** and **cough reflexes** run in through the glossopharyngeal and vagus and out through the vagus. They test the medulla.

In a patient in coma these reflexes answer two questions. The first is where the trouble lies: if all of them are intact, the brainstem is working and the cause is more likely to be a poison or a disturbance of the body's chemistry acting on the brain as a whole than a lesion within the brainstem. The second is how far a lesion extends, since the reflexes are lost in order as damage progresses downward. When every one of them is absent, and the patient makes no attempt to breathe as carbon dioxide rises, the conditions for the diagnosis of brainstem death described in the section on the medulla have been met.

16.4 Summary

The cranial nerves attach to the brain in numerical order from front to back: two to the forebrain, two to the midbrain, one to the pons, three to the junction of pons and medulla, three to the medulla, and one mainly to the upper spinal cord. Their nuclei are arranged with motor nuclei near the midline and sensory nuclei to the side, in columns that correspond to functions rather than to nerves, so that one nucleus may serve several nerves and one nerve may draw on several nuclei. Taken by function, the first two bring in smell and sight; the third, fourth, and sixth move the eyes, yoked together by the medial longitudinal fasciculus; the fifth and seventh serve the sensation and expression of the face; the eighth brings in hearing and balance; the ninth and tenth serve the throat and, through the vagus, connect the brainstem with the organs of the chest and abdomen; and the eleventh and twelfth move the neck and tongue. Because their reflexes pass through known levels, they allow the brainstem to be examined in a patient who cannot respond.

What these nerves show about the mind lies mostly in their size. Everything that will ever be seen by an eye reaches the brain through an optic nerve of about a million fibres, a compression of about a hundredfold on the number of receptor cells in the retina, and most of those fibres serve the central few degrees of the visual field. Everything that will ever be heard arrives in some thirty thousand fibres from each ear. Against those figures, the visual scene as it is experienced, wide, sharp from edge to edge, and continuous, cannot be a transcription of what the nerve delivers. The optic nerve even provides the standard demonstration. Where it leaves the eye there are no receptors, and each eye consequently has a blind region several degrees across, large enough to contain the full moon many times over. No one sees a hole there, and no one sees the patch as dark. The surrounding scene is simply continued across it. The interface between the brain and the world is narrow, and the claim made throughout the sections on the cortex, that the world as experienced is constructed rather than received, is in the first place a consequence of how little is coming in.

Table 7: The twelve cranial nerves, with the nuclei each draws on, the level at which it attaches to the brain, its main functions, and the principal sign of damage to it. The letter after each name gives the direction of traffic: S for sensory, M for motor, B for both.

Nerve	Nuclei	Attachment	Function	Sign of damage
I Olfactory (S)	None in the brainstem; ends in the olfactory bulb	Forebrain	Smell	Loss of smell
II Optic (S)	Lateral geniculate nucleus; also superior colliculus, pretectum, and suprachiasmatic nucleus	Diencephalon	Vision; light signal for the pupil and the body clock	Loss of part of the visual field, in a pattern that locates the lesion

Table 7 continued from previous page

Nerve	Nuclei	Attachment	Function	Sign of damage
III Oculomotor (M)	Oculomotor; Edinger–Westphal	Midbrain	Most movements of the eye; raising the lid; constricting the pupil; focusing	Eye turned down and out; drooping lid; large pupil
IV Trochlear (M)	Trochlear	Midbrain, from behind	Superior oblique muscle	Double vision on looking down
V Trigeminal (B)	Principal, spinal, and mesencephalic sensory; motor	Pons	Sensation from the face, mouth, and cornea; chewing	Numbness of the face; lost corneal reflex; weak jaw
VI Abducens (M)	Abducens	Pons–medulla junction	Lateral rectus muscle, turning the eye outward	Eye turned inward; double vision on looking sideways
VII Facial (B)	Facial; superior salivatory; solitary	Pons–medulla junction	Facial expression; taste from the front of the tongue; tears and saliva	Paralysis of one side of the face; loss of taste
VIII Vestibulocochlear (S)	Cochlear; vestibular	Pons–medulla junction	Hearing; balance	Deafness; vertigo; oscillating eyes
IX Glossopharyngeal (B)	Solitary; nucleus ambiguus; inferior salivatory	Medulla	Sensation and taste from the back of the tongue and the throat; arterial pressure and oxygen sensors; parotid gland	Loss of the incoming limb of the gag reflex
X Vagus (B)	Nucleus ambiguus; dorsal motor nucleus; solitary	Medulla	Swallowing and voice; parasympathetic supply to heart, lungs, and gut; sensation from the organs	Hoarseness; difficulty swallowing
XI Accessory (M)	Accessory nucleus of the upper cervical cord	Upper spinal cord	Turning the head; lifting the shoulder	Weakness of both movements
XII Hypoglossal (M)	Hypoglossal	Medulla	Movement of the tongue	Tongue deviates towards the weak side

17 The Spinal Cord

The spinal cord is the last stage of the descent that began at the cortex. The first section described it as doing two things at once: it is a highway between the brain and the body, and it is a simple decision-making device in its own right. Those two roles correspond, near enough, to its white matter and its grey matter, and after a look at its overall layout this section takes them in the reverse order, the grey matter first.

17.1 Gross Anatomy and Spinal Nerves

The cord is continuous with the medulla at the foramen magnum and runs down inside the vertebral column for about forty-five centimetres, at roughly the thickness of a finger. It does

not run the whole length of the column. The bones go on growing after the cord has stopped, so that in the adult the cord ends at about the level of the first lumbar vertebra, in a tapered tip called the **conus medullaris**. Below that the canal contains only the nerve roots of the lower segments, running steeply downward to reach their exits, a bundle known from its appearance as the **cauda equina**, the "horse's tail". This arrangement is what makes it safe to pass a needle into the canal in the lower back, between the third and fourth or fourth and fifth lumbar vertebrae, either to draw off cerebrospinal fluid or to deliver the local anaesthetic of an epidural, described in the section on neuron firing: the roots are pushed aside and there is no cord to injure.

The cord is wrapped in the same three membranes as the brain and bathed in the same fluid. It is divided into **thirty-one segments**, eight cervical, twelve thoracic, five lumbar, five sacral, and one coccygeal, each giving rise to a pair of spinal nerves. It is thicker in two places, the **cervical** and **lumbosacral enlargements**, where the additional neurons serving the arms and legs are housed.

Each spinal nerve is attached to the cord by two roots, and they do different things. The **dorsal root** is sensory. The cell bodies of its neurons lie outside the cord in a swelling on the root, the **dorsal root ganglion**, enclosed by the satellite cells described in the section on glial cells; each neuron sends one branch out to the skin, muscle, or organ it serves and another into the cord. The **ventral root** is motor, made of the axons of neurons whose cell bodies lie within the cord. That the two roots carry traffic in opposite directions was established by Bell and Magendie in the early nineteenth century, and it was the first demonstration that different nerve fibres carry different kinds of information. Just beyond the ganglion the two roots join to form the mixed spinal nerve.

The strip of skin supplied by the sensory root of one segment is a **dermatome**, and the dermatomes wrap the trunk in an orderly series of bands, with the nipple at about the fourth thoracic segment and the navel at the tenth. They can occasionally be seen directly. The virus of chickenpox lies dormant for decades in the dorsal root ganglia, and when it reactivates in one ganglion it travels out along the sensory fibres of that segment alone, producing the rash of **shingles** in a band around one side of the body that traces a single dermatome on the skin. The same segmental layout is why the level of an injury to the cord can be read from where sensation and movement are lost, as the first section noted. In the limbs the picture is complicated by the **plexuses**, in which the nerves of several segments are mixed and redistributed, so that each named nerve of the arm or leg contains fibres from several segments.

17.2 Grey Matter

In cross-section the cord shows a central region of grey matter in the shape of a butterfly or a letter H, surrounded by white matter: the reverse of the cerebrum, where the grey matter is on the outside. The two wings projecting backwards are the **dorsal horns**, which receive the incoming sensory fibres. The two projecting forwards are the **ventral horns**, which contain the motor neurons. Between them lies an intermediate zone made up of interneurons, which are far more numerous than the motor neurons and are where the cord does its own work. In the thoracic and upper lumbar segments a small **lateral horn** projects from the side of the intermediate zone. It contains the preganglionic neurons of the sympathetic system, the intermediolateral cell column to which the section on the hypothalamus traced that structure's descending commands.

The **dorsal horn** is layered, and the layers sort the incoming fibres by size. The thinnest fibres, which carry pain and temperature, end in the most superficial layers, in and around a pale zone called the **substantia gelatinosa**. The thicker fibres carrying touch end deeper. This superficial zone is the first synapse on the pain pathway, and it is a good deal more than a relay. The incoming pain fibres release glutamate and substance P onto the neurons that will carry the signal to the brain. Small interneurons in the same layer release enkephalin and GABA and

oppose them. The fibres descending from the raphe magnus and the locus coeruleus, described in the sections on the brainstem, end here and release serotonin and noradrenaline, which suppress transmission. Opioid receptors lie on both sides of the synapse, which is why morphine delivered around the cord relieves pain at a small fraction of the dose needed by mouth. And the large touch fibres entering at the same level excite the inhibitory interneurons, which is the reason that rubbing a knocked shin makes it hurt less. Melzack and Wall proposed in 1965 that the dorsal horn acts in this way as a **gate** on pain, and although the circuit they drew has been revised, the principle has held. How much of a painful signal leaves the cord is decided at the first synapse, by other sensory input and by the brain. The gate can also be forced wider. Under sustained input these synapses strengthen, by a mechanism dependent on the NMDA receptor and closely related to the long-term potentiation that the section on the limbic system describes, and the cord thereafter passes pain more readily. This is one of the ways in which pain comes to outlast the injury that caused it.

The **ventral horn** contains the **motor neurons**, which Sherrington called the final common pathway: whatever the rest of the nervous system decides, it can reach a muscle only through these cells. Each motor neuron and the muscle fibres it supplies form a single unit, and the neurons are arranged in an orderly way, with those for the muscles of the trunk lying medially and those for the hand and foot laterally.

The simplest thing the grey matter does is the **stretch reflex**, the one tested by tapping the tendon below the kneecap. Stretch of a muscle excites sensors within it, the muscle spindles; their fibres enter the dorsal root and synapse directly on the motor neurons of the same muscle, which contracts. One synapse lies between sensation and action, and the whole reflex takes a few hundredths of a second. A branch of the same sensory fibre excites an interneuron that inhibits the motor neurons of the opposing muscle, so that it relaxes as the first contracts. The transmitter of that interneuron is glycine, and its importance is shown by what happens without it. Strychnine blocks the glycine receptor, and tetanus toxin, carried backwards along motor axons into the cord, prevents the inhibitory interneurons from releasing their transmitter by cleaving the SNARE proteins on which release depends. In both cases every muscle contracts against its opponent at once. This is the answer to a question left open in the section on neuron structure. Botulinum toxin and tetanus toxin attack the same machinery, and the first paralyses while the second convulses, because the first disables the synapse onto muscle and the second disables the synapses that restrain it.

The **withdrawal reflex** is one step more elaborate. A painful stimulus to a limb excites, through several interneurons, the muscles that flex the whole limb away from it, and inhibits those that would extend it; in a leg, the interneurons also cross the cord and extend the opposite leg to take the weight. The circuit is complete within the cord.

More elaborate still are the circuits for walking. Networks of interneurons in the lumbar cord generate the alternating rhythm of flexion and extension in each leg and the alternation between the two legs. A cat whose spinal cord has been severed in the thoracic region, and which is supported over a moving treadmill, can be brought to step with its hind legs in a coordinated pattern that adapts to the speed of the belt. The evidence in humans is less direct but points the same way. The brain does not compose each step. It starts and stops a generator that the cord already contains, and steers it.

That the cord's circuits are normally held in check from above is shown by two contrasting patterns of weakness. When the motor neurons themselves or their axons are destroyed, the muscle is limp, wastes away, and has no reflexes: the **lower motor neuron** pattern. When the descending pathways from the brain are interrupted and the motor neurons are left intact, the muscle is weak but stiff, the reflexes are exaggerated, and stroking the sole of the foot makes the big toe turn upward instead of down, a response otherwise seen only in infants, whose descending pathways are not yet myelinated. This is the **upper motor neuron** pattern. The spinal reflexes have not been damaged; they have been released.

17.3 White Matter

The white matter surrounds the grey on all sides and is divided on each side into a **dorsal**, a **lateral**, and a **ventral column**. Within the columns the fibres are grouped into tracts, each with a known origin, destination, and content, and the important thing to know about each is where it crosses the midline. The principal tracts are listed in Table 8, and three of them carry most of the argument.

The **dorsal columns** carry discriminative touch, vibration, and the sense of position. Their fibres are the central branches of the dorsal root ganglion cells themselves, and they ascend without a synapse and without crossing as far as the gracile and cuneate nuclei of the medulla, where, as the section on the medulla described, the pathway relays and crosses. The **spinothalamic tract** carries pain and temperature. Its fibres arise from neurons in the dorsal horn, cross to the opposite side in front of the central canal within a segment or two of the level of entry, and ascend in the front and side of the cord to the thalamus. The **lateral corticospinal tract** carries voluntary movement. Having crossed in the pyramids, it descends in the lateral column and ends mostly on interneurons, but in humans and other primates some of its fibres end directly on the motor neurons of the hand, and it is on those direct connections that the independent movement of the fingers depends.

The other descending pathways arise in the brainstem. The vestibular nuclei, the reticular formation, and the superior colliculus send fibres down the ventral column to the motor neurons of the trunk and the near ends of the limbs, and between them these maintain posture and balance and turn the head and body towards events. The lateral column also carries the fibres by which the hypothalamus and the brainstem drive the autonomic neurons of the lateral horn, and the fibres that modulate pain in the dorsal horn.

The crossing levels are confirmed by a lesion. In 1850 Brown-Séquard described what follows when one half of the cord is cut across. Below the cut, on the same side, the limb is paralysed and has lost the sense of touch and position, because the corticospinal tract in that half has already crossed and the dorsal column in that half has not yet done so, and both therefore serve the side on which they lie. On the opposite side, strength and touch are normal and the sense of pain and temperature is gone, because the spinothalamic fibres from that side crossed into the damaged half soon after entering. The patient has one leg that cannot be moved and can still feel a burn, and another that moves normally and cannot. A second lesion makes the same point from the centre outward. In **syringomyelia** a cavity forms in the middle of the cord and destroys the pain fibres as they cross in front of the canal, while sparing the dorsal columns behind. The result is loss of pain and temperature across the shoulders and arms with normal touch, and the patients, who feel a hot object as merely an object, come to attention with painless burns of the hands.

Loss of the dorsal column pathway alone is in some ways the most instructive of all. Strength is normal, and yet the patient cannot stand with the eyes closed, and walks only by watching the feet, because the position of the limbs is no longer being reported. The extreme case is a man who at nineteen lost, through a disease of the large sensory fibres in his nerves, all sense of touch and position below the neck, with the motor fibres untouched. He was at first unable to make any controlled movement at all. Over years he taught himself to move again by watching his limbs and planning each movement deliberately, and he remains unable to stand if the lights go out. The body schema that the section on the parietal lobe described as constructed has an input, and this is what remains of ordinary movement when the input is removed.

17.4 Summary

The spinal cord runs from the foramen magnum to the first lumbar vertebra and gives off thirty-one pairs of spinal nerves, each formed from a sensory dorsal root and a motor ventral root and each supplying a band of the body. Its grey matter receives sensation in the dorsal

horn, where the first synapse of the pain pathway acts as a gate controlled by other sensory input and by the brain; holds the sympathetic preganglionic neurons in the lateral horn; and contains in the ventral horn the motor neurons through which every command to a muscle must pass. Its interneurons carry out the stretch and withdrawal reflexes and generate the rhythm of walking, and these circuits are restrained from above, as their release after damage to the descending pathways shows. Its white matter carries the long tracts. The dorsal columns ascend uncrossed and cross in the medulla, the spinothalamic tract crosses within the cord near the level of entry, and the corticospinal tract descends having crossed in the pyramids, an arrangement that the syndrome of hemisection displays exactly.

The first section remarked that by the time a burn is felt the hand has already moved, and this section has supplied the circuit. It is worth dwelling on what that means. The withdrawal is organised entirely within the cord and is under way within about a tenth of a second. The pain depends on a signal reaching the thalamus and cortex and being worked on there, and the slow, burning component travels in fibres that conduct at about a metre a second, so that from the hand it arrives most of a second later. It is natural to describe the episode afterwards as pulling the hand away because it hurt. The timing does not permit that description. The movement was not caused by the pain; both were caused by the stimulus, by different routes, and the movement came first. What is reported as a reason for the action is, in this simplest of cases, an account assembled after the action was over.

The cord separated from the brain makes the point more starkly. After a complete transection, once the initial shock has passed, a pinprick to the foot still produces a brisk withdrawal of the leg. The response is protective, coordinated, and appropriate to the stimulus, and the person to whom the leg belongs feels nothing and has done nothing. Together with the cat stepping on its treadmill, this is the same caution that the sections on the midbrain and pons arrived at, here in its most elementary form. Organised, apparently purposeful behaviour can be produced by circuits that no one supposes to be aware of anything, and a great deal of what a person does, including most of the detail of every movement, is done in this way and only its outline is ever known to them.

Table 8: The principal tracts of the spinal cord, with what each carries, where it crosses the midline, and where in the white matter it runs. The crossing level determines on which side of the body a lesion of the cord produces its effect.

Tract	**Direction**	**Carries**	**Crosses**	**Column**
Dorsal columns	Ascending	Fine touch, vibration, sense of position	In the medulla, after relay in the gracile and cuneate nuclei	Dorsal
Spinothalamic tract	Ascending	Pain, temperature, crude touch	In the cord, within a segment or two of entry	Ventral and lateral
Spinocerebellar tracts	Ascending	Position and movement of the limbs, to the cerebellum; not conscious	Effectively uncrossed	Lateral, at the surface
Lateral corticospinal tract	Descending	Voluntary movement, especially skilled movement of the limbs	In the decussation of the pyramids	Lateral
Anterior corticospinal tract	Descending	Voluntary movement of the trunk	Near the level at which the fibres end	Ventral

Table 8 continued from previous page

Tract	Direction	Carries	Crosses	Column
Rubrospinal tract	Descending	Flexion of the limbs; small in humans	In the midbrain	Lateral
Vestibulospinal and reticulospinal tracts	Descending	Posture, balance, muscle tone	Mostly uncrossed	Ventral
Tectospinal tract	Descending	Turning the head towards a stimulus	In the midbrain	Ventral, cervical segments only
Descending autonomic fibres	Descending	Control of the sympathetic and sacral parasympathetic neurons	Uncrossed	Lateral
Descending pain-modulating fibres	Descending	Serotonergic and noradrenergic control of the dorsal horn	Uncrossed	Lateral, dorsal part

18 The Basal Ganglia

With the central nervous system laid out from the top down, the remaining sections take up systems that run across it, and the first is the one that decides what the body does. The basal ganglia are not cortex. They are a set of nuclei buried deep beneath it, grey matter sunk into the white matter of the hemispheres and reaching down into the midbrain, and they are wired into a loop that begins and ends in the cortex we have already described. Their business is **action selection**. At any moment the cortex has many possible movements half-prepared; the basal ganglia are the machinery that lets one of them through and holds the rest back.

The most useful thing to fix in mind before any of the anatomy is that this circuit works by *subtraction*, not addition. Its output nuclei fire constantly, and what they do while firing is inhibit the thalamus. Movement is not released by switching something on; it is released by switching that inhibition off. Almost every confusion about the basal ganglia comes from forgetting that the resting state of the system is a brake held down, and that the pathways we are about to trace are competing over whether to lift it.

18.1 Components

The basal ganglia are made up of several nuclei, and the naming is genuinely awkward, because anatomists carved these structures up by appearance long before anyone knew what they did. The result is that a single functional unit can carry two names, and a single name can span two functional units. It is worth learning the components alongside their transmitters, because in this circuit the transmitter is what fixes the sign of every connection.

- The **striatum** is the input nucleus: almost everything the cortex sends to the basal ganglia arrives here. It is split by the **internal capsule**, the great sheet of white matter carrying fibres to and from the cortex, into the **caudate nucleus** medially and the **putamen** laterally. The split is anatomical rather than functional; the two are histologically identical, and where they remain joined underneath, the tissue is called the **nucleus accumbens**. The principal cells of the striatum are the **medium spiny neurons (MSNs)**, which are GABAergic and therefore inhibitory. They come in two populations, distinguished by which dopamine receptor they carry, and that distinction turns out to be the whole story.
- The **globus pallidus** sits medial to the putamen and has two segments. The **external segment (GPe)** is an internal relay used by only one of the two pathways. The **internal**

segment (GPi) is an output nucleus, one of the two structures through which the basal ganglia speak to the rest of the brain. Both are GABAergic.

- The **subthalamic nucleus (STN)** is a small lens-shaped nucleus lying beneath the thalamus. It is the exception in this circuit: alone among these nuclei it is *glutamatergic*, and therefore excitatory. Whenever a chain of reasoning about the basal ganglia produces a sign that seems wrong, the STN is usually the step that was missed.
- The **substantia nigra** lies in the midbrain and, confusingly, does two entirely different jobs. The **pars reticulata (SNr)** is GABAergic and is the second output nucleus, the functional twin of the GPi. The **pars compacta (SNc)** is dopaminergic and is not part of the loop at all; it projects onto the striatum from the side and modulates it, which is the subject of the last two subsections of this section.

One more structure has to be named even though it is not part of the basal ganglia: the **thalamus**, and specifically its **ventral anterior (VA)** and **ventral lateral (VL)** nuclei, met in the section on the thalamus. These are what the output nuclei act upon, and they project glutamate back up to the motor and premotor cortex. The thalamus is the thing being gated.

Putting these together, the flow of information is a loop:

$$\text{cortex} \to \text{striatum} \to \text{GPi/SNr} \to \text{thalamus} \to \text{cortex}$$

Two features of this loop do all the work in what follows. The first is that the output nuclei, GPi and SNr, are **tonically active**: they fire continuously, without needing to be told to, and being GABAergic they hold the thalamus under a permanent brake. The second is that when one inhibitory neuron inhibits another inhibitory neuron, the result at the far end is **disinhibition**, which is functionally indistinguishable from excitation. Two minus signs in series make a plus. The direct pathway is built out of exactly that arrangement.

18.2 The Direct Pathway

The direct pathway is the route by which a movement gets released. It runs from cortex to striatum to the output nuclei and out to the thalamus, and its entire trick is the double inhibition just described.

Tracing it synapse by synapse, naming the transmitter at each:

1. The **cortex** fires onto the striatum, releasing **glutamate**. This is excitatory, and it activates the MSNs that carry **D1 receptors**.
2. Those D1 MSNs project to the **GPi/SNr** and release **GABA**. This is inhibitory, and it silences output neurons that were, until this moment, firing tonically.
3. The **GPi/SNr** normally release **GABA** onto the thalamus. Now that they have been silenced, that inhibition stops. The brake is lifted.
4. The **thalamus**, released, fires up to the cortex on **glutamate**, exciting the motor areas that were preparing the movement.

Written as a chain, with the sign of each synapse marked:

$$\text{cortex} \xrightarrow{+} \text{striatum (D1)} \xrightarrow{-} \text{GPi/SNr} \xrightarrow{-} \text{thalamus} \xrightarrow{+} \text{cortex}$$

The arithmetic is the point. Steps two and three are both inhibitory, and inhibiting an inhibitor is excitation. The striatum does not excite the thalamus; it removes something that

was suppressing the thalamus, and the thalamus, which was ready to fire all along, does the rest. The cortex ends up exciting itself, by way of a detour that takes the brake off.

The net effect of activity in the direct pathway is therefore to **release the selected movement**. When a motor plan wins through this route, the thalamus is disinhibited, the loop back to cortex closes with excitation, and the action proceeds to execution.

18.3 The Indirect Pathway

The indirect pathway is the mirror image: it exists to hold movements back. It starts from the same cortical input and ends at the same output nuclei, but it reaches them by a longer route that inserts an extra inhibitory step and, crucially, an excitatory one. The extra step flips the sign.

Tracing it as before:

1. The **cortex** again fires onto the striatum with **glutamate**, but this time it activates the MSNs carrying **D2 receptors**, a separate population from those of the direct pathway.
2. Those D2 MSNs project to the **GPe**, not to the output nuclei, releasing **GABA** and silencing it.
3. The **GPe** was itself holding down the subthalamic nucleus with tonic **GABA**. Silencing the GPe therefore *releases* the STN.
4. The **STN**, now disinhibited, fires onto the GPi/SNr with **glutamate**. This is the exception noted earlier, and it means the output nuclei are now being driven *harder* than at rest.
5. The **GPi/SNr**, excited, pour still more **GABA** onto the thalamus. The brake is pressed down more firmly than before.

As a signed chain:

$$\text{cortex} \xrightarrow{+} \text{striatum (D2)} \xrightarrow{-} \text{GPe} \xrightarrow{-} \text{STN} \xrightarrow{+} \text{GPi/SNr} \xrightarrow{-} \text{thalamus}$$

Follow the signs and the contrast with the direct pathway becomes clear. Here too there is a double inhibition (striatum onto GPe onto STN), and here too it produces disinhibition. But the structure being disinhibited is excitatory. So instead of the released structure exciting the thalamus, the released structure excites the very nuclei that inhibit the thalamus. The disinhibition is spent on strengthening the brake rather than lifting it.

The net effect of activity in the indirect pathway is therefore to **suppress movement**. The two pathways begin at the same place and converge on the same place, and they arrive with opposite sign: the direct pathway subtracts from the thalamic brake, the indirect pathway adds to it. What the cortex actually gets back depends on the balance struck between them. That balance is not fixed, and what sets it is dopamine.

18.4 Dopamine

Dopamine has appeared already in these notes, listed among the neuromodulators and given the brief gloss of "movement, reward, motivation". It is now worth being precise about what it is and, more importantly, about how it acts, because dopamine does not work anything like the glutamate and GABA we have been tracing through the two pathways.

Dopamine is a **catecholamine**, synthesised from the amino acid **tyrosine** by way of an intermediate that will be familiar from pharmacology: tyrosine is converted to **L-DOPA**, and L-DOPA is decarboxylated to dopamine. It is loaded into synaptic vesicles, released by the calcium-triggered mechanism described earlier, and cleared from the cleft mainly by reuptake

through the **dopamine transporter (DAT)**, along the lines set out in the discussion of reuptake and degradation.

The important difference lies in what happens after dopamine binds. Every synapse traced through the direct and indirect pathways used a ligand-gated ion channel, where the receptor is itself the pore and the membrane potential moves within a millisecond. Dopamine receptors are not ion channels at all. They are **G protein-coupled receptors**, working by the cascade set out when we first met graded potentials: binding throws a switch on a G protein, a GDP is swapped for a GTP, an effector enzyme is engaged, and a second messenger goes on to tune ion channels the cell already has. The consequences established there are exactly the ones that matter here. Dopamine acts slowly, in hundreds of milliseconds to seconds; its effect is amplified far beyond the amount of transmitter released; and it modulates rather than commands, changing how a striatal neuron answers its cortical input rather than firing it directly.

18.4.1 The D1 and D2 Receptor Families

Dopamine receptors come in two families, distinguished by which G protein they couple to, and therefore by which direction they push adenylyl cyclase.

- The **D1 family** (D1 and D5) couples to $\mathbf{G_s}$, the stimulatory G protein. Its α subunit *activates* adenylyl cyclase, cAMP rises, PKA is switched on, and the net effect on the neuron is **increased excitability**: it becomes more responsive to the glutamate arriving from cortex.
- The **D2 family** (D2, D3, and D4) couples to $\mathbf{G_{i/o}}$, the inhibitory G protein. Its α subunit *suppresses* adenylyl cyclase, cAMP falls, PKA activity drops, and the net effect is **decreased excitability**: the neuron becomes less responsive to that same cortical input.

This is the crux, and it is worth stating plainly because it is the single fact on which the next subsection depends. Dopamine is one molecule with one shape, and it has opposite effects on two different cells. Nothing about the transmitter decides this. What decides it is which receptor the cell happens to express, and therefore which G protein sits behind that receptor. The message is identical; the interpretation is not.

18.5 The Modulatory Pathway

We now have everything needed to see what the substantia nigra does. Recall from the components that the pars compacta is dopaminergic and stands outside the cortex-striatum-thalamus loop. Its neurons send axons up into the striatum, spreading through it diffusely, in a projection called the **nigrostriatal pathway**.

This projection carries no motor command. It does not tell the striatum which movement to make, and it is not a third route from cortex to thalamus alongside the other two. What it does is release dopamine broadly across the striatum, where it lands on both MSN populations at once. The direct and indirect pathways select; the nigrostriatal projection sets the terms on which they compete.

The consequence follows directly from the receptor pharmacology of the previous subsection, because the two MSN populations were defined by exactly that difference:

- Direct pathway MSNs carry **D1** receptors. Dopamine acting through G_s raises cAMP and makes them *more* excitable, so the pathway that lifts the thalamic brake is amplified.
- Indirect pathway MSNs carry **D2** receptors. Dopamine acting through $G_{i/o}$ lowers cAMP and makes them *less* excitable, so the pathway that presses the brake down is damped.

These two actions are opposite in mechanism and identical in outcome. Dopamine excites the pathway that promotes movement and inhibits the pathway that prevents it, and both effects push the balance the same way. This is the elegance of the arrangement: a single diffuse chemical signal, requiring no wiring specificity whatsoever, can shift the entire system toward action, because the two competing populations have been pre-labelled with receptors of opposite sign.

The result is a **gain control** on movement. Dopamine does not choose the action; the cortex does that, by deciding which striatal neurons receive glutamate. Dopamine sets how readily whatever has been chosen gets released. When nigrostriatal dopamine is high, the threshold for releasing a movement falls and actions come easily; when it is low, the direct pathway is under-driven and the indirect pathway is under-damped, the brake stays down, and movement becomes harder to initiate.

This is what fails in **Parkinson's disease**, in which the dopaminergic neurons of the pars compacta progressively die. Symptoms usually appear only once most of the striatum's dopamine has been lost, and they are the ones the arithmetic predicts: **bradykinesia** (slowness and difficulty in starting movement), rigidity, and a characteristic tremor at rest. The standard treatment is **L-DOPA**, the precursor met above, which unlike dopamine itself can cross the blood-brain barrier and is converted to dopamine by the surviving cells. **Huntington's disease** is close to the opposite case: the striatal neurons of the indirect pathway are among the first to degenerate, the brake on unwanted movement fails, and the result is **chorea**, a stream of involuntary, dance-like movements.

It is worth separating the two halves of the substantia nigra once more, since the shared name invites confusion. The **pars reticulata** is a member of the loop, an output nucleus inhibiting the thalamus exactly as the GPi does. The **pars compacta** is not in the loop at all; it stands to one side and adjusts the gain of the structure that feeds it. Same anatomical name, two entirely different roles.

18.6 Summary

The basal ganglia are the brain's mechanism for selecting actions, and they work by releasing a brake rather than by issuing a command. Cortex feeds glutamate into the striatum; the striatum's GABAergic medium spiny neurons act on output nuclei, the GPi and SNr, which fire tonically and inhibit the thalamus without pause. The direct pathway runs striatum to GPi/SNr, and because two inhibitory synapses in series produce disinhibition, its activity lifts the thalamic brake and releases the selected movement. The indirect pathway runs striatum to GPe to subthalamic nucleus to GPi/SNr, and because the disinhibited subthalamic nucleus is glutamatergic, its activity drives the output nuclei harder and presses the brake down, suppressing movement. The two routes converge with opposite sign, and the balance between them is set by dopamine, which acts through neither of the fast ionotropic mechanisms used elsewhere in the circuit but through G protein-coupled receptors, where binding triggers a GDP-for-GTP exchange, an effector enzyme, a second messenger, and a slow, amplified, modulatory change in how the cell answers its other inputs. Because direct pathway neurons carry stimulatory D1 receptors and indirect pathway neurons carry inhibitory D2 receptors, dopamine from the substantia nigra pars compacta amplifies the pathway that permits movement while damping the one that prevents it, and a single diffuse chemical broadcast is enough to set the threshold at which the whole system consents to act.

The capacity this section adds is choosing. At any moment the cortex has many actions partly prepared, and something has to permit one and withhold the others, since a body has only one set of limbs and cannot execute competing plans at once. That is a real problem with a real solution, and the solution turns out to be a brake that is lifted rather than an instruction that is issued. It is worth noticing how little of this resembles the folk picture of deciding to move. There is no stage at which a command is composed; there is a set of candidates, a tonic

inhibition holding all of them back, and a competition whose winner is disinhibited. Volition, whatever else is true of it, has at its base a mechanism of exactly this kind, and the mechanism is visible in what happens when it degrades: too little dopamine and the brake stays down, so that movements become hard to initiate despite intact strength and intact intention, which is a dissociation between wanting to move and being able to start that no account of action as simple command can easily accommodate.

19 The Limbic System

Every section so far has taken a piece of the nervous system and asked what it does. This one works in the opposite direction. The **limbic system** is not a region that can be pointed to on a specimen, and almost nothing in it is new: the hypothalamus and the thalamus have each had a section to themselves, the amygdala and the hippocampus were met in passing in the temporal lobe, and the entorhinal cortex was introduced there as the olfactory association area. What the term names is not fresh tissue but a *set of connections* among tissue already described, together with a set of functions that no one of the participating structures could carry out alone.

The name comes from Broca, who in 1878 described *le grand lobe limbique*, the "bordering lobe", after the Latin *limbus*, a hem or a margin. What he was pointing at was a ring of cortex on the medial surface of each hemisphere that curls around the corpus callosum and the upper brainstem like a collar, dividing the neocortex outside it from the diencephalon within. Broca meant the description anatomically and supposed the ring to be concerned with smell. It was **Papez** in 1937 who proposed that part of it formed a circuit for emotion, and **MacLean** in the 1950s who broadened that proposal into the "limbic system" as it is now generally taught.

It is worth saying plainly that the term is contested. There is no agreed membership list, no boundary that anatomy itself supplies, and no single function that all the candidate structures share; several authors have argued the phrase should be retired altogether on the grounds that it groups together structures whose connections are more varied than the label suggests. The reason it survives, and the reason it is worth a section here, is that the structures conventionally included are in fact densely interconnected with one another, and that they are jointly responsible for a set of capacities that the rest of the nervous system does not supply. The components taken up below are the ones most consistently included; the nucleus accumbens, the ventral tegmental area, and the periaqueductal grey appear in the functional subsections as close associates rather than as members.

The organising idea of the section is this. The cortex described in earlier sections tells the organism what is present: an object, a face, a sound, a smell, a limb in a particular position. That is not enough to act on. Two further questions have to be answered, and the limbic system is where each is answered. The first is *what happened here before*, which is memory, and it is the business of the hippocampus and the circuit running out of it. The second is *what does this mean for me*, which is emotion and value, and it is the business of the amygdala and the cortex it reports to. A third structure, the hypothalamus, converts whatever answer is arrived at into bodily change. Perception, memory, valuation, and response: the limbic system supplies the middle two and commands the last.

19.1 Components

19.1.1 Limbic Lobe

The **limbic lobe** is Broca's original ring, and it is the cortical component of the system. It is best appreciated on a midsagittal section, where it appears as a continuous band of cortex sweeping around the corpus callosum: forward and up from beneath its front end, back along its upper surface, down around its posterior end, and then forward again along the medial temporal lobe, closing the circle beneath. Three named regions make up the band.

The **cingulate gyrus** runs immediately above the corpus callosum and forms the upper half of the ring. Beneath it lies the **cingulum bundle**, the white matter tract that carries traffic along the ring and connects its ends. The gyrus is not functionally uniform, and the division between its front and back halves matters:

- The **anterior cingulate cortex (ACC)** is concerned with conflict, effort, error, and the affective weighting of experience. It is the cortical partner of the anterior insula in the **salience network** described in the insula section, and it is the destination of the medial pain pathway traced in the thalamus section, which is why it carries the unpleasantness of pain rather than its location. It also has direct descending connections to autonomic centres, making it one of the few pieces of cortex that acts on the body without an intermediary.
- The **posterior cingulate cortex** and the **retrosplenial cortex** behind it are concerned with spatial orientation, the retrieval of autobiographical memory, and self-referential thought. The retrosplenial cortex in particular is a waypoint between the hippocampus and the parietal cortex, and it is where a memory anchored to the body's own frame of reference is translated into one anchored to the world.

The **parahippocampal gyrus** forms the lower half of the ring, on the medial surface of the temporal lobe, and is continuous with the cingulate gyrus behind by way of the narrow **isthmus**. Its anterior part is the **entorhinal cortex** (Brodmann area 28), already met as the olfactory association cortex, and it is more than that: it is the principal gateway through which the neocortex reaches the hippocampus and the route by which the hippocampus answers. Its posterior part contributes to the recognition of scenes and spatial context. At the front the gyrus hooks sharply backwards to form the **uncus**, which overlies the amygdala and carries the primary olfactory cortex.

Two smaller regions complete the lobe. The **subcallosal** or **paraterminal gyrus** lies beneath the front end of the corpus callosum and is continuous with the septal area described below. The **hippocampal formation** itself is rolled into the medial temporal lobe beneath the parahippocampal gyrus, and although it is buried it belongs to this ring rather than to the neocortex outside it.

Histologically the ring is distinct from the neocortex it borders. Neocortex has six layers; the cortex of the limbic lobe is transitional, and the hippocampal formation at its core has only three. This is one of the reasons the ring was recognised as a unit long before anyone knew what it did, and the reason it is sometimes called **allocortex** and **mesocortex** in contrast to the **neocortex** outside it.

19.1.2 Hippocampus

The **hippocampus** lies in the floor of the inferior horn of the lateral ventricle, curled into the medial temporal lobe, and takes its name from the seahorse it resembles in cross-section. It is the structure most closely associated with the formation of new memories, and it is the best-understood piece of the limbic system by a considerable margin.

Strictly, the term **hippocampal formation** covers three parts that differ in structure and are worth distinguishing: the **dentate gyrus**, the **hippocampus proper** (subdivided into the fields **CA1** to **CA3**, the initials standing for *cornu ammonis*, Ammon's horn), and the **subiculum**, which is the transition back towards the entorhinal cortex. Unlike the six-layered neocortex, all of these are three-layered, with a single dense band of principal cells, which is what makes the internal circuitry so unusually easy to trace.

That circuitry is the **trisynaptic circuit**, and it is essentially unidirectional:

$$\text{entorhinal cortex} \rightarrow \text{dentate gyrus} \rightarrow \text{CA3} \rightarrow \text{CA1} \rightarrow \text{subiculum}$$

Highly processed information from association cortex throughout the brain converges on the entorhinal cortex, which sends it along the **perforant path** to the dentate gyrus. The dentate projects by **mossy fibres** to CA3, CA3 projects by the **Schaffer collaterals** to CA1, and CA1 passes to the subiculum, from which output leaves the formation. The projection is glutamatergic at every step. CA3 is additionally distinguished by a dense network of recurrent connections onto itself, an architecture well suited to completing a whole pattern from a fragment of it, which is a reasonable description of what recollection feels like from the inside.

Output leaves by two routes. The **fornix**, described in the hypothalamus section, is the great arching bundle carrying subicular and CA1 fibres forward to the mammillary bodies and the septal area. A second, quieter route runs back to the entorhinal cortex and from there to the same association cortices that supplied the input, which is the anatomical form of the hippocampus talking back to the cortex about what the cortex has just sent it.

Two properties of hippocampal neurons deserve naming here because the memory subsection depends on them. The first is **long-term potentiation (LTP)**, the lasting strengthening of a synapse that has been repeatedly and strongly driven, which was discovered in this pathway and remains the leading candidate mechanism for memory at the cellular level. The second is spatial coding: hippocampal **place cells** fire when the animal occupies a particular location, and **grid cells** in the entorhinal cortex fire in a regular hexagonal lattice across an environment, so that the structure carries a metric map of space as well as a record of events.

The hippocampus is also conspicuously fragile. CA1 is among the most hypoxia-sensitive tissue in the brain and is characteristically lost after cardiac arrest; the entorhinal cortex and hippocampus are the earliest cortical casualties of Alzheimer's disease, which is why its first symptom is a failure to lay down new memories; and the medial temporal lobe is the commonest origin of focal epilepsy in adults, where the associated **mesial temporal sclerosis** is a scarring and shrinkage of exactly these fields.

19.1.3 Amygdala

The **amygdala** is an almond-shaped mass of grey matter, hence the name, sitting in the medial temporal lobe immediately in front of the hippocampus and beneath the uncus. If the hippocampus asks what happened here before, the amygdala asks what it is worth, and its output is not a record but a response.

It is a collection of nuclei rather than a single structure, and three groups matter:

- The **basolateral group** is the input side and is cortex-like in its cellular organisation. It receives highly processed sensory information from the association cortices and, importantly, a second and much cruder stream directly from the thalamus. It is where the association between a stimulus and its significance is actually learned.
- The **centromedial group**, and in particular the **central nucleus**, is the output side. It projects to the hypothalamus and brainstem and is the structure that converts a judgment of significance into a bodily response.
- The **cortical group** receives direct input from the olfactory bulb, one of very few places in the brain to do so, and is part of the reason smell reaches emotion so directly.

The existence of two input routes is worth dwelling on, because it explains a familiar experience. A sensory signal can reach the basolateral amygdala directly from the thalamus in a single synapse, fast but carrying little detail, or it can travel the longer way through primary and association cortex and arrive slower but fully analysed. The short route is what starts the body reacting to a shape on the path before the cortex has established that it is a stick rather than a snake; the long route is what subsequently calls the reaction off. The cost of a false alarm is trivial and the cost of a missed threat is not, and the wiring reflects that asymmetry.

Output from the central nucleus fans out to a set of targets each of which produces one component of an emotional response, and the pattern we recognise as fear is the sum of them:

- To the **lateral hypothalamus**, driving the sympathetic response of raised heart rate and blood pressure.
- To the **paraventricular nucleus**, driving **CRH** release and with it the whole endocrine stress axis described in the hypothalamus section.
- To the **periaqueductal grey** of the midbrain, which organises defensive behaviour such as freezing and flight.
- To the **locus coeruleus**, raising noradrenaline and with it vigilance, and to the **basal forebrain** cholinergic neurons, sharpening attention.
- To the **parabrachial nucleus**, altering the pattern of breathing.

Its two major tracts have already been named in the hypothalamus section: the long, arcing **stria terminalis** and the short, direct **ventral amygdalofugal pathway**. Connections with the **mediodorsal thalamus** and, through it, the prefrontal cortex are the route by which an emotionally significant event captures deliberation rather than merely producing a reflex, and the reciprocal projection from the **ventromedial prefrontal cortex** back onto the amygdala is the route by which deliberation restrains it.

Clinically the amygdala is best known through two syndromes. Bilateral destruction produces **Klüver–Bucy syndrome**, in which animals and, rarely, patients become placid and lose the capacity for fear or appropriate aggression, along with hyperorality, indiscriminate sexual behaviour, and a failure to recognise objects visually. **Urbach–Wiethe disease**, which calcifies the amygdalae bilaterally, has yielded the striking case of a patient who could recognise every other emotion in a face but not fear, and who reported no fear herself in situations that reliably produce it in others. In the opposite direction, an overactive amygdala under weak prefrontal restraint is one of the most consistent findings in anxiety disorders and post-traumatic stress disorder.

19.1.4 Hypothalamus

The hypothalamus has a section of its own, and the point to carry into this one is the role it was given there: it is the **output stage** of the limbic system. The amygdala and the hippocampus determine what a situation means and what it resembles, and neither can do anything about it. The hypothalamus is the only member of the system with the connections to act, commanding the autonomic nervous system through the brainstem and spinal cord and the endocrine system through the pituitary, and it is therefore the point at which emotional significance becomes bodily change. This is the anatomical reason emotion is felt in the body rather than merely registered in the head.

Two of its parts belong to this section specifically. The **mammillary bodies** at its posterior underside are not autonomic or endocrine at all: they receive the fornix from the hippocampus and pass its signal on through the mammillothalamic tract, which makes them a memory relay sitting inside a homeostatic organ. The **lateral hypothalamic area**, through which the medial forebrain bundle runs, is the main recipient of amygdalar output and a central node in motivated behaviour.

19.1.5 Thalamus

The thalamus likewise has its own section, and three of its nuclear groups are limbic members rather than sensory relays.

The **anterior nuclear group** is the thalamic station of the Papez circuit, receiving the mammillothalamic tract and projecting to the cingulate cortex. It is a memory structure, and damage to it or to the mammillary bodies feeding it produces the dense anterograde amnesia of Korsakoff's syndrome.

The **mediodorsal nucleus** is the thalamic partner of the prefrontal cortex, and it receives from the amygdala, the olfactory cortex, and the hypothalamus. It is the point at which limbic information reaches the executive machinery of the frontal lobe, and it is also where olfaction takes its deferred thalamic step on the way to the orbitofrontal cortex.

The **midline** and **intralaminar nuclei** connect with the hippocampus, amygdala, and hypothalamus and carry arousal, visceral, and affective signals; the intralaminar group in particular delivers the unpleasantness of pain to the anterior cingulate and insula.

The general point is that the thalamus is the limbic system's route to the cortex in exactly the same way it is the sensory systems' route to the cortex, and it gates the traffic in the same way. What a limbic structure can make the cortex attend to is subject to the same thalamic permission as what the retina can.

19.1.6 Septal Area

The **septal area** is a small region of grey matter lying beneath the front end of the corpus callosum, in front of the anterior commissure and continuous above with the septum pellucidum, the thin partition between the lateral ventricles. It is easy to overlook because of its size, and it is the hinge on which two of the system's circuits turn.

Its central relationship is with the hippocampus, and it is reciprocal. The **septohippocampal pathway** runs through the fornix in both directions: the hippocampus projects to the septal nuclei, and the **medial septal nucleus** projects back, sending cholinergic and GABAergic fibres that pace hippocampal activity. What they pace is the **theta rhythm**, a four to eight hertz oscillation that dominates the hippocampus during exploration and attentive behaviour and that appears to be the timing signal under which encoding takes place. Silence the medial septum and the theta rhythm stops and hippocampal learning is impaired, which is a rare case of a memory structure being disabled by removing a clock rather than a component.

The septal area also connects forward to the prefrontal cortex, downward to the hypothalamus and midbrain through the medial forebrain bundle, and back to the **habenula** by way of the **stria medullaris thalami**. It sits immediately above the **nucleus accumbens**, with which it is often discussed, and this vicinity is where Olds and Milner in 1954 found that rats would repeatedly press a lever to stimulate their own brains, the experiment that opened the study of reward.

Two further points are worth carrying. Septal lesions in animals produce a marked increase in defensive aggression, the phenomenon known as **septal rage**, which suggests the region normally exerts a restraining influence on the amygdala and hypothalamus. And the cholinergic cells of the medial septum belong to the same **basal forebrain** population as the nucleus basalis of Meynert, whose degeneration is a feature of Alzheimer's disease and the target of the cholinesterase inhibitors used to treat it. The role of acetylcholine in attention and memory, noted when the transmitters were first introduced, has its anatomical seat here.

19.1.7 Habenula

The **habenula** is a pair of small nuclei in the **epithalamus**, sitting at the posterior end of the roof of the third ventricle just in front of the pineal gland, joined across the midline by the habenular commissure. It is the smallest structure given its own heading in these notes and, until fairly recently, among the least studied. It is included because it is the system's negative pole, and because it makes direct contact with the dopamine machinery described in the basal ganglia section.

Its connections are unusually tidy. Input arrives chiefly along the **stria medullaris thalami** from the septal area and basal forebrain, and output leaves along the **fasciculus retroflexus** to the midbrain. The two habenular nuclei then diverge:

- The **medial habenula** projects to the interpeduncular nucleus and is largely cholinergic. It is densely supplied with nicotinic receptors and is implicated in the aversive state of nicotine withdrawal.
- The **lateral habenula** is the functionally important one. It projects, mostly by way of a small GABAergic relay in the midbrain, onto the dopaminergic neurons of the ventral tegmental area and the substantia nigra pars compacta, and onto the serotonergic raphe nuclei, and its effect on all of them is inhibitory.

What makes the lateral habenula interesting is the signal it carries. Midbrain dopamine neurons fire in response to reward, and more precisely to reward that is better than expected. Lateral habenular neurons do the exact opposite: they fire when an expected reward fails to arrive, and when something aversive occurs, and by inhibiting the dopamine neurons they impose the dip in dopamine that follows disappointment. Read against the section on the basal ganglia, where dopamine was shown to set the gain on whether an action is released at all, the habenula is the structure that turns that gain down when the evidence says the action was not worth taking.

The clinical interest follows from that description. A structure whose job is to suppress dopaminergic and serotonergic activity in response to failure is an obvious candidate for involvement in depression, and hyperactivity of the lateral habenula is now among the better supported physiological findings in depression in both animal models and human imaging. It has become a subject of considerable interest as a target, both for deep brain stimulation and in accounting for the action of drugs whose antidepressant effect is faster than the autoreceptor adaptation described earlier can explain.

19.2 Function

19.2.1 Olfaction

Olfaction is where the limbic system began, both in Broca's description of it and, on the usual reading of the comparative anatomy, in evolutionary terms. The older literature calls these structures the **rhinencephalon**, the "nose brain", and although the name is now too narrow to be useful it records something real about how the system is put together.

The peculiarity of smell was set out in the temporal lobe section: alone among the senses it reaches cortex without relaying in the thalamus. What that section did not emphasise is that the cortex it reaches is *limbic* cortex. The olfactory tract projects to the piriform and periamygdaloid cortex on the uncus, to the entorhinal cortex, and directly to the cortical nucleus of the amygdala. Every one of those targets is a structure in the present section. A smell is therefore two synapses from the hippocampus and one from the amygdala, while a sight or a sound must pass through a thalamic relay, a primary cortex, and an association cortex before it arrives anywhere comparable.

This is the anatomy behind an experience everyone recognises: that a smell can produce a specific autobiographical memory, complete with its emotional colouring, more abruptly and more completely than any other sensory cue. It is also why olfactory hallucinations are a characteristic aura of medial temporal lobe seizures, the so-called uncinate fits, and why the loss of smell is an early sign in Alzheimer's disease and in Parkinson's disease, both of which affect these structures early.

The evolutionary reading, which should be marked as an interpretation rather than a fact, is that the machinery came first and the functions were layered onto it. A chemical arriving at the

nose of a simple vertebrate had to be classified as food, threat, or mate, remembered, and acted on, and that is a requirement for valuation, memory, and autonomic output built around an olfactory input. On this account emotion and memory as we experience them are elaborations of a system that originally existed to decide what a smell meant, which is why the structures that serve them are still wrapped around the olfactory pathway.

19.2.2 Memory

Memory is not one faculty, and the first step is to divide it, because the divisions map onto different structures. **Declarative** (or explicit) memory is memory for facts and events that can be brought to mind and stated, and it divides into **episodic** memory for experienced events and **semantic** memory for knowledge detached from the occasion of learning it. **Non-declarative** (or implicit) memory covers everything that is expressed in performance rather than in recollection: motor skills and habits, which depend on the basal ganglia and cerebellum; conditioned emotional responses, which depend on the amygdala; and priming, which depends on the cortex directly. Only the declarative branch is the hippocampus's business, and that is why hippocampal damage produces such a strange clinical picture.

The circuit that carries it is the **Papez circuit**, met twice already and worth setting out once as a whole:

$$\text{hippocampus} \xrightarrow{\text{fornix}} \text{mammillary bodies} \xrightarrow{\text{mammillothalamic tract}} \text{anterior thalamus}$$
$$\rightarrow \text{cingulate} \xrightarrow{\text{cingulum}} \text{parahippocampal cortex} \rightarrow \text{hippocampus}$$

Papez proposed it as a circuit for emotion and it is no longer read that way, but as a memory circuit it has held up remarkably well, and the clinical evidence is that damage at any point along it produces amnesia. Lesions of the hippocampus itself do so, and so do lesions of the fornix, the mammillary bodies, and the anterior thalamus, which is what makes **Korsakoff's syndrome** such a clean demonstration: thiamine deficiency damages the mammillary bodies and medial thalamus specifically, sparing the hippocampus, and produces a dense amnesia nonetheless.

The classical case at the hippocampal end is the patient known during his lifetime as **H.M.**, whose medial temporal lobes were removed bilaterally in 1953 to control intractable epilepsy. The result was a profound **anterograde amnesia**: he could not form new declarative memories at all, and forgot a conversation within minutes of having it. What was spared is as informative as what was lost. His intelligence and language were intact. His working memory was intact, so he could hold a number in mind while attending to it. His memories from well before the surgery were largely intact. And he could learn new motor skills perfectly well while denying each time that he had ever practised them. Three conclusions were drawn, and they have survived: that declarative memory is separable from other kinds, that the hippocampus is required for laying down new memories but is not where old ones are stored, and that immediate holding of information is a different operation altogether.

That second conclusion is the process called **systems consolidation**. On the standard account the hippocampus binds together the scattered cortical fragments of an experience, the sight and sound and place and feeling of it, and by repeatedly reactivating them, including during sleep, gradually strengthens the direct cortical connections among them until the binding structure is no longer required. A memory therefore migrates, in effect, from being hippocampus-dependent to being cortically self-sufficient. The prediction this makes is that retrograde amnesia after hippocampal damage should be **temporally graded**, with recent memories lost and remote ones spared, and that is broadly what is observed.

At the cellular level the mechanism is **long-term potentiation**, and it draws directly on the receptor material of the firing section. The key element is the **NMDA receptor**, which is a coincidence detector: it is a glutamate-gated channel, but its pore is physically plugged by a

magnesium ion at resting potential, and the block is only expelled when the membrane is already depolarised. It therefore opens only when two conditions are met at once: the presynaptic cell has released glutamate, and the postsynaptic cell is already active. When both hold, calcium enters through the NMDA receptor and triggers a cascade whose short-term effect is to insert more AMPA receptors into the postsynaptic membrane, and whose long-term effect, by way of the activity-dependent gene expression noted when the neuron's nucleus was described, is structural change in the synapse itself. A synapse that has been part of a successful coincidence is thereafter stronger. Hebb's much-quoted formulation, that cells which fire together wire together, turns out to have a molecule behind it.

Finally, memory and emotion are not independent, and the anatomy says why. The amygdala projects to the hippocampus, and emotional arousal releases adrenaline peripherally and cortisol through the axis described in the hypothalamus section, both of which act back on these structures. Emotionally significant events are consequently encoded more strongly and retained better than neutral ones, which is adaptive in most circumstances and pathological in a few: the intrusive, vivid, involuntary memories of post-traumatic stress disorder are, on one widely held view, the same consolidation mechanism running at excessive gain.

19.2.3 Emotional Responses

An emotional response can be decomposed into four stages, and each has an anatomical home in the structures above. There is **appraisal**, the judgment that something matters and in what way. There is **output**, the bodily and behavioural change that follows. There is **feeling**, the conscious experience that accompanies it. And there is **regulation**, the modification of all three in the light of context.

Appraisal is the amygdala's, working with the orbitofrontal and ventromedial prefrontal cortex. The basolateral amygdala learns which stimuli predict which outcomes, and the best-worked-out case is **fear conditioning**: a neutral tone paired with an aversive event acquires, within a few trials, the capacity to produce the full defensive response by itself, and the synaptic change underlying it occurs in the basolateral amygdala by the same NMDA-dependent potentiation described for memory. The amygdala is not exclusively a fear structure, and it responds to appetitive and socially significant stimuli too, but fear is where the circuit has been traced most completely.

Output is the central nucleus's, by way of the projections listed earlier, and it has three limbs which are worth keeping distinct because they operate on different timescales. The autonomic limb runs through the hypothalamus and brainstem and acts within a second. The endocrine limb runs through CRH, ACTH, and cortisol and acts over minutes to hours. The behavioural limb runs through the periaqueductal grey, which organises the coordinated patterns of freezing, flight, and defensive aggression, and which is also the origin of the descending analgesia that makes injuries sustained in an emergency go unnoticed.

Feeling is the least settled of the four and is where this section makes contact with the insula. The classical dispute is between the **James–Lange** position, that the bodily change comes first and the feeling is the perception of it, and the **Cannon–Bard** position, that the two arise in parallel from a central structure. Neither is now held in its original form, but the insula section supplied something that reads as a partial vindication of the first: the posterior insula constructs an interoceptive map of the body's state and the anterior insula re-represents it as a felt quality. On that account the racing heart is not merely a symptom of fear but part of what is read to produce the experience of it, and the anterior insula and anterior cingulate are where the reading happens.

Regulation runs from the prefrontal cortex back onto the amygdala, chiefly from the ventromedial and orbitofrontal regions described in the frontal lobe section. Its clearest experimental expression is **extinction**: a conditioned fear response that is no longer reinforced does not fade because the original association has been erased, but because a second, inhibitory association is

learned, and it is the ventromedial prefrontal cortex that holds it and imposes it on the amygdala. This has two consequences worth knowing. It explains why extinguished fears return under stress or in a new context, since the inhibitory memory is more fragile and more context-bound than the original. And it explains why the combination of an overactive amygdala with an underactive ventromedial prefrontal cortex recurs so persistently in the imaging literature on anxiety disorders and post-traumatic stress disorder: the appraisal is intact and the brake is not.

19.2.4 Motivation and Behaviours

The behaviours the limbic system organises are traditionally summarised as feeding, fighting, fleeing, and reproduction, and although the mnemonic is crude it identifies the right set. What unifies them is that each is a case of the organism acting to change its own state rather than responding to the world, and each requires the same three things: a signal of what is needed, a valuation of what is available, and a mechanism for selecting among the options.

The valuation machinery is the **reward system**, and its core is dopaminergic. The **ventral tegmental area (VTA)** of the midbrain projects along the medial forebrain bundle to the **nucleus accumbens** in the ventral striatum (the **mesolimbic pathway**) and to the prefrontal cortex (the **mesocortical pathway**). These cells are close neighbours of the pars compacta neurons whose motor role was traced in the basal ganglia section, they release the same transmitter, and the parallel is not a coincidence. There is a **limbic loop** through the basal ganglia with exactly the architecture of the motor loop set out there: limbic and prefrontal cortex projects to the ventral striatum, the ventral striatum to the ventral pallidum, the ventral pallidum to the mediodorsal thalamus, and the thalamus back to the anterior cingulate and orbitofrontal cortex. The same disinhibitory arithmetic, the same dopaminergic gain control, applied not to the selection of a movement but to the selection of a goal.

Two refinements of the popular account are worth having. The first is that dopamine does not signal pleasure. What midbrain dopamine neurons encode is closer to a **reward prediction error**, the difference between the reward obtained and the reward expected: they fire to an unexpected reward, fall silent when an expected one fails to arrive, and come to fire to the cue that predicts a reward rather than to the reward itself. It is a teaching signal, not a pleasure signal, and the lateral habenula described above supplies its negative half. The second, related, refinement is the distinction between **wanting** and **liking**. Dopamine supports the incentive salience of a goal, the pull towards it and the effort spent pursuing it, while the hedonic impact of actually obtaining it depends on opioid and endocannabinoid signalling in small regions of the accumbens and pallidum. The two can be dissociated experimentally, and the dissociation is the most useful single fact about addiction: a drug can come to be wanted enormously while being liked very little.

Addiction follows from that architecture rather than from any special property of the drugs. The agents described in the mimicry subsection reach the same endpoint by different doors: nicotine as an agonist at nicotinic receptors sitting on VTA dopamine neurons, opioids as agonists at μ receptors that disinhibit them, and cannabinoids at CB1. Each raises accumbens dopamine more reliably and more steeply than any natural reward, and the teaching signal, which evolved on the assumption that such a discrepancy indicates something of real value, duly does its job.

The remaining behaviours are best traced as circuits already assembled. **Feeding and drinking** run from the arcuate, ventromedial, and lateral hypothalamic nuclei described earlier, and the lateral area's role in seeking food is inseparable from its position on the medial forebrain bundle: it is a hunger centre and a reward pathway at the same time. **Defence and aggression** run from the amygdala to the medial hypothalamus and thence to the periaqueductal grey, with the septal area exerting the restraining influence whose removal produces septal rage. **Sexual behaviour** depends on the medial preoptic area and the amygdala together, under endocrine control through the axes of the hypothalamus section.

The last point to make is the one the frontal lobe section anticipated. The limbic system

generates drives that are, in themselves, indifferent to circumstance: hunger does not know whether it is an appropriate moment to eat, and fear does not know whether the room contains people whose opinion matters. What supplies that knowledge is the prefrontal cortex, through its reciprocal connections with the amygdala, the hypothalamus, and the ventral striatum. Self-control, in the anatomical sense, is not the absence of limbic drive but the presence of a projection capable of modulating it, and most of the conditions in which behaviour goes conspicuously wrong (addiction, impulsivity after orbitofrontal damage, the disinhibition of frontotemporal dementia) are failures of that projection rather than excesses of the drive itself.

19.3 Summary

The limbic system is a set of structures defined by their connections rather than their location, forming a ring of transitional cortex around the corpus callosum together with a group of subcortical nuclei that ring encloses. The **limbic lobe** supplies the cortical component: the cingulate gyrus, whose anterior part weights experience for salience and affective pain and whose posterior part serves spatial and autobiographical memory, and the parahippocampal gyrus, whose entorhinal cortex is the gateway to the hippocampus. The **hippocampus**, three-layered and wired as an almost unidirectional trisynaptic circuit, is required for laying down new declarative memories and carries a map of space. The **amygdala** appraises significance, learning associations in its basolateral group and issuing them from its central nucleus to the hypothalamus, brainstem, and endocrine axis. The **hypothalamus** is the output stage that turns those judgments into bodily change; the **thalamus**, through its anterior, mediodorsal, midline, and intralaminar nuclei, is the route by which limbic signals reach the cortex and are gated on the way; the **septal area** paces the hippocampal theta rhythm and restrains the amygdala; and the **habenula** supplies the system's negative pole, inhibiting midbrain dopamine when an expected reward fails to arrive.

Functionally, the system is built around olfaction in the sense that smell reaches it directly and without a thalamic relay, which is why odour recruits memory and emotion more immediately than any other sense. Its memory role runs through the Papez circuit, damage anywhere along which produces amnesia, and rests cellularly on NMDA-dependent long-term potentiation, a coincidence detector requiring transmitter and depolarisation together. Its emotional role decomposes into appraisal by the amygdala, output through autonomic, endocrine, and behavioural limbs of quite different timescales, feeling constructed interoceptively in the insula, and regulation imposed from the ventromedial prefrontal cortex. Its motivational role runs through a limbic loop of the basal ganglia with precisely the architecture of the motor loop, in which dopamine from the ventral tegmental area encodes not pleasure but the discrepancy between reward expected and reward obtained, and sets the gain on whether a goal is pursued. The components and their principal connections are gathered in Table 9. Taken together, the limbic system is where the nervous system stops asking what is present and starts asking what it is worth and what happened last time, and it is the point at which those two questions are converted into a body doing something about it.

What remains is the route by which these judgments reach the body. The hypothalamus was called the output stage of this system, and the two sections that follow trace that output through the autonomic and enteric nervous systems to the organs, and trace the returning signals by which the state of the body is reported back.

Table 9: The principal components of the limbic system, with their major connections and the role each plays. The named tracts are those visible in dissection and used throughout this section; several structures appear here and in earlier sections under other headings, and the cross-references in the prose identify where.

Component	Main input	Main output	Role
Cingulate gyrus	Anterior thalamus; intralaminar nuclei	Cingulum to parahippocampal gyrus; autonomic centres	Salience, conflict, affective pain (anterior); spatial and autobiographical memory (posterior)
Parahippocampal gyrus (entorhinal)	Association cortex; olfactory tract	Perforant path to hippocampus	Gateway to and from the hippocampus; olfactory association; grid cells
Hippocampus	Entorhinal cortex (perforant path)	Fornix to mammillary bodies and septal area	Consolidation of declarative memory; spatial map
Amygdala	Thalamus (direct); association cortex; olfactory bulb	Stria terminalis and ventral amygdalofugal pathway to hypothalamus and brainstem	Appraisal of significance; fear conditioning; emotional output
Hypothalamus	Fornix; amygdalar pathways; prefrontal cortex	Brainstem and spinal autonomic centres; pituitary	Output stage: autonomic, endocrine, and behavioural response
Thalamus (anterior, MD, midline, intralaminar)	Mammillothalamic tract; amygdala; hypothalamus	Cingulate; prefrontal cortex; insula	Route to cortex for memory, emotion, and affective pain
Septal area	Fornix from hippocampus; hypothalamus	Septohippocampal pathway; stria medullaris to habenula	Hippocampal theta rhythm; restraint of aggression; reward
Habenula	Stria medullaris from septal area and basal forebrain	Fasciculus retroflexus to VTA, SNc, raphe	Inhibition of dopamine on reward omission; aversive signalling

20 The Autonomic Nervous System

The section on the limbic system ended at the hypothalamus, which it called the output stage: the point at which a judgment about what a situation means is converted into a change in the body. This section follows that output the rest of the way. The **autonomic nervous system** is the set of nerves that carries the brain's commands to the heart, the blood vessels, the airways, the gut, the bladder, the glands, and the pupils, and it was introduced in the first section as having two divisions with broadly opposite effects. The name, given by Langley in 1898, means "self-governing", and records the fact that it does its work without being directed and mostly without being noticed.

Its construction differs from that of the somatic motor system described in the section on the spinal cord in three ways, and they explain a good deal of its behaviour.

The first is that it uses **two neurons in series** where the somatic system uses one. A **preganglionic** neuron has its cell body in the brainstem or the spinal cord and sends a thin myelinated axon out to a **ganglion**, a cluster of nerve cell bodies outside the central nervous system. There it synapses on a **postganglionic** neuron, whose unmyelinated axon runs on to the target. The transmitter at the ganglion is always acetylcholine acting on nicotinic receptors,

in both divisions.

The second is that its targets are already active. A skeletal muscle does nothing unless its motor neuron fires, and the motor neuron can only excite it. The heart beats, the gut contracts, and the blood vessels hold their tone without any nerve supply at all. Autonomic nerves do not start these activities; they turn them up or down, and a given nerve may excite one tissue and inhibit another.

The third is the manner of release. A postganglionic axon does not end in a single tight synapse. It runs across its target bearing a long string of swellings, each of which releases transmitter into the surrounding tissue, where it spreads to many cells. The signal is slower and more diffuse than at a synapse in the brain or at a muscle, closer in character to the neuromodulators of the section on neuron structure than to the point-to-point transmitters.

20.1 The Sympathetic Nervous System

The preganglionic neurons of the sympathetic division lie in the lateral horn of the spinal cord, from the first thoracic segment to the second lumbar, and the division is accordingly said to have a **thoracolumbar outflow**. Their axons leave by the ventral root and at once pass, by a short connecting branch, into the **sympathetic chain**, a string of ganglia lying alongside the vertebral column on each side from the neck to the pelvis.

Once in the chain a fibre has three options. It may synapse in the ganglion at its own level. It may run up or down the chain and synapse in a ganglion at another level, which is how a division fed from only fourteen segments of the cord comes to supply every part of the body from the scalp to the feet. Or it may pass straight through the chain without synapsing and continue to one of the **prevertebral ganglia**, which lie in front of the aorta and supply the organs of the abdomen and pelvis. From the ganglia of the chain, postganglionic fibres rejoin the spinal nerves and travel with them to the blood vessels, sweat glands, and hair follicles of the skin. From the uppermost ganglion, in the neck, they follow the carotid artery into the head, where they dilate the pupil and help to hold up the eyelid.

That last route gives the division its most useful clinical sign. The pathway to the eye runs from the hypothalamus down through the brainstem to the first thoracic segment, out to the chain, up to the top of the neck, and back along the artery to the orbit. Interruption anywhere along it produces **Horner's syndrome**: on the affected side the pupil is small, the lid droops, and the face may not sweat. Because the route is so long, the finding can point to a lesion in the lateral medulla, in the cervical cord, at the top of the lung, or in the wall of the carotid artery, and the accompanying signs decide which.

Two features give the division its reach. One is **divergence**: each preganglionic neuron synapses on many postganglionic neurons, so a modest central command is distributed widely. The other is the **adrenal medulla**. Some preganglionic fibres pass through the chain and through the prevertebral ganglia and end directly on the cells of the core of the adrenal gland. Those cells are postganglionic neurons that have lost their axons, and instead of releasing their transmitter onto a target they release it into the blood, mostly as **adrenaline**. The arrangement is the one met in the posterior pituitary, a neuron turned into an endocrine cell. Circulating adrenaline reaches every tissue that carries a receptor for it, including tissues with no sympathetic nerve supply, and it lasts for minutes rather than seconds, which is why the body takes some time to settle after a fright that is over in an instant.

The classical picture, due to Cannon, is of a division that discharges as a whole in an emergency, and the phrase "fight or flight" is his. It is right about emergencies and misleading about everything else. The sympathetic supply to different organs is controlled separately: the vessels of the skin are adjusted to regulate temperature without any change in the heart, and the tonic drive from the medulla that holds the blood vessels partly constricted, described in the section on the medulla, never stops. Standing up would be impossible without it.

The postganglionic transmitter is **noradrenaline**, with one exception, the sweat glands, which are supplied by sympathetic fibres that release acetylcholine. What noradrenaline and adrenaline do to a tissue depends on which **adrenergic receptor** it carries, and these are the receptors listed in Table 1.

- α_1 receptors couple to G_q, a third type of G protein, which raises calcium inside the cell. In smooth muscle that means contraction: the blood vessels of the skin and gut constrict, and the muscle that dilates the pupil shortens.
- α_2 receptors couple to G_i and are found mainly on the sympathetic nerve terminals themselves, where they are the autoreceptors of the section on neuron firing, limiting further release.
- β_1 receptors couple to G_s and are the receptors of the heart, which beats faster and more forcefully.
- β_2 receptors also couple to G_s, but in smooth muscle the rise in cyclic AMP causes relaxation: the airways widen and the vessels supplying skeletal muscle dilate. They respond mainly to circulating adrenaline, and they also release glucose from the liver.

The redistribution of blood in an emergency, away from the skin and gut and towards the muscles, is therefore achieved by one chemical signal read by two receptors. The vessels of the skin carry mostly α_1 receptors and constrict; those of the muscles carry mostly β_2 and dilate. This is the principle stated at the end of the section on neuron firing, that the meaning of a signal is decided by the receiver, operating on the scale of the whole body.

20.2 The Parasympathetic Nervous System

The preganglionic neurons of the parasympathetic division lie in two widely separated places, the brainstem and the sacral segments of the spinal cord, and it is accordingly said to have a **craniosacral outflow**. Its anatomy is the reverse of the sympathetic in almost every respect. The preganglionic fibres are long and run all the way to the organ they serve; the ganglia lie in the wall of the organ or close beside it; the postganglionic fibres are short. There is little divergence, and no equivalent of the adrenal medulla. The division therefore acts on one organ at a time, and its effects are local and brief.

The cranial part travels in four of the cranial nerves, from the parasympathetic nuclei listed in the previous sections. Fibres in the oculomotor nerve constrict the pupil and focus the lens for near vision. Fibres in the facial and glossopharyngeal nerves supply the glands that produce tears and saliva. The remainder, about three quarters of the whole parasympathetic outflow, travels in the **vagus**, which supplies the heart, the airways, and the gut as far as the far end of the transverse colon. The sacral part, from the second to fourth sacral segments, supplies the rest of the colon, the bladder, and the genital organs. There is no parasympathetic supply to the skin, to the limbs, or to most blood vessels.

The transmitter at the target is **acetylcholine**, acting on **muscarinic receptors**, the metabotropic family introduced in the section on neuron firing. Two subtypes do most of the work. The $\mathbf{M_2}$ receptor is the receptor of the heart. It couples to G_i, and its $\beta\gamma$ subunits open the GIRK potassium channels described in that section; the pacemaker cells are hyperpolarised and take longer to reach threshold, and the heart slows. The $\mathbf{M_3}$ receptor couples to G_q and is the receptor of glands and of smooth muscle: saliva and digestive juices flow, the airways narrow, the gut contracts, and the pupil constricts. Atropine, the muscarinic antagonist met in the section on neuron firing, blocks all of these at once, which is why it dries the mouth, widens the pupils, and speeds the heart. The nerve agents mentioned in the same section do the opposite: by blocking the enzyme that destroys acetylcholine they flood the muscarinic receptors, and

the victim has pinpoint pupils, streaming secretions, narrowed airways, and a slow heart. The antidote is atropine.

The heart shows how constantly the division is at work. Left to itself, the pacemaker of the human heart fires about a hundred times a minute, which is the resting rate of a transplanted heart, whose nerves have been cut. The resting rate of sixty or seventy in a healthy person is the result of continuous braking by the vagus, known as **vagal tone**. The brake is also fast. Acetylcholine acts on the pacemaker by the direct $\beta\gamma$ route, with no second messenger involved, and is destroyed within a fraction of a second, so the vagus can change the interval between one beat and the next, whereas the sympathetic effect on the heart, which depends on a second messenger, takes several seconds to develop. Quick changes in heart rate are almost entirely vagal.

One such change occurs with every breath. The heart speeds up slightly during breathing in and slows during breathing out, because the respiratory generator of the medulla inhibits the cardiac neurons of the vagus during inspiration and releases them during expiration. The size of this fluctuation is a measure of vagal tone that can be taken from outside the body, and it is a main component of what is reported as **heart rate variability**.

20.3 Balance and Cooperation

Most organs of the chest and abdomen are supplied by both divisions, and where they are, the effects are usually opposed: the heart is sped by one and slowed by the other, the airways widened and narrowed, the gut quietened and activated, the pupil dilated and constricted. The opposition is not a single dial with two labels. The two divisions are controlled separately, each has a tonic level of activity, and the state of an organ at any moment is the outcome of both.

There are two kinds of exception. Some tissues are supplied by the sympathetic division alone: most blood vessels, the sweat glands, the muscles that raise the hairs, and the adrenal medulla. These are controlled by raising and lowering a single tonic drive. And in some functions the two divisions cooperate in sequence instead of opposing one another. In sexual function the parasympathetic division produces erection, by releasing nitric oxide, the gaseous messenger of the section on neuron structure, which relaxes the smooth muscle of the vessels; the sympathetic division produces ejaculation. In the bladder, the sympathetic division relaxes the wall and closes the outlet during filling, and the parasympathetic contracts the wall to empty it, with the changeover made by the centre in the pons described in the section on that structure and permitted by the prefrontal cortex.

Standing up shows the whole system working in a second or two. Blood shifts towards the legs, the pressure in the arteries of the neck falls, and the baroreflex of the medulla responds on both limbs at once: vagal tone is withdrawn and the heart speeds within a beat, and sympathetic drive to the vessels rises and they constrict. Nothing is felt. In diseases that destroy the autonomic neurons the response fails, and the patient loses consciousness on rising from a chair, which is the plainest evidence that the regulation is continuous and not reserved for emergencies. The effects of the two divisions on each organ are set out in Table 11.

20.4 Central Control and Visceral Afferents

The preganglionic neurons are the bottom of a hierarchy that has been assembled piece by piece in earlier sections. Reflexes are organised in the spinal cord and the medulla, around the solitary nucleus and the ventrolateral medulla. Above them, the parabrachial nuclei and the periaqueductal grey tie autonomic changes to breathing and to patterns of defensive behaviour. Above those, the hypothalamus supplies the set points and projects directly to the preganglionic neurons of both divisions. And the hypothalamus is in turn driven by the central nucleus of the amygdala, the anterior cingulate cortex, the insula, and the ventromedial prefrontal cortex. This chain is how something that exists only as an appraisal comes to change the body. A

remembered embarrassment reddens the face and the prospect of speaking in public dries the mouth, and in neither case has anything happened outside the head.

The traffic is not one way. Every autonomic nerve carries sensory fibres, and in the vagus they are the great majority. Stretch of the stomach and gut, the filling of the heart, the inflation of the lungs, and the chemistry of the blood are reported continuously to the solitary nucleus and passed up through the parabrachial nuclei to the hypothalamus, the amygdala, and, through the thalamus, the insula. The sympathetic nerves carry the fibres for pain from the internal organs into the dorsal horn. These last are responsible for a curious error. The dorsal horn neurons that receive pain fibres from the heart also receive them from the skin of the left arm and chest, which enter at the same segments, and the brain, having no other means of telling the two apart, attributes the signal to the surface. This is **referred pain**, and it is why a heart starved of blood is felt as a pain in the arm. Where a pain is felt is an inference, and here it is the wrong one.

The body is therefore not only the place where an emotion is expressed. Its state is fed back to the structures that produced the response and to the cortex that, according to the section on the insula, reads it as feeling. The section on the limbic system set out the old dispute about whether that feedback is part of what an emotion is, and the autonomic nervous system is where the evidence is to be found. It is mixed, and it is worth stating exactly.

- If feeling depends on feedback from the body, then cutting the feedback should blunt it. In 1966 Hohmann reported that men with spinal cord injuries described their emotions as less intense than before, and the more so the higher the injury. Later studies have mostly failed to confirm this. Patients whose peripheral autonomic neurons have degenerated, so that the body no longer responds at all, do show some dulling of emotional experience, but it is slight.

- If feeling depends on feedback, then producing the bodily state artificially should produce the feeling. Injections of adrenaline, studied by Marañón in the 1920s, produce a pounding heart and trembling, and most subjects report either no emotion or a detached sense of feeling "as if" afraid. The bodily state alone is not enough.

- Blocking the bodily response from outside does reduce some kinds of distress. **Beta blockers** such as propranolol, which block the β receptors of the heart, are widely used by performers against stage fright. They do not act on the circuits of fear; they stop the heart from pounding and the hands from shaking, and with those gone many performers report that the fear itself is less.

- Stimulating the afferent side has effects on mood. Electrical stimulation of the vagus in the neck, through an implanted device, is an established treatment for epilepsy and is used in depression that has not responded to drugs. It acts through the sensory fibres, by way of the solitary nucleus, on the locus coeruleus and the raphe nuclei. Its effect is modest and slow to develop.

The fair conclusion is that feedback from the body is one of the inputs from which feeling is made, and neither the whole of it nor dispensable.

Breathing gives this loop a voluntary handle, as the section on the medulla anticipated. Since the vagal brake on the heart is released during inspiration and applied during expiration, slow breathing with a long outward breath increases vagal influence on the heart, and the increase can be measured in the heart rate. That much is physiology. Whether practices built on it produce lasting changes in anxiety or mood is a separate question, on which the trials are encouraging but mostly small. An influential account known as the polyvagal theory makes much larger claims for the vagus in emotion and social behaviour; its anatomical and evolutionary premises have been strongly criticised, and it should not be taken as established physiology.

20.5 Summary

The autonomic nervous system carries the brain's commands to the heart, vessels, airways, gut, bladder, glands, and pupils through a chain of two neurons with a ganglion between them, and it modulates tissues that are already active. The sympathetic division arises from the thoracic and upper lumbar cord, relays in the sympathetic chain and the prevertebral ganglia, diverges widely, and is extended by the adrenal medulla, which releases adrenaline into the blood. Its transmitter is noradrenaline, and the effect on each tissue is set by whether that tissue carries α or β receptors. The parasympathetic division arises from the brainstem and the sacral cord, relays in ganglia at the organs themselves, and acts locally through acetylcholine on muscarinic receptors; three quarters of it travels in the vagus, and its continuous braking of the heart sets the resting pulse. The two usually oppose one another, sometimes act alone, and sometimes act in sequence, and both are commanded through a hierarchy that runs from the medulla through the hypothalamus to the amygdala, insula, and prefrontal cortex. The sensory fibres that accompany them return the state of the organs to the same structures. The divisions are compared in Table 10.

The first section said that a frightened mind produces a sympathetically driven body and that a calm body feeds back into a calmer mind. Both halves of that sentence now have an anatomy. The descending half runs from the amygdala and prefrontal cortex through the hypothalamus and brainstem to the preganglionic neurons; the ascending half runs from the organs through the vagus and the solitary nucleus to the insula. What this arrangement implies is that an emotion is not an event in the brain that has consequences in the body. It is a loop through both, in which appraisal alters the body, the altered body is sensed, and the sensing is part of what is felt. The pounding heart seems, from the inside, to be the result of the fear. The anatomy says that it is also one of the things the fear is made from, and the evidence, mixed as it is, says that this contribution is real and partial.

There is a further oddity worth registering. The autonomic nervous system is the least voluntary part of the nervous system and the part whose operation is furthest from awareness: no one knows what their blood vessels are doing. Yet its effects supply some of the most vivid and intimate contents of experience, the lurch of the stomach, the heat in the face, the tightness in the chest. What reaches awareness is not the regulation but a summary of its results, constructed, as referred pain shows, by inference, and capable of being wrong.

Table 10: The two divisions of the autonomic nervous system compared.

Feature	Sympathetic	Parasympathetic
Origin in the central nervous system	Lateral horn of the spinal cord, first thoracic to second lumbar segment	Brainstem nuclei of cranial nerves III, VII, IX, and X; second to fourth sacral segments
Ganglia	Sympathetic chain beside the vertebral column; prevertebral ganglia in front of the aorta	In or close to the organ supplied
Fibre lengths	Short preganglionic, long postganglionic	Long preganglionic, short postganglionic
Divergence	High: one preganglionic neuron drives many postganglionic neurons	Low: organs are controlled individually
Transmitter at the ganglion	Acetylcholine, on nicotinic receptors	Acetylcholine, on nicotinic receptors

Table 10 continued from previous page

Feature	Sympathetic	Parasympathetic
Transmitter at the target	Noradrenaline, on α and β receptors; acetylcholine at sweat glands	Acetylcholine, on muscarinic receptors
Hormonal arm	Adrenal medulla, releasing adrenaline into the blood	None
Distribution	Whole body, including the skin and blood vessels	Head, and the organs of the chest, abdomen, and pelvis
General effect	Mobilisation for exertion	Rest, digestion, and repair

Table 11: The effects of the two autonomic divisions on the main organs, with the receptor responsible where it is discussed in the text.

Organ	Sympathetic effect	Parasympathetic effect
Pupil	Dilates (α_1)	Constricts (M_3); lens focused for near vision
Heart	Faster and more forceful (β_1)	Slower (M_2)
Airways	Widen (β_2)	Narrow, with more secretion (M_3)
Vessels of skin and gut	Constrict (α_1)	No supply
Vessels of skeletal muscle	Dilate (β_2, mainly to circulating adrenaline)	No supply
Gut	Movement and secretion reduced; sphincters closed	Movement and secretion increased (M_3), acting through the enteric nervous system
Salivary glands	A little thick saliva	Copious watery saliva
Sweat glands	Sweating (acetylcholine, on muscarinic receptors)	No supply
Liver	Releases glucose (β_2)	Little direct effect
Bladder	Wall relaxed and outlet closed: storage	Wall contracts: emptying
Genital organs	Ejaculation	Erection, through nitric oxide
Adrenal medulla	Releases adrenaline and noradrenaline	No supply

21 The Enteric Nervous System

The first section mentioned a third component of the autonomic system, a network of neurons in the wall of the gut that runs digestion largely by itself. Langley, who gave the autonomic system its name, counted this network in 1921 as a division in its own right, alongside the sympathetic and the parasympathetic, and he had a good reason. The **enteric nervous system** is the only part of the peripheral nervous system that can carry out complete reflexes, from sensation to response, with every connection to the brain and spinal cord cut.

It is also large. It contains somewhere between two hundred and six hundred million neurons, about as many as the spinal cord, spread along the whole length of the gut from the oesophagus to the anus. It has been called the "second brain", a phrase that overstates the case and still points at something real: this is a nervous system with its own sensory neurons, interneurons, and motor neurons, its own glia, and most of the transmitters found in the head. This last section of the neurology sets out what it consists of, what it can do unaided, and how it is connected to the brain.

21.1 Structure and Components

The wall of the gut is built in layers. Innermost is the **mucosa**, the lining that secretes and absorbs. Outside it is a layer of connective tissue, the **submucosa**, and outside that are two layers of smooth muscle: an inner layer whose fibres run around the tube and narrow it when they contract, and an outer layer whose fibres run along the tube and shorten it.

The enteric neurons lie in two sheets within this wall, each a mesh of small ganglia joined by strands of nerve fibres.

- The **myenteric plexus** lies between the two layers of muscle and runs the whole length of the gut. It controls movement.
- The **submucosal plexus** lies in the submucosa and is best developed in the intestines. It controls secretion, absorption, and the local flow of blood.

The two are interconnected, and between them they contain every element needed for a reflex. **Intrinsic sensory neurons** have endings in the mucosa and the muscle and respond to stretch of the wall and to the chemistry of what the gut contains. **Interneurons** run up and down the gut within the myenteric plexus, linking one level to the next. **Motor neurons** are of two kinds: excitatory neurons, which release acetylcholine and substance P and make the muscle contract, and inhibitory neurons, which release nitric oxide, a peptide called VIP, and ATP and make it relax. Further neurons in the submucosal plexus drive the secreting cells and dilate the local blood vessels. Something like twenty types of enteric neuron have been distinguished, and nearly every transmitter introduced in the section on neuron structure is found among them, including the unconventional ones: nitric oxide, ATP, and the neuropeptides are all in routine use here.

Three other kinds of cell work alongside the neurons. **Enteric glia** surround the neurons in the ganglia and resemble the astrocytes of the brain far more than they resemble the Schwann cells of other peripheral nerves. The **interstitial cells of Cajal** are pacemakers. They generate a slow electrical rhythm that spreads through the muscle, about three cycles a minute in the stomach and about twelve in the upper small intestine, and this rhythm sets the times at which the muscle is able to contract. Whether it does contract on any given cycle is decided by the neurons. And scattered through the lining are the **enterochromaffin cells**, which are sensors: they respond to pressure on the mucosa and to chemicals in the gut by releasing serotonin onto the nerve endings beneath them. The cells and their transmitters are gathered in Table 12.

21.2 Reflexes Without the Brain

The basic act of the gut is **peristalsis**, the movement that carries its contents onward. When a section of intestine is stretched by something inside it, the muscle just above the stretched point contracts and the muscle just below it relaxes, and the contents are pushed forward into the relaxed segment, where the same thing happens again. Bayliss and Starling described this in 1899 and called it the law of the intestine. They showed that it survives the cutting of every nerve connecting the gut to the brain and spinal cord, and it was later shown to occur in a length of intestine removed from the body and kept alive in a dish.

The circuit is the one the inventory above would suggest. Sensory neurons detect the stretch, helped by serotonin released from the enterochromaffin cells. They excite interneurons running in both directions. Those running upward excite the excitatory motor neurons, and the muscle behind the contents contracts; those running downward excite the inhibitory motor neurons, and the muscle ahead relaxes. Sensation, integration, and a coordinated motor response in two directions are all accomplished within a few centimetres of gut wall.

Peristalsis is one programme among several, and the enteric nervous system chooses among them according to what the gut contains. After a meal the small intestine switches to **segmentation**, in which rings of contraction appear and disappear without travelling, mixing the contents and holding them against the absorbing surface. During fasting it switches to the **migrating motor complex**, a band of strong contractions that begins in the stomach and sweeps slowly down the whole small intestine about every ninety minutes, clearing out debris and bacteria; the rumbling of an empty stomach is this programme at work. In vomiting the direction of travel is reversed. The selection among these is made locally, on the basis of what the sensory neurons report.

That these neurons are necessary, and what they chiefly do, is shown by two diseases in which they are missing. The enteric neurons arrive in the gut during development by migrating down it from the head end, and in **Hirschsprung's disease** the migration stops short, leaving the last part of the colon without ganglia. That segment does not go limp. It remains tightly and permanently contracted, nothing can pass it, and the bowel above swells enormously. In **achalasia** the inhibitory neurons at the lower end of the oesophagus degenerate, and the ring of muscle there fails to open on swallowing. In both, the muscle is perfectly able to contract; what it has lost is the ability to relax. The dominant action of the enteric nervous system on gut muscle is inhibitory, and removing it reveals, as with the released reflexes of the spinal cord, not stillness but unrestrained activity.

21.3 Serotonin in the Gut

More than nine tenths of the serotonin in the body is in the gut. Most of it is in the enterochromaffin cells and a smaller part in enteric neurons, which use it as a transmitter. Released by the enterochromaffin cells, it acts on the intrinsic sensory neurons to help start the reflexes just described, and on the endings of vagal sensory fibres, which carry the signal to the brainstem. It is cleared, as in the brain, by the serotonin transporter, here in the cells of the gut lining.

Several facts noted in the section on neuron firing follow from this. The SSRIs block the transporter in the gut as they do in the brain, serotonin rises around the nerve endings of the mucosa, and the result is the nausea and looseness of the bowels that are common in the first weeks of treatment. The vagal sensory endings carry 5-HT_3 receptors, and when the lining of the gut is damaged, as it is by many of the drugs used in chemotherapy, the enterochromaffin cells release serotonin in bulk and the vagus reports it to the solitary nucleus and the area postrema as a reason to vomit. This is the signal that ondansetron blocks.

One caution is needed, because the figure for gut serotonin is often quoted as though it meant that mood is manufactured in the bowel. Serotonin does not cross the blood-brain barrier. The serotonin of the brain is made in the brain, by the raphe neurons, from the amino acid tryptophan, and none of the serotonin in the gut ever reaches a receptor in the cortex. Whatever influence the gut has on the brain, it does not exert it by supplying the transmitter.

21.4 The Gut–Brain Axis

The brain's control over the gut is real but indirect. The fibres of the vagus do not, for the most part, end on the muscle or the glands of the gut. They end on enteric neurons, and so do the sympathetic fibres from the prevertebral ganglia, which mainly inhibit them. The brain addresses the gut in much the way the cortex addresses the spinal cord: it does not operate

the muscles but starts, stops, and biases programmes that the local circuits already contain. The sight and smell of food set the stomach secreting through the vagus before anything has been swallowed. Fear and exertion shut digestion down through the sympathetic division. Stress acting over a long period alters the movement and the sensitivity of the bowel, and the common condition called **irritable bowel syndrome**, in which pain and disordered bowel habit occur without any structural disease, is now classified as a disorder of the interaction between gut and brain; the low doses of antidepressant drugs used to treat it act on that interaction.

The traffic in the other direction is heavier. As the section on the cranial nerves noted, the great majority of the fibres in the vagus are sensory, and a large share of them come from the gut. They report its stretch and its contents to the solitary nucleus, from which the information passes to the hypothalamus, the amygdala, and the insula. Alongside the nerves there is a hormonal channel. Cells in the gut lining release hormones in response to food, some of which act on the vagal endings and some of which travel in the blood to the hypothalamus and brainstem. **Ghrelin**, from the empty stomach, promotes hunger, and several others released after a meal, among them **GLP-1**, promote the sense of having had enough. These are the circulating signals of energy state that the section on the hypothalamus described as acting on the arcuate nucleus, and the drugs now widely used to treat obesity are mimics of GLP-1, agonists at its receptor in the sense defined in the section on neuron firing. What a person experiences of all this is very little: hunger, fullness, nausea, and occasionally pain.

Two further claims about this axis have attracted a great deal of attention, and both need to be held with some care.

The first concerns the bacteria of the gut. The colon contains tens of trillions of them, and they produce compounds that act on enteric neurons, on enterochromaffin cells, on the immune system, and on the vagus. The evidence from animals that they influence the brain is strong. Mice raised without any gut bacteria show an exaggerated hormonal response to stress, which is corrected if bacteria are introduced early in life, and some patterns of behaviour can be transferred from one animal to another by transferring the bacteria. In humans the composition of the gut bacteria differs on average between people with and without depression and several other conditions, but those findings are correlations, in populations that also differ in diet and medication, and trials of treatments aimed at the bacteria have so far been small and inconsistent. That the bacteria of the gut affect the human mind is a reasonable hypothesis and not yet an established result.

The second concerns Parkinson's disease. Constipation commonly precedes the disorder of movement by many years, and the abnormal protein deposits that characterise the disease are found in enteric neurons as well as in the brain. Braak proposed in 2003 that in many patients the disease process begins outside the brain, in the gut and the olfactory system, and spreads along the vagus to its nucleus in the medulla, then upward through the pons to the midbrain, reaching the substantia nigra only late. The proposal has the merit of accounting for three observations recorded separately in these notes: the early loss of smell mentioned in the section on the limbic system, the REM sleep behaviour disorder that the section on the pons described as preceding the disease by years, and the constipation noted here. Each would be a sign of the process at an earlier station. Two large registry studies of people whose vagus nerves had been cut surgically found a lower rate of Parkinson's disease in later years, which is what the hypothesis predicts, but the effect was modest and in one of the two it was not statistically secure, and the proposed sequence does not fit every patient. It remains a hypothesis.

21.5 Summary

The enteric nervous system is a network of several hundred million neurons in the wall of the gut, arranged in a myenteric plexus between the muscle layers, which controls movement, and a submucosal plexus, which controls secretion and blood flow. It contains sensory neurons, interneurons, and excitatory and inhibitory motor neurons, supported by glia, paced by the

interstitial cells of Cajal, and informed by the serotonin-releasing enterochromaffin cells of the lining. With these it carries out the peristaltic reflex and selects among programmes of movement without any help from the brain or spinal cord, and the diseases in which its neurons are absent show that its main action on the muscle is to relax it. Most of the body's serotonin is here, which explains the effects of the SSRIs on the gut, but none of it reaches the brain. The brain modulates the system through the vagus and the sympathetic nerves, acting on enteric neurons and not on the gut directly, and it is informed in return by vagal sensory fibres and by hormones. Claims that the bacteria of the gut shape the mind, and that Parkinson's disease begins in the gut, are plausible and unproven.

This system is the limiting case of something that has come up repeatedly since the section on the midbrain. Judged by the terms in which these notes began, it lacks nothing. It has receptors and channels, the same transmitters as the brain, synapses that excite and inhibit, sensory, integrating, and motor neurons, glia, reflexes, and a repertoire of programmes among which it chooses according to circumstances. It is unmistakably a nervous system doing what nervous systems do. And nothing in anyone's experience corresponds to what it is doing. The contractions of the intestine, the wave that sweeps it clean every ninety minutes, the adjustment of secretion to each meal: none of this is felt. What reaches awareness is a thin abstract, carried up the vagus and worked over by the insula, consisting of hunger, fullness, nausea, and the vaguer states that go by the name of gut feelings.

Set beside the cerebellum, the spinal cord, and the vital networks of the medulla, the point becomes general. Being made of neurons, using the brain's chemistry, and carrying out a computation, even an elaborate and adaptive one, is not enough for a process to form part of experience. Most of what the nervous system does is of this silent kind, and the part that is experienced is the exception. Which neural processes are accompanied by awareness, and what distinguishes them from the far larger number that are not, is therefore a question with real contrasts to work from, and it is a question about mechanism, of the kind the introduction called easy. It is a different question from why any neural process should be accompanied by awareness at all, and that one has been left throughout exactly where the introduction put it.

Table 12: The cells of the enteric nervous system and the cells that work with it, with their principal transmitters and what each contributes.

Cell	**Transmitters**	**Role**
Intrinsic sensory neurons	Acetylcholine, substance P	Detect stretch and the chemistry of the contents; begin local reflexes
Interneurons	Acetylcholine; also serotonin, nitric oxide, ATP	Carry the reflex up and down the gut and set the pattern of movement
Excitatory motor neurons	Acetylcholine, substance P	Contract the muscle behind the contents
Inhibitory motor neurons	Nitric oxide, VIP, ATP	Relax the muscle ahead; open the sphincters
Secretomotor neurons	Acetylcholine, VIP	Drive secretion; widen local blood vessels
Enteric glia	—	Support the neurons, as astrocytes do in the brain
Interstitial cells of Cajal	—	Pacemakers for the slow rhythm of the muscle
Enterochromaffin cells	Serotonin	Sense the contents; signal to sensory neurons and the vagus